1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ZONE	ECN	DESCRIPTION OF CHANGE	CK APPD DATE	ENG APPD DATE
?		?	?	?	?

SCHEM,CORNHOLE,K19

PVT 04/24/2009

ALIASES RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-7892	1	SCHEM,CORNHOLE,K19	SCH	CRITICAL	
820-2523	1	PCBF,CORNHOLE,K19	PCB	CRITICAL	

DRAWING TITLE=MLB
ABBREV=DRAWING
LAST_MODIFIED=Fri Apr 24 15:23:24 2009

DIMENSIONS ARE IN MILLIMETERS
XX ±
X.XX ±
X.XXX ±
ANGLES ±
DO NOT SCALE DRAWING
THIRD ANGLE PROJECTION

METRIC

DRAFTER | DESIGN CK | ENG APPD | MFG APPD | QA APPD | DESIGNER | RELEASE | SCALE NONE

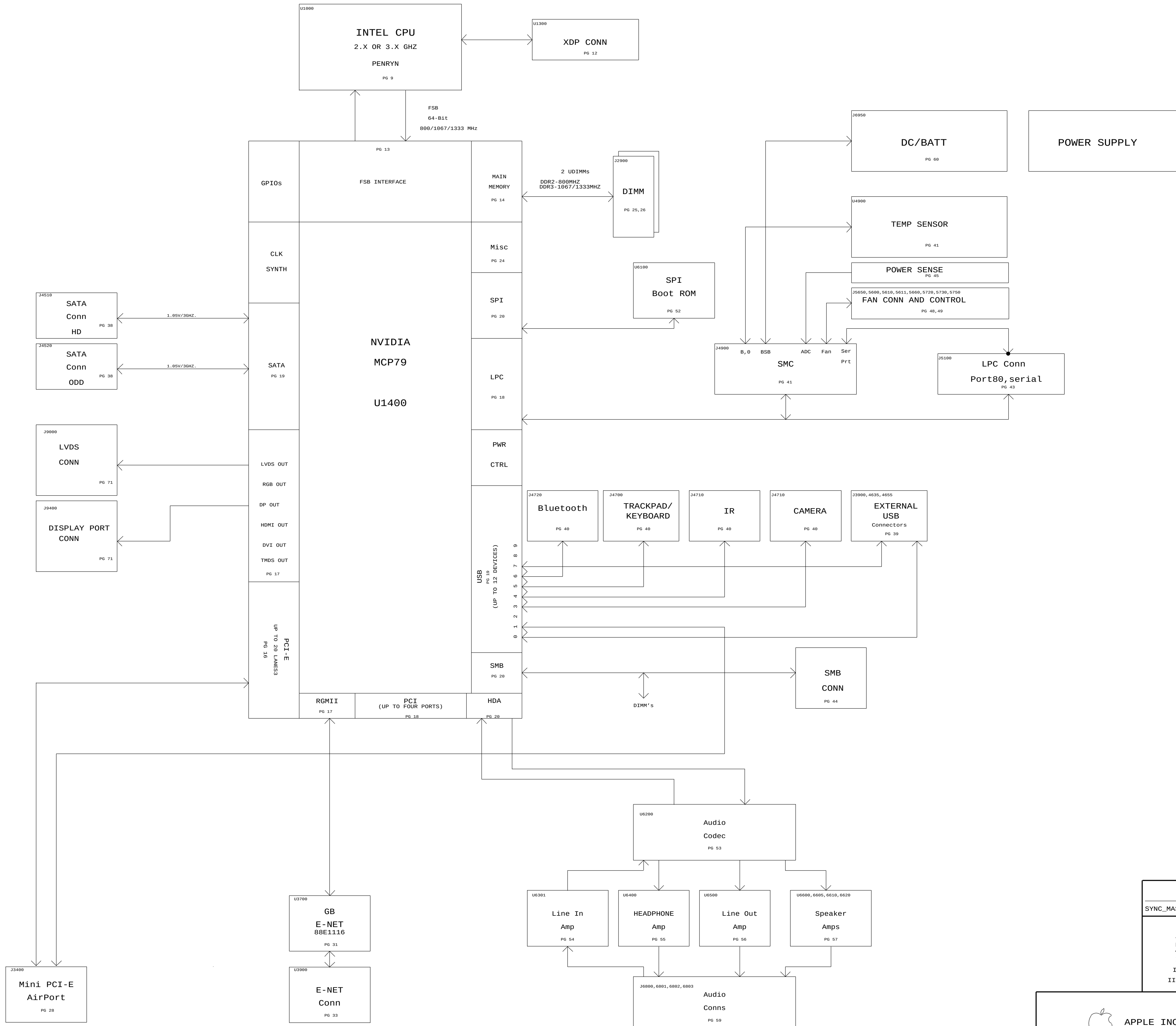
MATERIAL/FINISH NOTED AS APPLICABLE

APPLE INC.

NOTICE OF PROPRIETARY PROPERTY

TITLE SCHEM,MBP 15MLB

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0



System Block Diagram

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2007

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

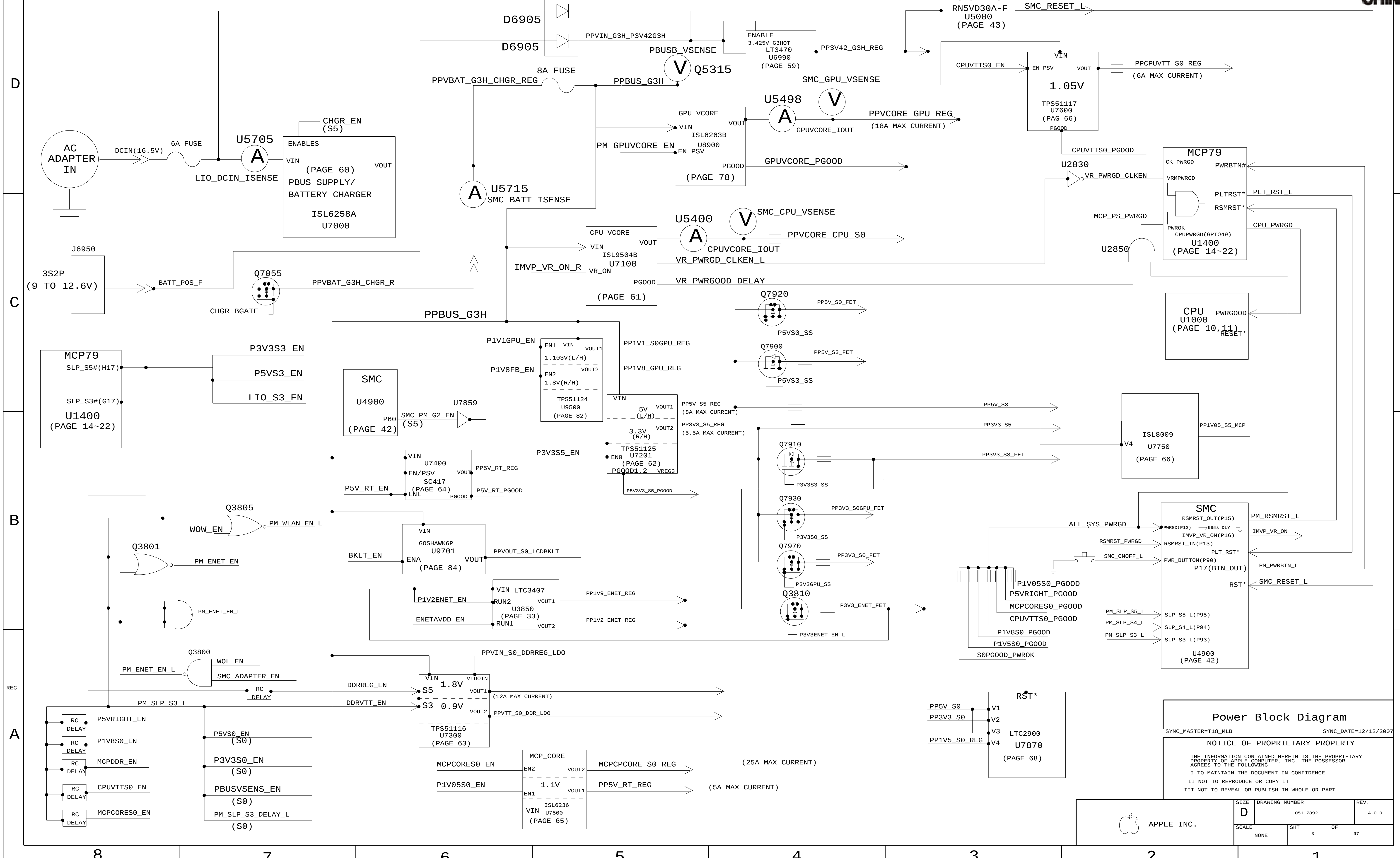
II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 2	OF 97

M98 POWER SYSTEM ARCHITECTURE



Power Block Diagram

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2007

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 3	OF 97

Power Block Diagram

SYNC_MASTER=N/A SYNC_DATE=N/A

APPLE INC.	SIZE D	DRAWING NUMBER 051-7892	REV. A.0.0
	SCALE NONE	SHT 4	OF 97

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
630-9965	PCBA,2.66GHZ,256SAM_VRAM,HB_AUDIO,K19	K19_COMMON,DEVEL_BOM,EEE_6XN,CPU_2_66GHZ,FB_256_SAMSUNG
630-9966	PCBA,2.66GHZ,256HYN_VRAM,HB_AUDIO,K19	K19_COMMON,DEVEL_BOM,EEE_6XP,CPU_2_66GHZ,FB_256_HYNIX
630-9967	PCBA,2.80GHZ,512SAM_VRAM,HB_AUDIO,K19	K19_COMMON,DEVEL_BOM,EEE_6XQ,CPU_2_80GHZ,FB_512_SAMSUNG
630-9968	PCBA,2.80GHZ,512HYN_VRAM,HB_AUDIO,K19	K19_COMMON,DEVEL_BOM,EEE_6XR,CPU_2_80GHZ,FB_512_HYNIX
630-9969	PCBA,3.06GHZ,512SAM_VRAM,HB_AUDIO,K19	K19_COMMON,DEVEL_BOM,EEE_6XS,CPU_3_06GHZ,FB_512_SAMSUNG
630-9970	PCBA,3.06GHZ,512HYN_VRAM,HB_AUDIO,K19	K19_COMMON,DEVEL_BOM,EEE_6XT,CPU_3_06GHZ,FB_512_HYNIX
085-0736	K19 MLB DEVELOPMENT	K19_DEVEL_PVT

K19 BOM Groups

BOM GROUP	BOM OPTIONS
K19_COMMON	ALTERNATE,COMMON,K19,K19_COMMON1,K19_COMMON2,K19_PROGPARTS
K19_COMMON1	BOOT_MODE_USER,DPMUX_EN_S0,DP_CA_DET_EG_PLD,DP_ESD,EG_PWRSEQ_HW,EXTRACT_BUFF
K19_COMMON2	GMUX_1V8,GPUVID_1P00V,GPU_SS_INT,ISL6258A,MCP_B03,MCPSEQ_SMC,MIKEY,MUXGFX,SMC_DEBUG_YES,XDP
K19_DEVEL_ENG	BMON_ENG,DEBUG_ADC,GMUX_JTAG,LPCPLUS,VREFMRGN,XDP_CONN
K19_DEVEL_PVT	BMON_PROD,LPCPLUS,NO_VREFMRGN,XDP_CONN
K19_PROD	BMON_PROD,LPCPLUS_NOT,NO_VREFMRGN
K19_PROGPARTS	GMUX_PROG,BOOTROM_PROG,SMC_PROG,TPAD_PROG

BOM GROUP	BOM OPTIONS
FB_256_SAMSUNG	VRAM4,VRAM_256_SAMSUNG
FB_256_HYNIX	VRAM4,VRAM_256_HYNIX
FB_512_SAMSUNG	VRAM4,VRAM_512_SAMSUNG
FB_512_HYNIX	VRAM4,VRAM_512_HYNIX

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEE:6XN]	CRITICAL	EEE_6XN
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEE:6XP]	CRITICAL	EEE_6XP
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEE:6XQ]	CRITICAL	EEE_6XQ
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEE:6XR]	CRITICAL	EEE_6XR
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEE:6XS]	CRITICAL	EEE_6XS
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEE:6XT]	CRITICAL	EEE_6XT

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3761	1	IC,PDC,SLGLA,PRQ,2.66G,25W,1066,R0,3M,BGA	U1000	CRITICAL	CPU_2_66GHZ
337S3682	1	IC,PDC,SLGEM,PRQ,2.80G,35W,1066,E0,6M,BGA	U1000	CRITICAL	CPU_2_80GHZ
337S3744	1	IC,PDC,SLGKH,QS,3.06G,35W,1066,E0,6M,BGA	U1000	CRITICAL	CPU_3_06GHZ
338S0710	1	IC,MCP79MXT-B3,35X35MM,BGA1437	U1400	CRITICAL	MCP_B03
338S0694	1	IC,RTL8251CA-VB-GR,GIGE TRANSCEIVER,48P LQFP	U3700	CRITICAL	
338S0654	1	IC,FW643-E,1394B PHY/OHCI LINK/PCI-E,12	U4100	CRITICAL	
341S2384	1	IR,ENCORE II, CY7C63803-LQXC	U4800	CRITICAL	
338S0563	1	IC,SMC,HS8/2117,9MMX9MM,TLP	U4900	CRITICAL	SMC_BLANK
341S2462	1	IC,SMC,DEVELOPMENT,K19	U4900	CRITICAL	SMC_PROG
341S2503	1	IC,PSOC +W/USB,56PIN,MLF,K19	U5701	CRITICAL	TPAD_PROG
335S0384	1	IC,32MBIT 8-PIN SPI SERIAL FLASH,SOIC8	U6100	CRITICAL	BOOTROM_BLANK
341S2456	1	IC,EFI ROM,DEVELOPMENT,K19	U6100	CRITICAL	BOOTROM_PROG
338S0554	1	IC,GPU,55nm,NV G96-GS,BGA969,LF	U8000	CRITICAL	
333S0507	4	IC,SGRAM,GDDR3,16Mx32,1000MHZ,136 FBGA	U8400,U8450,U8500,U8550	CRITICAL	VRAM_256_SAMSUNG
333S0483	4	IC,SGRAM,GDDR3,16Mx32,900MHZ,136 FBGA	U8400,U8450,U8500,U8550	CRITICAL	VRAM_256_HYNIX
333S0511	4	IC,SGRAM,GDDR3,32Mx32,800MHZ,136 FBGA	U8400,U8450,U8500,U8550	CRITICAL	VRAM_512_SAMSUNG
333S0506	4	IC,SGRAM,GDDR3,32Mx32,900MHZ,136 FBGA	U8400,U8450,U8500,U8550	CRITICAL	VRAM_512_HYNIX

Development BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-0736	1	K19 MLB DEVELOPMENT	DEVEL	CRITICAL	DEVEL_BOM

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0603	138S0602		ALL	Murata alt to Samsung
353S1681	353S1294		ALL	LMV2011,OPAMP. GBW
152S0276	152S0683		ALL	Maglayers alt to Dale/Vishay
341S2367	341S2366		ALL	Macronix alt to SST
152S1034	152S0867		ALL	Toko alt to Delta
157S0058	157S0055		ALL	Delta alt to TDK Magnetics
152S0915	152S0796		ALL	Maglayers alt to Cyntec IND
128S0220	128S0262		ALL	KEMET ALT TO SANYO
127S0062	127S0108		ALL	ROHM ALT TO KEMET
152S0968	152S0966		ALL	Maglayer alt to Delta
311S0447	311S0406		ALL	NXP alt to TI
338S0714	338S0554		ALL	Low Leakage G96 GPU
107S0138	107S0074		ALL	CYNTEC alt to YDS
107S0139	107S0075		ALL	CYNTEC alt to YDS

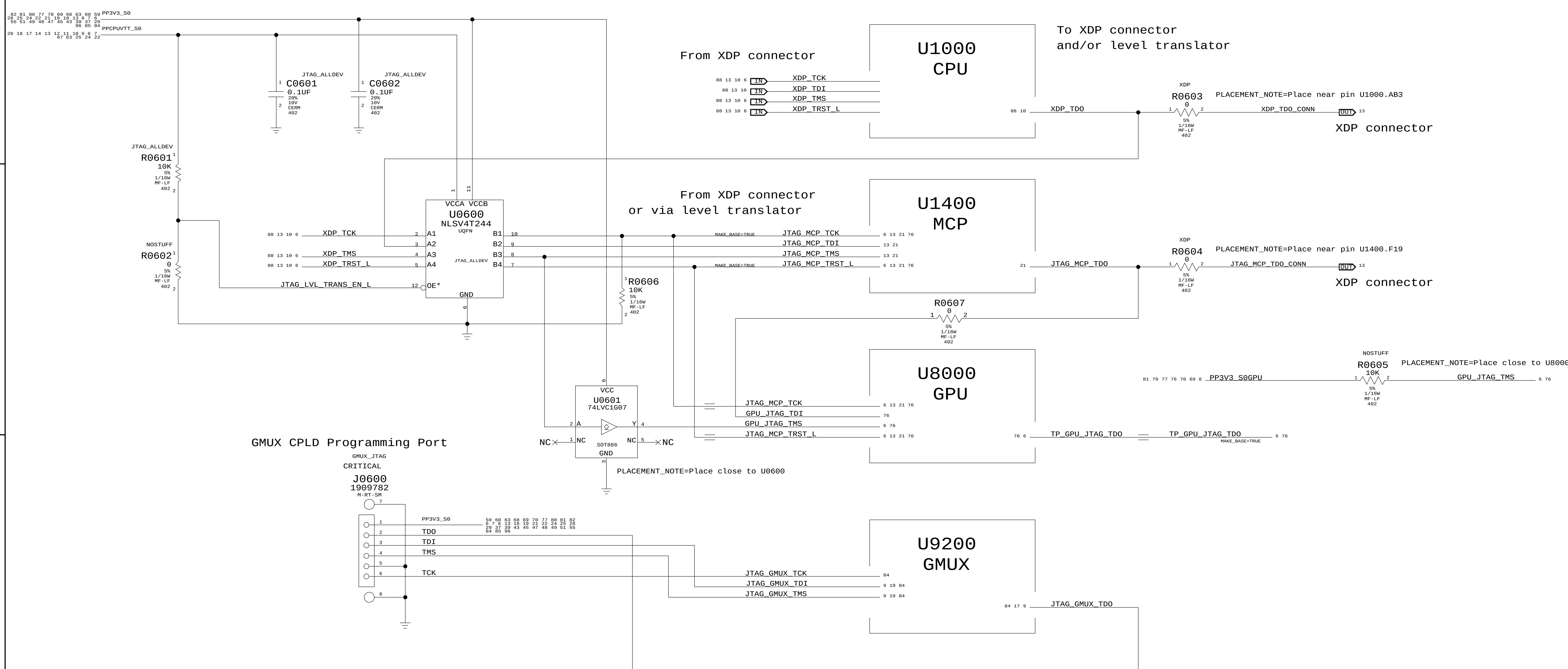
BOM Configuration

SYNC_MASTER=DDR SYNC_DATE=12/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 5	OF 97

1.05V TO 3.3V LEVEL TRANSLATOR (M98: ON ICT FIXTURE)



JTAG Scan Chain

SYNC_MASTER=DDR SYNC_DATE=07/22/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 6	OF 97

Functional Test Points

Fan Connectors

FUNC_TEST

- TRUE PP5V_S0 — 3 TPs
- TRUE FAN_LT_PWM
- TRUE FAN_LT_TACH
- TRUE FAN_RT_PWM
- TRUE FAN_RT_TACH
- TRUE GND — 5 TPs

LVDS Connector

FUNC_TEST

- TRUE PP3V3_S0
- TRUE PP3V3_SW_LCD
- TRUE PPVOUT_S0_LCDBKLT
- TRUE LVDS_DDC_CLK
- TRUE LVDS_DDC_DATA
- TRUE LVDS_CONN_A_DATA_N<0>
- TRUE LVDS_CONN_A_DATA_P<0>
- TRUE LVDS_CONN_A_DATA_N<1>
- TRUE LVDS_CONN_A_DATA_P<1>
- TRUE LVDS_CONN_A_DATA_N<2>
- TRUE LVDS_CONN_A_DATA_P<2>
- TRUE LVDS_CONN_A_CLK_F_N
- TRUE LVDS_CONN_A_CLK_F_P
- TRUE LVDS_CONN_B_DATA_N<0>
- TRUE LVDS_CONN_B_DATA_P<0>
- TRUE LVDS_CONN_B_DATA_N<1>
- TRUE LVDS_CONN_B_DATA_P<1>
- TRUE LVDS_CONN_B_DATA_N<2>
- TRUE LVDS_CONN_B_DATA_P<2>
- TRUE LVDS_CONN_B_CLK_F_N
- TRUE LVDS_CONN_B_CLK_F_P
- TRUE LED_RETURN_1
- TRUE LED_RETURN_2
- TRUE LED_RETURN_3
- TRUE LED_RETURN_4
- TRUE LED_RETURN_5
- TRUE LED_RETURN_6
- TRUE BKL_ISEN1
- TRUE BKL_ISEN2
- TRUE BKL_ISEN3
- TRUE BKL_ISEN4
- TRUE BKL_ISEN5
- TRUE BKL_ISEN6
- TRUE GND — 5 TPs

IPD Flex Connector

FUNC_TEST

- TRUE PP3V3_S3_LDO
- TRUE PP18V5_S3
- TRUE Z2_CS_L
- TRUE Z2_DEBUG3
- TRUE Z2_MOSI
- TRUE Z2_MISO
- TRUE Z2_SCLK
- TRUE Z2_BOOST_EN
- TRUE Z2_HOST_INTN
- TRUE Z2_CLKIN
- TRUE Z2_KEY_ACT_L
- TRUE Z2_RESET
- TRUE PSOC_MISO
- TRUE PSOC_MOSI
- TRUE PSOC_SCLK
- TRUE SMBUS_SMC_A_S3_SDA
- TRUE SMBUS_SMC_A_S3_SCL
- TRUE PSOC_F_CS_L
- TRUE PICKB_L
- TRUE GND — 2 TPs

SD Card Connector

FUNC_TEST

- TRUE SD_D<7..0>
- TRUE SD_CMD
- TRUE SD_CLK
- TRUE SD_CD_L
- TRUE SD_WP
- TRUE GND — 2 TPs

Speaker Connectors

FUNC_TEST

- TRUE BI_MIC_LO
- TRUE BI_MIC_SHIELD
- TRUE BI_MIC_HI
- TRUE SPKRCONN_L_OUT_P
- TRUE SPKRCONN_L_OUT_N
- TRUE SPKRCONN_R_OUT_P
- TRUE SPKRCONN_R_OUT_N
- TRUE SPKRCONN_S_OUT_P
- TRUE SPKRCONN_S_OUT_N
- TRUE GND — 6 TPs

SATA ODD Connectors

FUNC_TEST

- TRUE PP5V_SW_ODD — 4 TPs
- TRUE SMC_ODD_DETECT
- TRUE SATA_ODD_R2D_P
- TRUE SATA_ODD_R2D_N
- TRUE SATA_ODD_D2R_C_N
- TRUE SATA_ODD_D2R_C_P
- TRUE GND — 3 TPs

Keyboard Connector

FUNC_TEST

- TRUE PP3V3_S3
- TRUE PP3V42_G3H
- TRUE WS_KBD1
- TRUE WS_KBD2
- TRUE WS_KBD3
- TRUE WS_KBD4
- TRUE WS_KBD5
- TRUE WS_KBD6
- TRUE WS_KBD7
- TRUE WS_KBD8
- TRUE WS_KBD9
- TRUE WS_KBD10
- TRUE WS_KBD11
- TRUE WS_KBD12
- TRUE WS_KBD13
- TRUE WS_KBD14
- TRUE WS_KBD15_CAP
- TRUE WS_KBD16_NUM
- TRUE WS_KBD17
- TRUE WS_KBD18
- TRUE WS_KBD19
- TRUE WS_KBD20
- TRUE WS_KBD21
- TRUE WS_KBD22
- TRUE WS_KBD23
- TRUE WS_KBD_ONOFF_L
- TRUE WS_LEFT_SHIFT_KBD
- TRUE WS_LEFT_OPTION_KBD
- TRUE WS_CONTROL_KBD
- TRUE KBDLED_ANODE
- TRUE GND — 2 TPs

Airport/BT/Camera Conn.

FUNC_TEST

- TRUE PCIE_MINI_D2R_P
- TRUE PCIE_MINI_D2R_N
- TRUE PCIE_MINI_R2D_P
- TRUE PCIE_MINI_R2D_N
- TRUE PCIE_CLK100M_MINI_CONN_P
- TRUE PCIE_CLK100M_MINI_CONN_N
- TRUE MINI_CLKREQ_Q_L
- TRUE PCIE_WAKE_L
- TRUE MINI_RESET_CONN_L
- TRUE PP5V_WLAN
- TRUE PP3V3_S3_BT_F
- TRUE PP5V_S3_BTCAMERA_F
- TRUE SMBUS_SMC_A_S3_SDA
- TRUE SMBUS_SMC_A_S3_SCL
- TRUE USB_CAMERA_CONN_P
- TRUE USB_CAMERA_CONN_N
- TRUE CONN_USB2_BT_P
- TRUE CONN_USB2_BT_N
- TRUE GND — 10 TPs

SATA HDD Connector

FUNC_TEST

- TRUE PP5V_S0_HDD_FLT — 4 TPs
- TRUE PP5V_S3_IR_R
- TRUE SATA_HDD_R2D_P
- TRUE SATA_HDD_R2D_N
- TRUE SATA_HDD_D2R_C_P
- TRUE SATA_HDD_D2R_C_N
- TRUE IR_RX_OUT
- TRUE SYS_LED_ANODE_R
- TRUE GND — 6 TPs

KBD Backlight Conn.

FUNC_TEST

- TRUE KBDLED_ANODE — 2 TPs
- TRUE SMC_KDBLED_PRESENT_L
- TRUE GND

DC Power Connector

FUNC_TEST

- TRUE PP18V5_DCIN_FUSE — 3 TPs
- TRUE ADAPTER_SENSE
- TRUE GND — 3 TPs

Battery Connector

FUNC_TEST

- TRUE PPVBAT_G3H_CONN — 3 TPs
- TRUE SMBUS_SMC_BSA_SCL
- TRUE SMBUS_SMC_BSA_SDA
- TRUE SYS_DETECT_L
- TRUE GND — 6 TPs

Batt Signal Connector

FUNC_TEST

- TRUE PP3V42_G3H — 3 TPs
- TRUE SMBUS_SMC_BSA_SCL
- TRUE SMBUS_SMC_BSA_SDA
- TRUE SMC_BIL_BUTTON_L
- TRUE SMC_LID_R
- TRUE GND — 3 TPs

Power Nets

FUNC_TEST

- TRUE PPVCORE_S0_CPU
- TRUE PPVCORE_S0_MCP_REG
- TRUE PP0V9R0V75_S0_DDRVTT
- TRUE PPCPUVTT_S0
- TRUE PP1V8R1V5_S0_FET
- TRUE PP1V8_S0
- TRUE PP3V3_S0
- TRUE PP1V8R1V5_S3
- TRUE PP3V3_S3
- TRUE PP1V2R1V05_S5
- TRUE PP3V3_S5
- TRUE PP3V42_G3H
- TRUE PPBUS_G3H
- TRUE PP3V3_ENET_PHY
- TRUE PP1V2R1V05_ENET
- TRUE PP5V_S3
- TRUE PP3V3_S5_AVREF_SMC
- TRUE PP18V5_S3
- TRUE PP3V3_S3_LDO
- TRUE PPVOUT_S0_LCDBKLT
- TRUE PP4V5_AUDIO_ANALOG
- TRUE SMC_PM_G2_EN
- TRUE PM_SLP_S4_L
- TRUE PM_SLP_S3_L
- TRUE PP1V05_S0_MCP_PLL_UF
- TRUE PP5V_SW_ODD
- TRUE PP5V_S0_HDD_FLT
- TRUE BKL_VLDO
- TRUE GND — 6 TPs

ICT Test Points

NO_TEST

NC_AUD_LO1_N_L	NC_AUD_LO1_N_L	MAKE_BASE=TRUE	TRUE
NC_AUD_LO1_P_L	NC_AUD_LO1_P_L	MAKE_BASE=TRUE	TRUE
NC_USB_10N	NC_USB_10N	MAKE_BASE=TRUE	TRUE
NC_USB_10P	NC_USB_10P	MAKE_BASE=TRUE	TRUE
NC_ENET_INTR_L	NC_ENET_INTR_L	MAKE_BASE=TRUE	TRUE
NC_ENET_PWRDWN_L	NC_ENET_PWRDWN_L	MAKE_BASE=TRUE	TRUE
NC_LPC_DRQ0_L	NC_LPC_DRQ0_L	MAKE_BASE=TRUE	TRUE
TP_MEM_A_CKE<3..2>	NC_MEM_A_CKE<3..2>	MAKE_BASE=TRUE	TRUE
NC_MEM_A_CLK2N	NC_MEM_A_CLK2N	MAKE_BASE=TRUE	TRUE
NC_MEM_A_CLK3N	NC_MEM_A_CLK3N	MAKE_BASE=TRUE	TRUE
NC_MEM_A_CLK3P	NC_MEM_A_CLK3P	MAKE_BASE=TRUE	TRUE
NC_MEM_A_CLK4P	NC_MEM_A_CLK4P	MAKE_BASE=TRUE	TRUE
NC_MEM_A_CS_L<3>	NC_MEM_A_CS_L<3>	MAKE_BASE=TRUE	TRUE
TP_MEM_A_ODT<3..2>	NC_MEM_A_ODT<3..2>	MAKE_BASE=TRUE	TRUE
NC_MEM_B_CKE<2>	NC_MEM_B_CKE<2>	MAKE_BASE=TRUE	TRUE
NC_MEM_B_CLK3P	NC_MEM_B_CLK3P	MAKE_BASE=TRUE	TRUE
NC_MEM_B_CLK4N	NC_MEM_B_CLK4N	MAKE_BASE=TRUE	TRUE
NC_MEM_B_CLK4P	NC_MEM_B_CLK4P	MAKE_BASE=TRUE	TRUE
NC_MEM_B_CLK5N	NC_MEM_B_CLK5N	MAKE_BASE=TRUE	TRUE
NC_MEM_B_ODT<2>	NC_MEM_B_ODT<2>	MAKE_BASE=TRUE	TRUE
NC_MLB_RAM_SIZE	NC_MLB_RAM_SIZE	MAKE_BASE=TRUE	TRUE
NC_P7_7	NC_P7_7	MAKE_BASE=TRUE	TRUE
TP_PCI_AD<31..8>	NC_PCI_AD<31..8>	MAKE_BASE=TRUE	TRUE
TP_PCI_C_BE_L<3..0>	NC_PCI_C_BE_L<3..0>	MAKE_BASE=TRUE	TRUE
NC_PCI_CLK0	NC_PCI_CLK0	MAKE_BASE=TRUE	TRUE
NC_PCI_CLK1	NC_PCI_CLK1	MAKE_BASE=TRUE	TRUE
NC_PCI_DEVSEL_L	NC_PCI_DEVSEL_L	MAKE_BASE=TRUE	TRUE
NC_PCI_FRAME_L	NC_PCI_FRAME_L	MAKE_BASE=TRUE	TRUE
NC_PCI_GNT0_L	NC_PCI_GNT0_L	MAKE_BASE=TRUE	TRUE
NC_PCI_GNT1_L	NC_PCI_GNT1_L	MAKE_BASE=TRUE	TRUE
NC_PCI_INTW_L	NC_PCI_INTW_L	MAKE_BASE=TRUE	TRUE
NC_PCI_INTX_L	NC_PCI_INTX_L	MAKE_BASE=TRUE	TRUE
NC_PCI_INTZ_L	NC_PCI_INTZ_L	MAKE_BASE=TRUE	TRUE
NC_PCI_IRDY_L	NC_PCI_IRDY_L	MAKE_BASE=TRUE	TRUE
NC_PCI_PERR_L	NC_PCI_PERR_L	MAKE_BASE=TRUE	TRUE
NC_PCI_RESET1_L	NC_PCI_RESET1_L	MAKE_BASE=TRUE	TRUE
NC_PCI_SERR_L	NC_PCI_SERR_L	MAKE_BASE=TRUE	TRUE
NC_PCI_STOP_L	NC_PCI_STOP_L	MAKE_BASE=TRUE	TRUE
NC_PCI_TRDY_L	NC_PCI_TRDY_L	MAKE_BASE=TRUE	TRUE
NC_PCIE_CLK100M_PE4N	NC_PCIE_CLK100M_PE4N	MAKE_BASE=TRUE	TRUE
NC_PCIE_CLK100M_PE4P	NC_PCIE_CLK100M_PE4P	MAKE_BASE=TRUE	TRUE
NC_PCIE_CLK100M_PE5N	NC_PCIE_CLK100M_PE5N	MAKE_BASE=TRUE	TRUE
NC_PCIE_CLK100M_PE5P	NC_PCIE_CLK100M_PE5P	MAKE_BASE=TRUE	TRUE
NC_PCIE_CLK100M_PE6P	NC_PCIE_CLK100M_PE6P	MAKE_BASE=TRUE	TRUE
NC_PCIE_PE4_D2RN	NC_PCIE_PE4_D2RN	MAKE_BASE=TRUE	TRUE
NC_PCIE_PE4_R2D_CN	NC_PCIE_PE4_R2D_CN	MAKE_BASE=TRUE	TRUE
NC_PE4_PRSNT_L	NC_PE4_PRSNT_L	MAKE_BASE=TRUE	TRUE
NC_PSOC_P1_3	NC_PSOC_P1_3	MAKE_BASE=TRUE	TRUE
NC_PSOC_SDA	NC_PSOC_SDA	MAKE_BASE=TRUE	TRUE
NC_SATA_C_D2RP	NC_SATA_C_D2RP	MAKE_BASE=TRUE	TRUE
NC_SATA_C_R2D_CN	NC_SATA_C_R2D_CN	MAKE_BASE=TRUE	TRUE
NC_SATA_C_R2D_CP	NC_SATA_C_R2D_CP	MAKE_BASE=TRUE	TRUE
NC_SATA_D_D2RN	NC_SATA_D_D2RN	MAKE_BASE=TRUE	TRUE
NC_SATA_D_D2RP	NC_SATA_D_D2RP	MAKE_BASE=TRUE	TRUE
NC_SB_A20GATE	NC_SB_A20GATE	MAKE_BASE=TRUE	TRUE

NO_TEST

- TRUE FSB_A_L<31..3>
- TRUE FSB_ADS_L
- TRUE FSB_ADSTB_L<1..0>
- TRUE FSB_D_L<63..0>
- TRUE FSB_DINV_L<3..0>
- TRUE FSB_DSTB_L_N<3..0>
- TRUE FSB_DSTB_L_P<3..0>
- TRUE FSB_HIT_L
- TRUE FSB_HITM_L
- TRUE FSB_LOCK_L
- TRUE USB_BT_N
- TRUE USB_BT_P
- TRUE USB_CAMERA_N
- TRUE USB_CAMERA_P
- TRUE SATA_ODD_D2R_UF_N
- TRUE SATA_ODD_D2R_UF_P
- TRUE DP_ML_C_P<3..0>

Note.
NO_TEST properties are also on page9,26,43,50

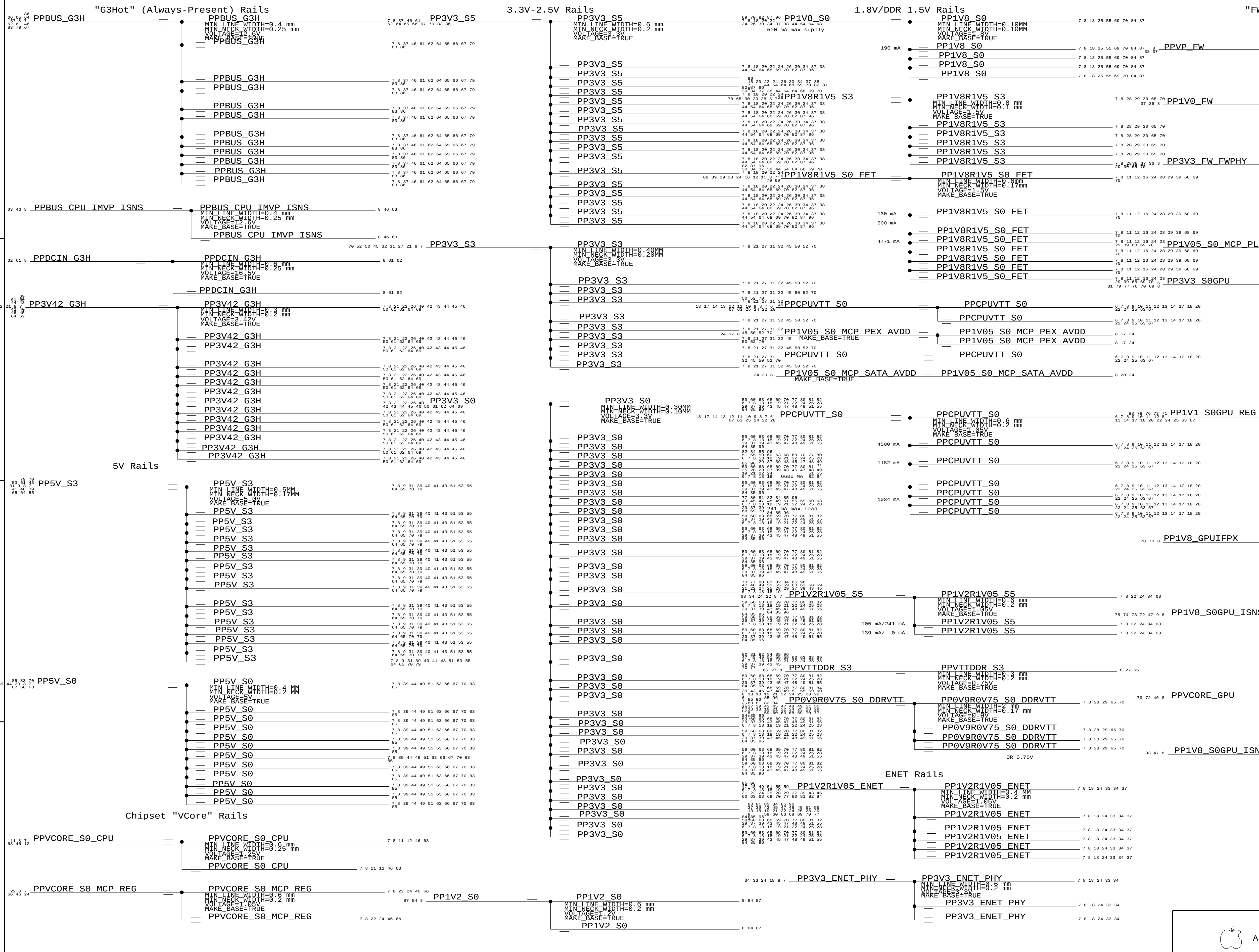
Functional / ICT Test

SYNC_MASTER=N/A SYNC_DATE=N/A

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE | SHT 7 | OF 97



Power Aliases

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

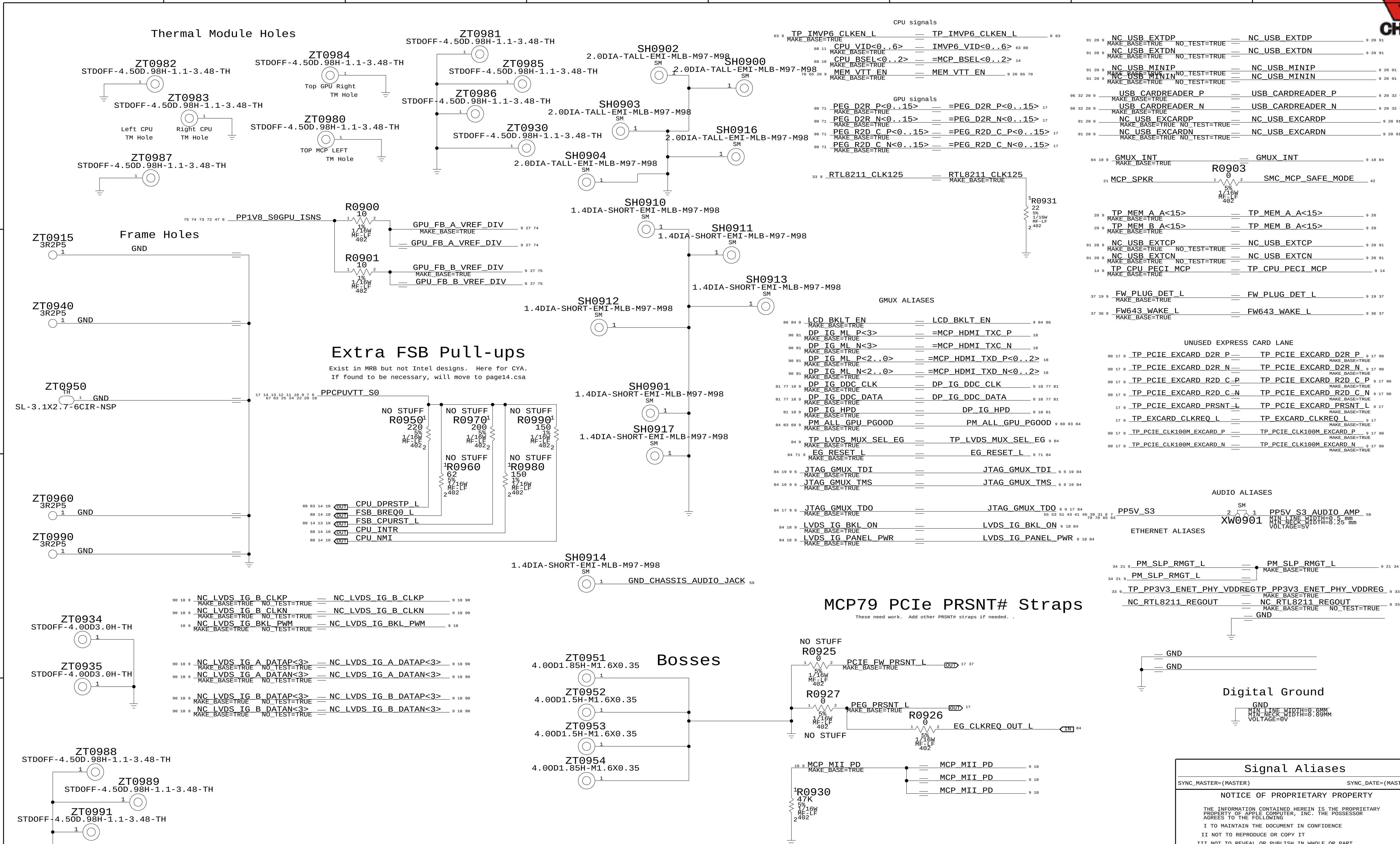
NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 8	OF 97



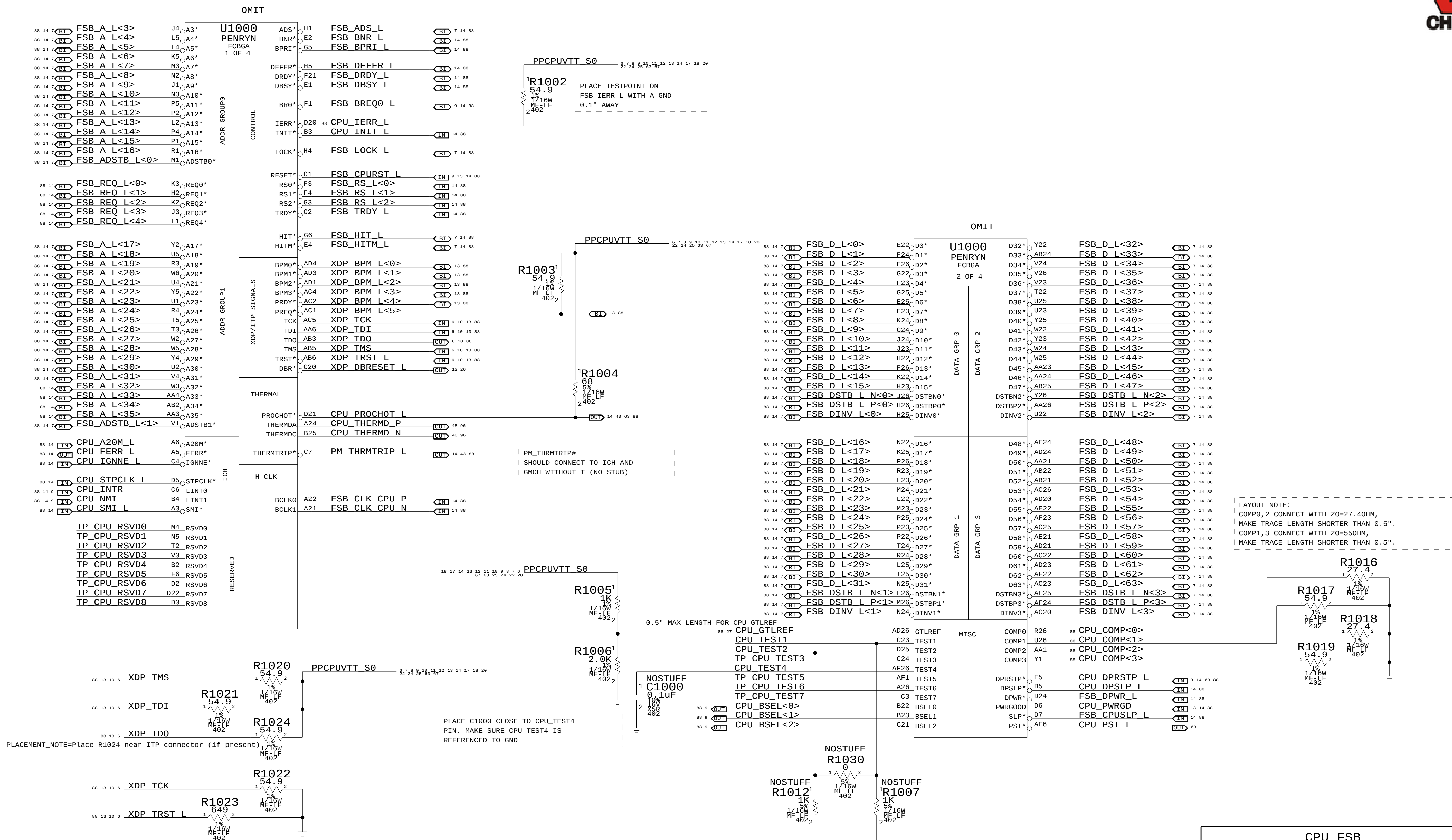
Signal Aliases

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 9	OF 97



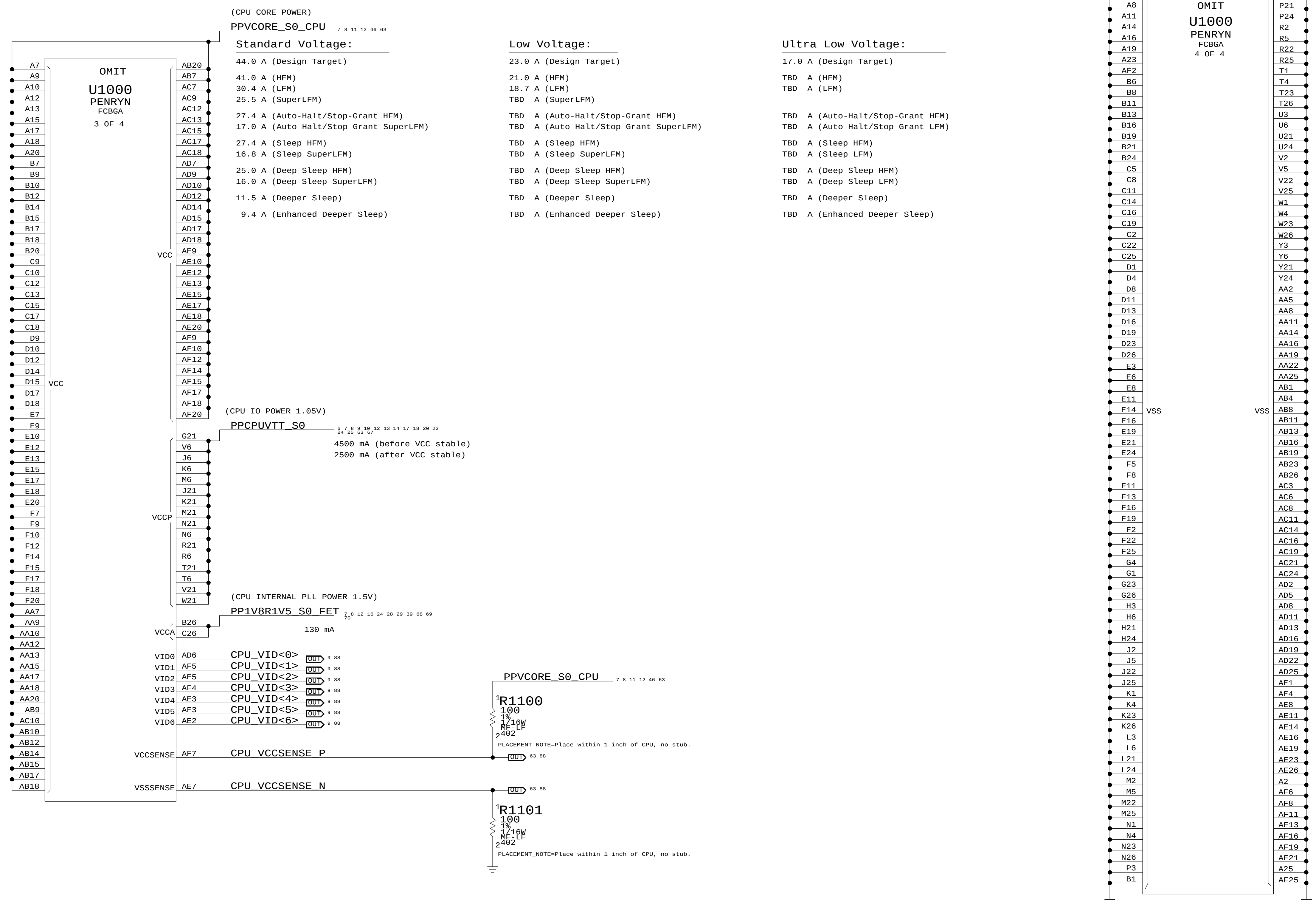
CPU FSB

SYNC_MASTER=M98_MLB SYNC_DATE=11/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 10	OF 97

D C B A



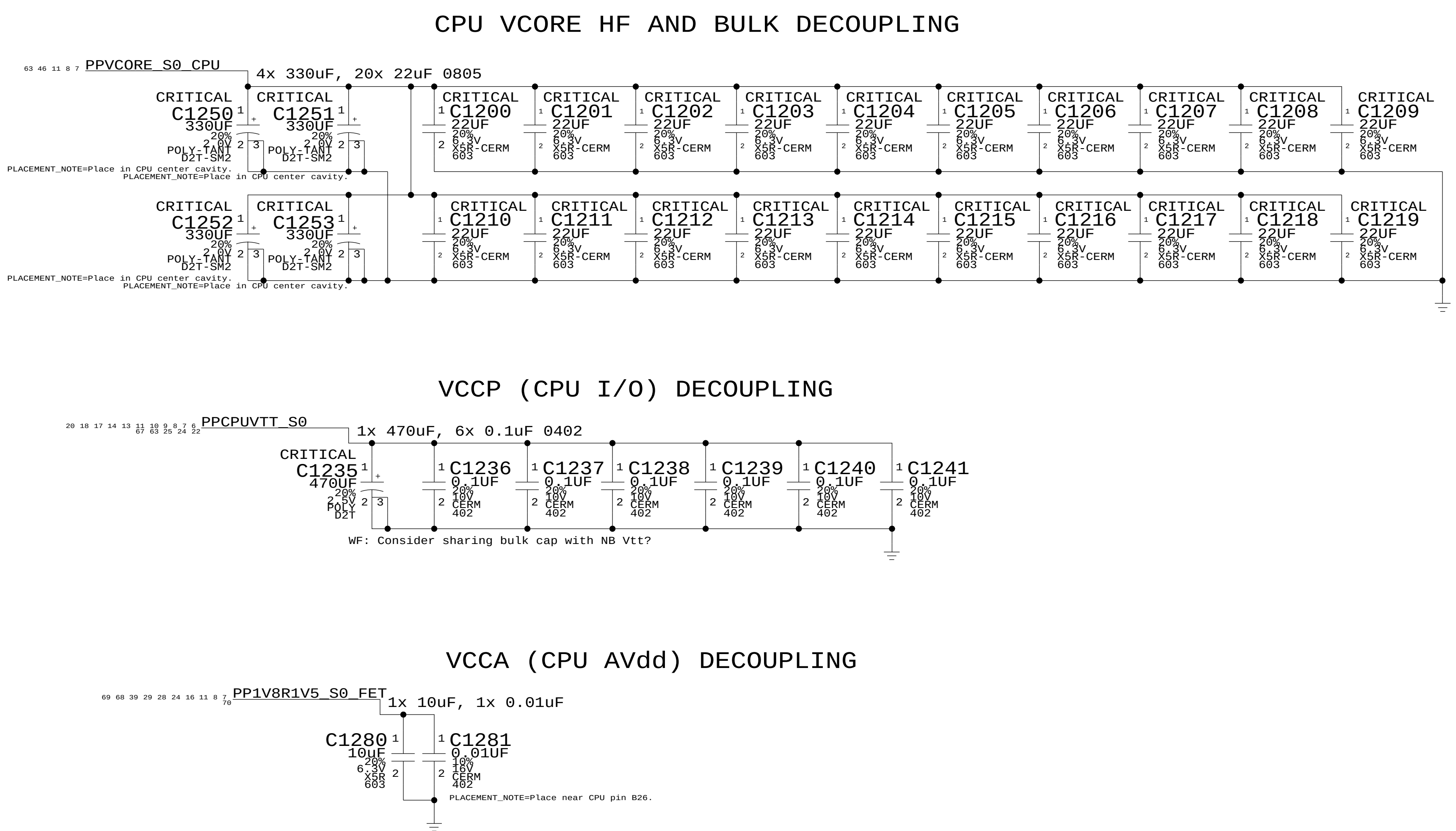
CPU Power & Ground

SYNC_MASTER=M98_MLB SYNC_DATE=11/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 11	OF 97

Current numbers from Merom for Santa Rosa EMTS, doc #22221.



CPU Decoupling & VID

SYNC_MASTER=M87_MLB SYNC_DATE=10/17/2007

APPLE INC.

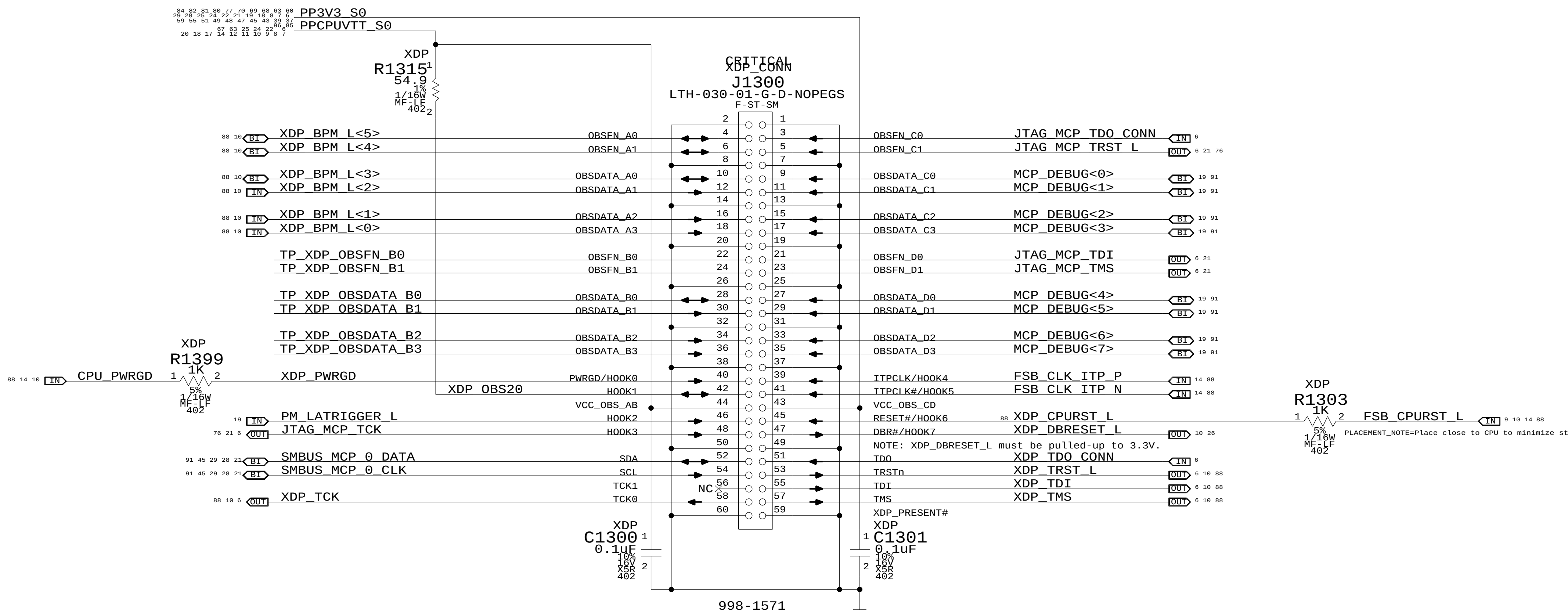
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 12	OF 97

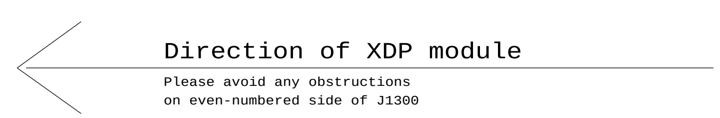
Mini-XDP Connector

NOTE: This is not the standard XDP pinout.

Use with 920-0620 adapter board to support CPU, MCP debugging.

MCP79-specific pinout



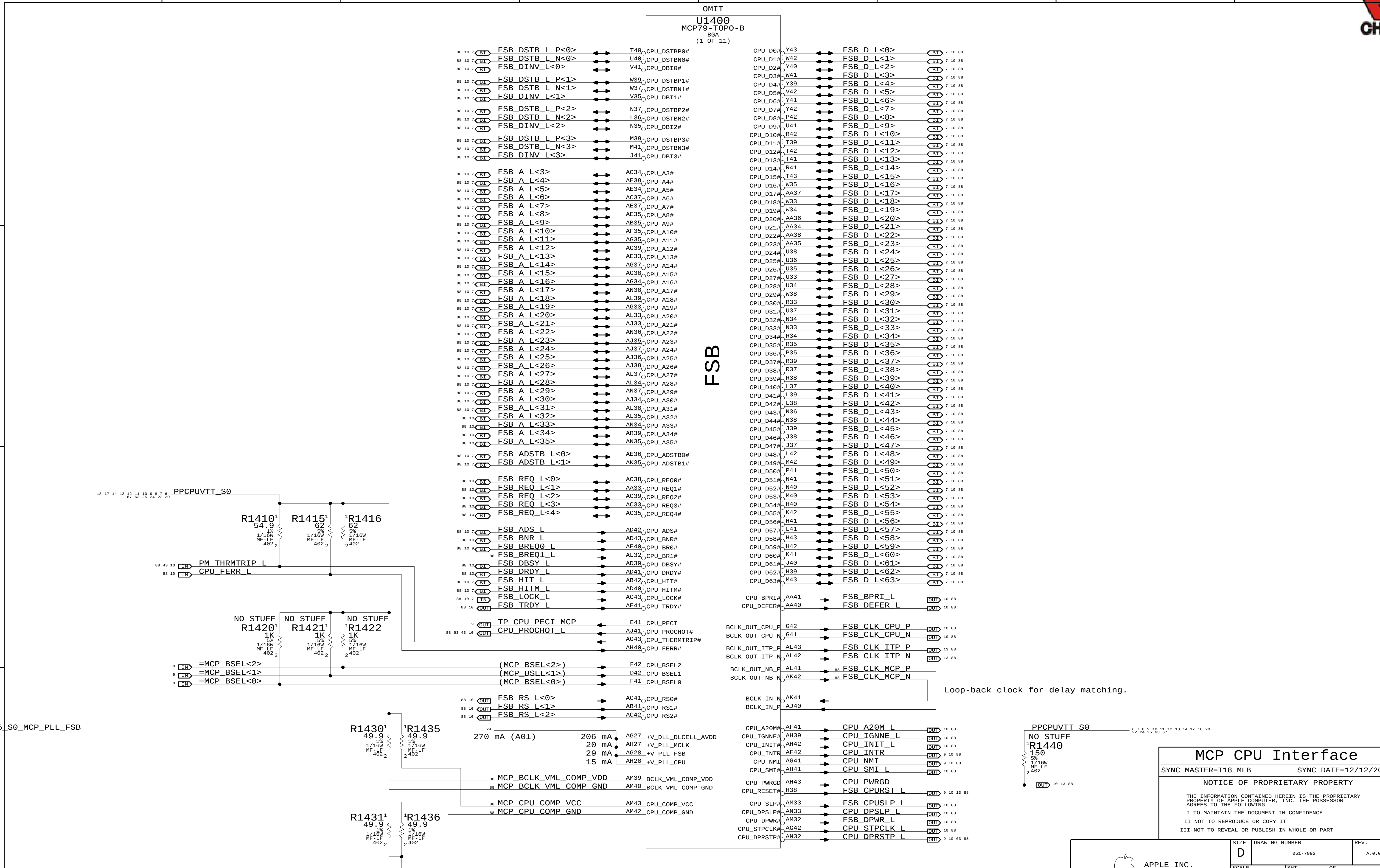


eXtended Debug Port(MiniXDP)

SYNC_MASTER=M98_MLB SYNC_DATE=11/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 13	OF 97



Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

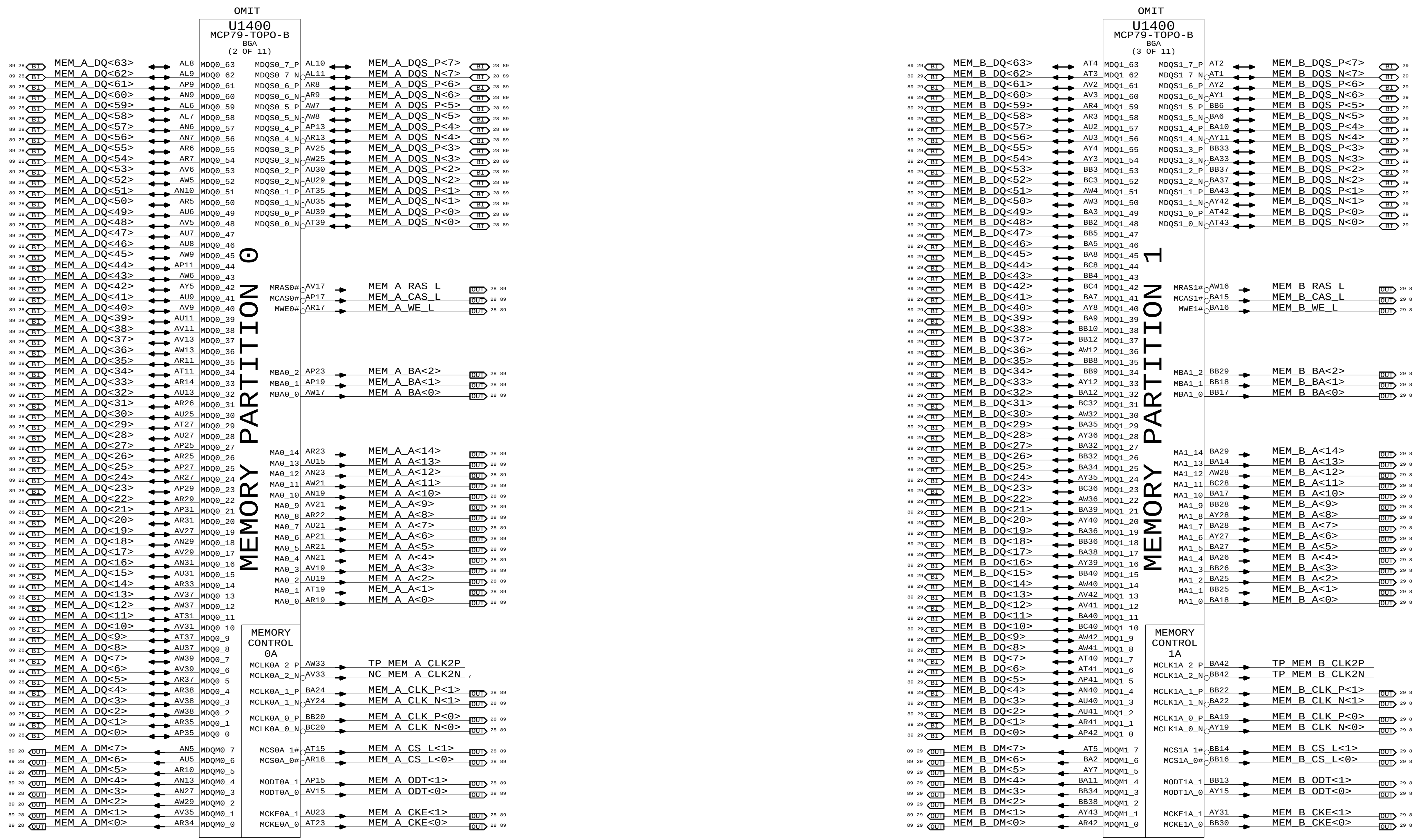
MCP CPU Interface

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE | SHT 14 OF 97



MCP Memory Interface

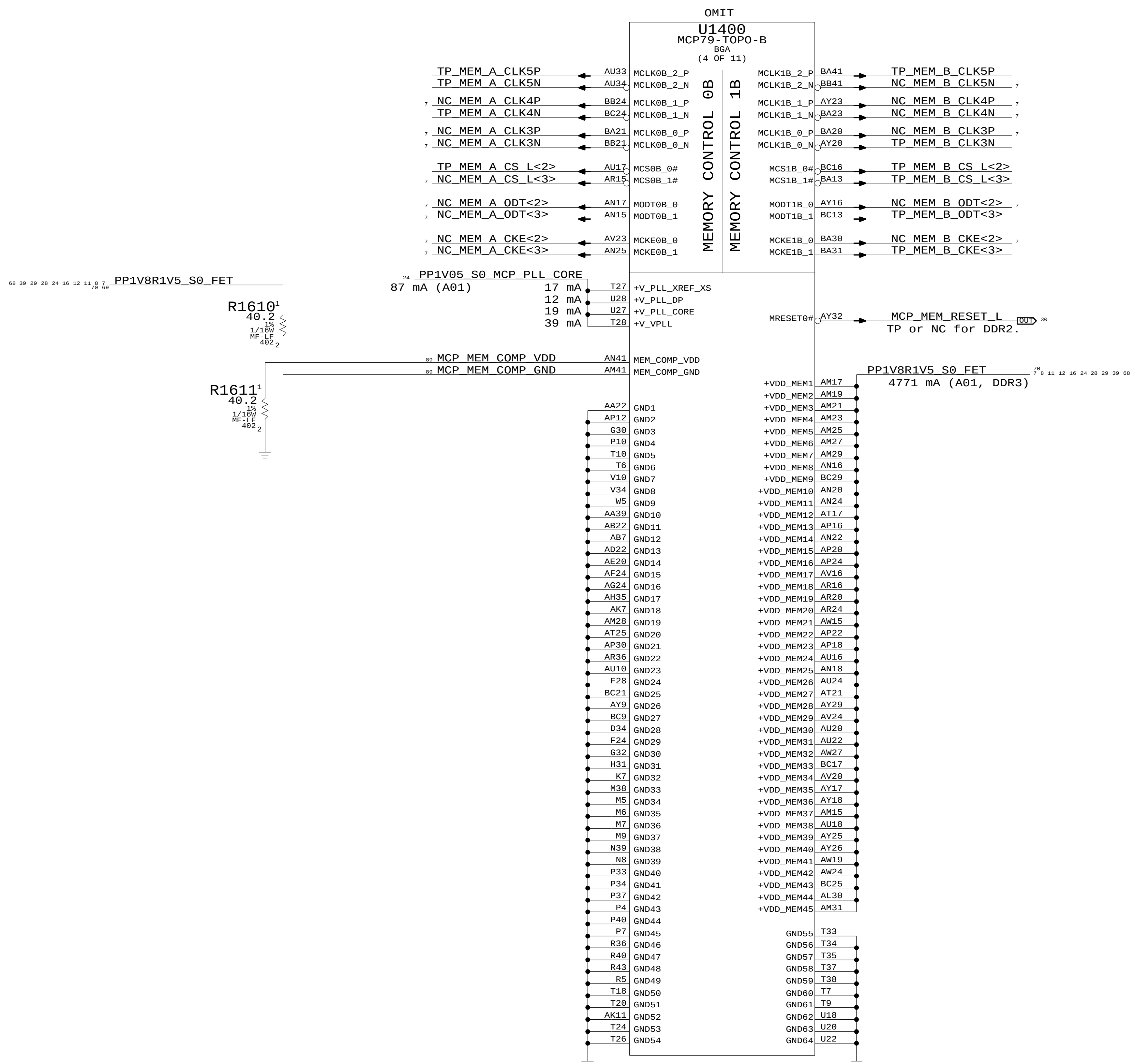
SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 15	OF 97

8 7 6 5 4 3 2 1

D C B A



Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

MCP Memory Misc

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

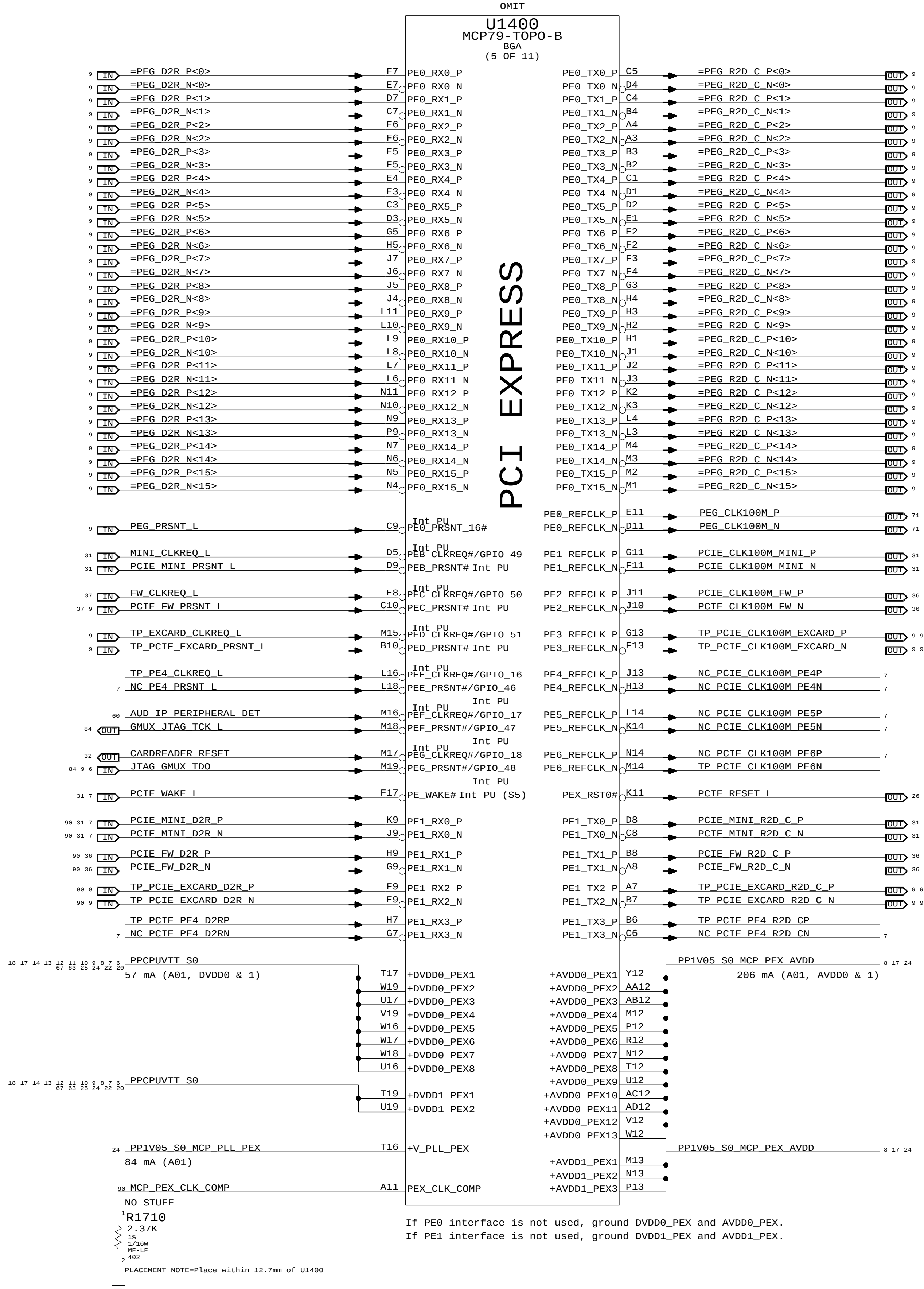
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 16	OF 97



If PE0 interface is not used, ground DVDD0_PEX and AVDD0_PEX.
If PE1 interface is not used, ground DVDD1_PEX and AVDD1_PEX.

Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

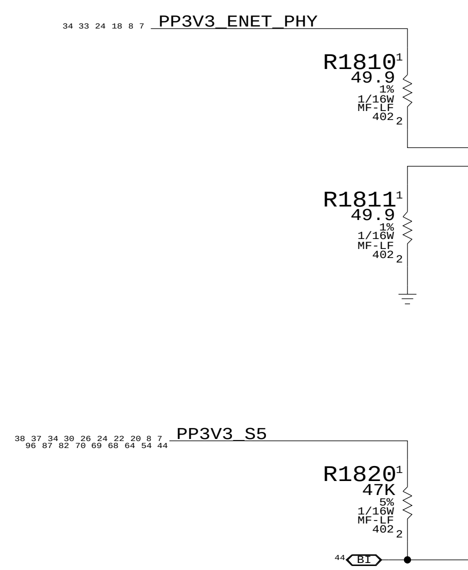
MCP PCIe Interfaces

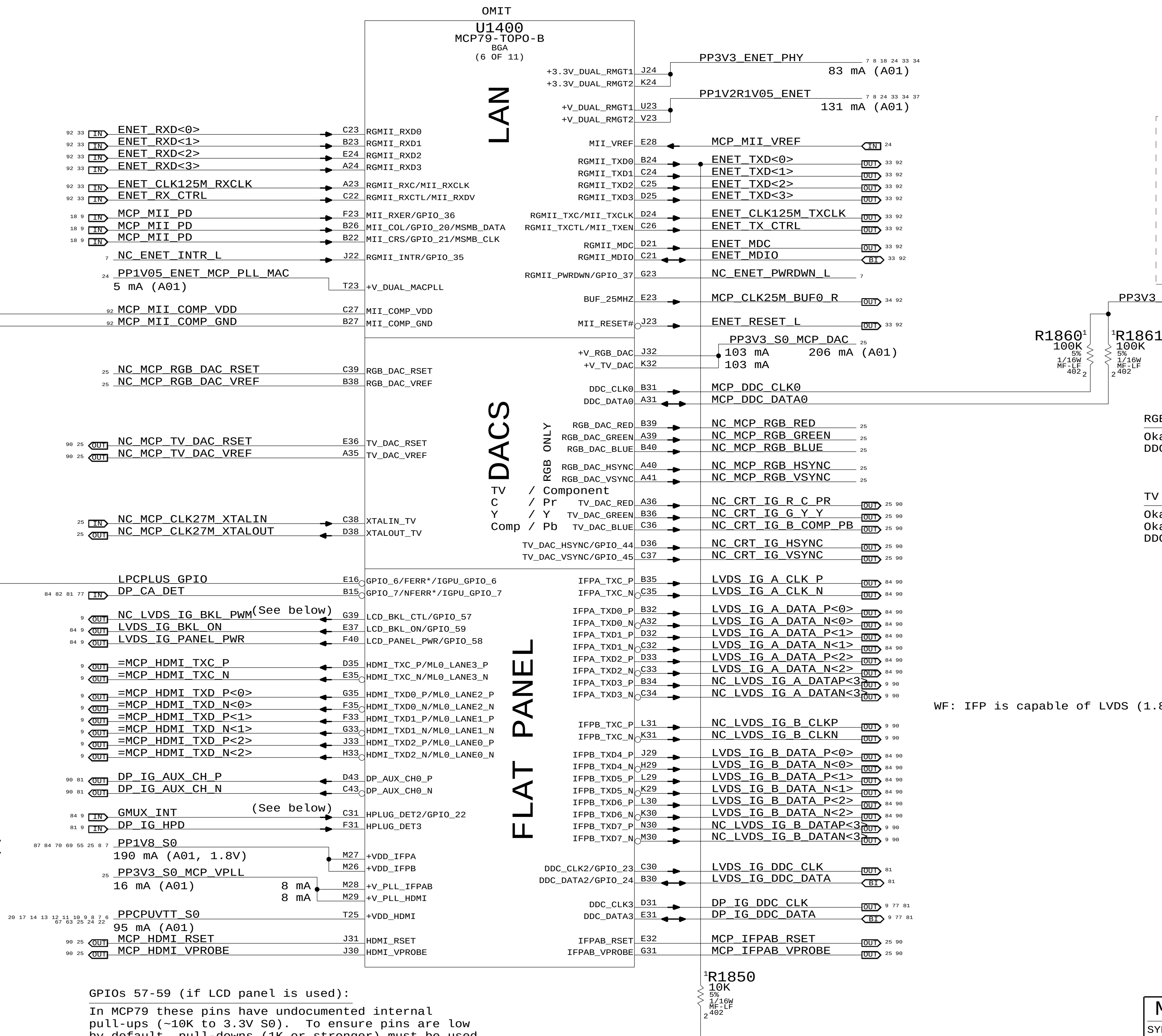
SYNC_MASTER=T18_MLB SYNC_DATE=04/04/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 17	OF 97





Network Interface Select

Interface	ENET_TXD<0>
RGMII	1
MII	0

NOTE: All Apple products set strap to MII, RGMII products will enable feature via software. This avoids a leakage issue since MCP79 requires a S5 pull-up.

RGB DAC Disable:

Okay to float all RGB_DAC signals.
DDC_CLK0/DDC_DATA0 pull-ups still required.

TV DAC Disable:

Okay to float all TV_DAC signals.
Okay to float XTALIN_TV and XTALOUT_TV.
DDC_CLK0/DDC_DATA0 pull-ups still required.

WF: IFP is capable of LVDS (1.8V) or TMDS (3.3V), need aliases

Interface Mode

MCP Signal	TMDS/HDMI	DisplayPort
=MCP_HDMI_TXC_P/N	TMDS_IG_TXC_P/N	DP_IG_ML_P/N<3>
=MCP_HDMI_TXD_P/N<0>	TMDS_IG_TXD_P/N<0>	DP_IG_ML_P/N<2>
=MCP_HDMI_TXD_P/N<1>	TMDS_IG_TXD_P/N<1>	DP_IG_ML_P/N<1>
=MCP_HDMI_TXD_P/N<2>	TMDS_IG_TXD_P/N<2>	DP_IG_ML_P/N<0>
=MCP_HDMI_DDC_CLK	TMDS_IG_DDC_CLK	DP_IG_DDC_CLK
=MCP_HDMI_DDC_DATA	TMDS_IG_DDC_DATA	DP_IG_DDC_DATA
=MCP_HDMI_HPD	TMDS_IG_HPD	DP_IG_HPD
DP_IG_AUX_CH_P/N	TP_DP_IG_AUX_CHP/N	DP_IG_AUX_CH_P/N

NOTE: 1M pull-down required on DP_IG_CA_DET if DP not used.
NOTE: 20K pull-down required on DP_HPD_DET.
NOTE: 1K pull-down required on DP_IG_AUX_CH_N if DP is used.

NOTE: HDMI port requires level-shifting. IFP interface can be used to provide HDMI or dual-channel TMDS without level-shifters.

LVDS: Power +VDD_IFPx at 1.8V
Dual-channel TMDS: Power +VDD_IFPx at 3.3V

GPIOs 57-59 (if LCD panel is used):

In MCP79 these pins have undocumented internal pull-ups (~10K to 3.3V S0). To ensure pins are low by default, pull-downs (1K or stronger) must be used.

=DVI_HPD_GMUX_INT:

Alias to DVI_HPD for systems using IFP for DVI.
Alias to GMUX_INT for systems with GMUX.
Alias to HPLUG_DET2 for other systems.
Pull-down (20k) required in all cases.

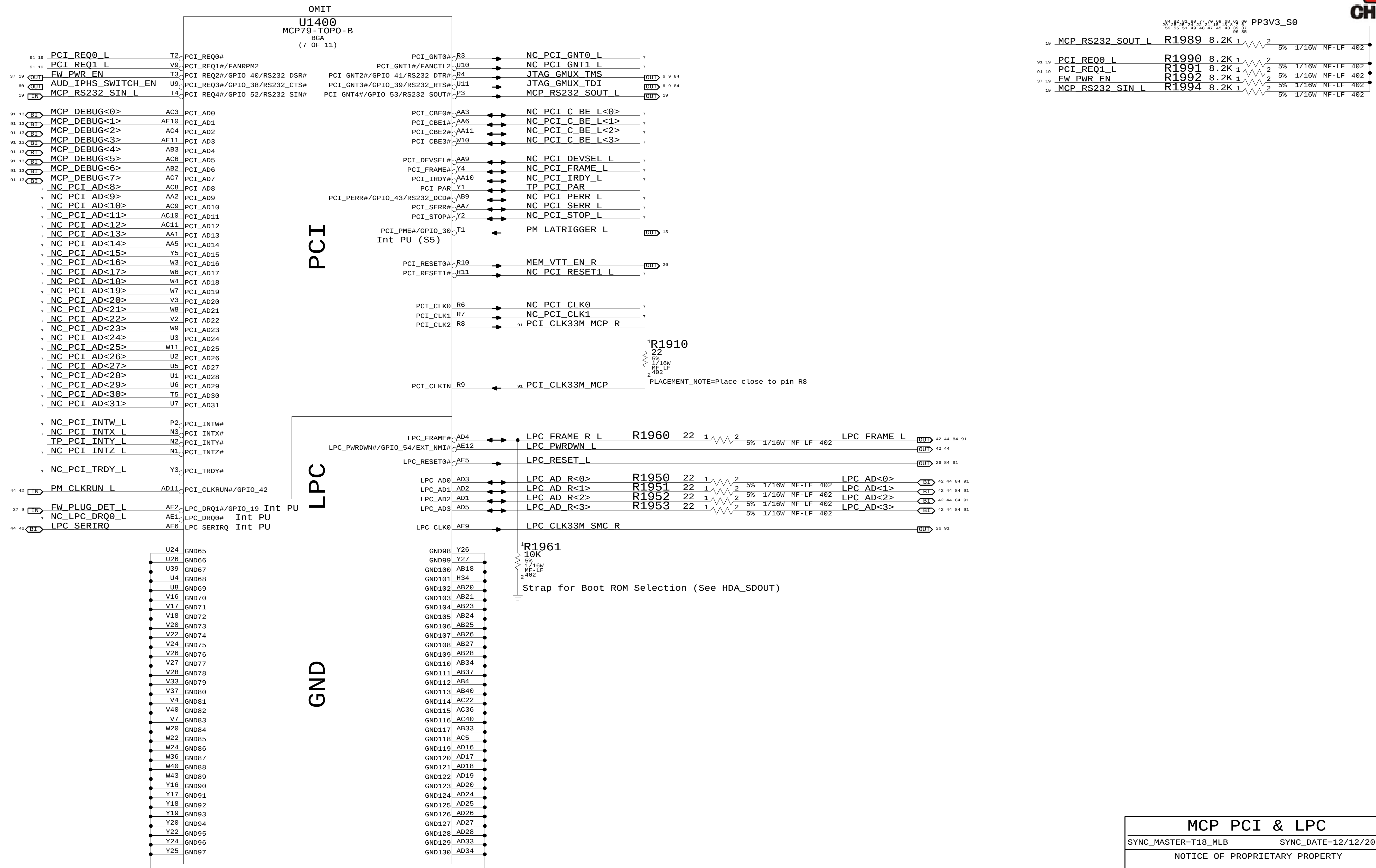
Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

MCP Ethernet & Graphics

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0
SCALE NONE | SHT 18 | OF 97



MCP PCI & LPC

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

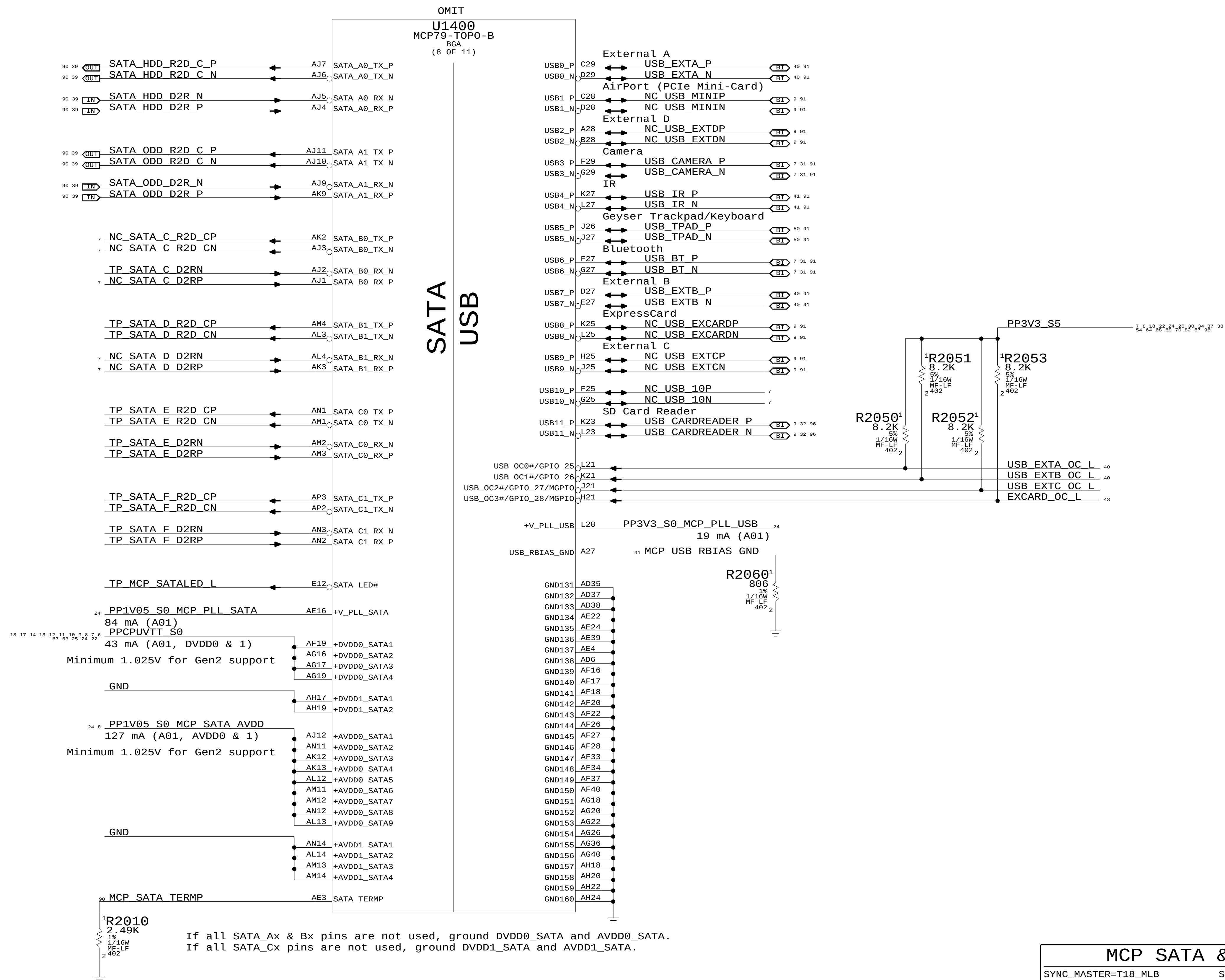
NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

SIZE D	DRAWING NUMBER 051-7892	REV. A.0.0
SCALE NONE	SHT 19	OF 97

APPLE INC.



If all SATA_Ax & Bx pins are not used, ground DVDD0_SATA and AVDD0_SATA.
If all SATA_Cx pins are not used, ground DVDD1_SATA and AVDD1_SATA.

Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

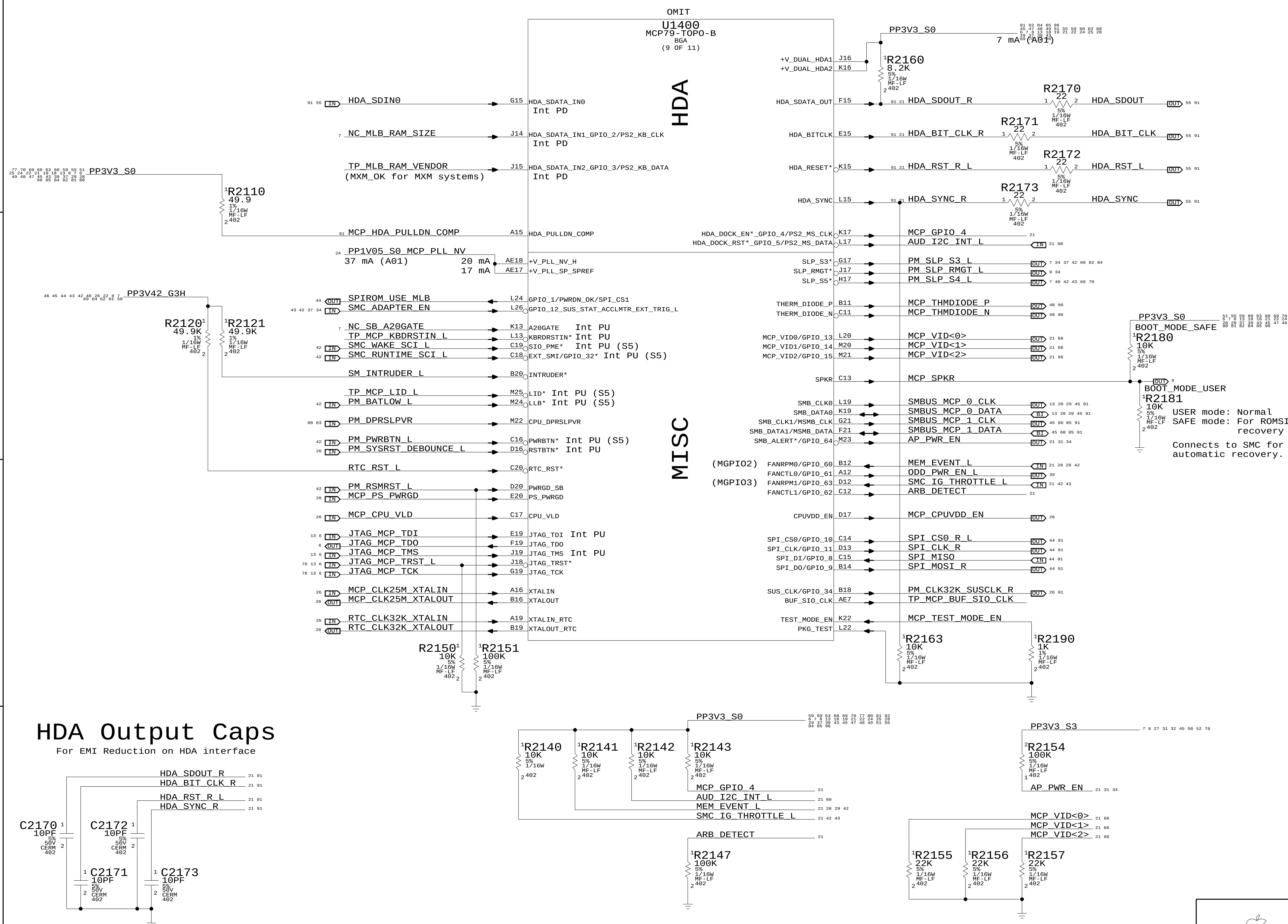
MCP SATA & USB

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 20	OF 97

8 7 6 5 4 3 2 1



BIOS Boot Select

I/F	HDA_SDOUT	LPC_FRAME#
LPC	0	0
PCI	0	1
SPI0	1	0
SPI1	1	1

SPI0 = SPI_CS0_L, SPI1 = SPI_CS1_L

R1961 and R2160 selects SPI0 ROM by default, LPC+ debug card pulls LPC_FRAME# high for SPI1 ROM override.

NOTE: MCP79 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

NOTE: MCP79 rev A01 does not support SPI1 option. Rev B01 will.

BUF_SIO_CLK Frequency

Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

SPI Frequency Select

Frequency	SPI_DO	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

NOTE: Straps not provided on this page.

Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

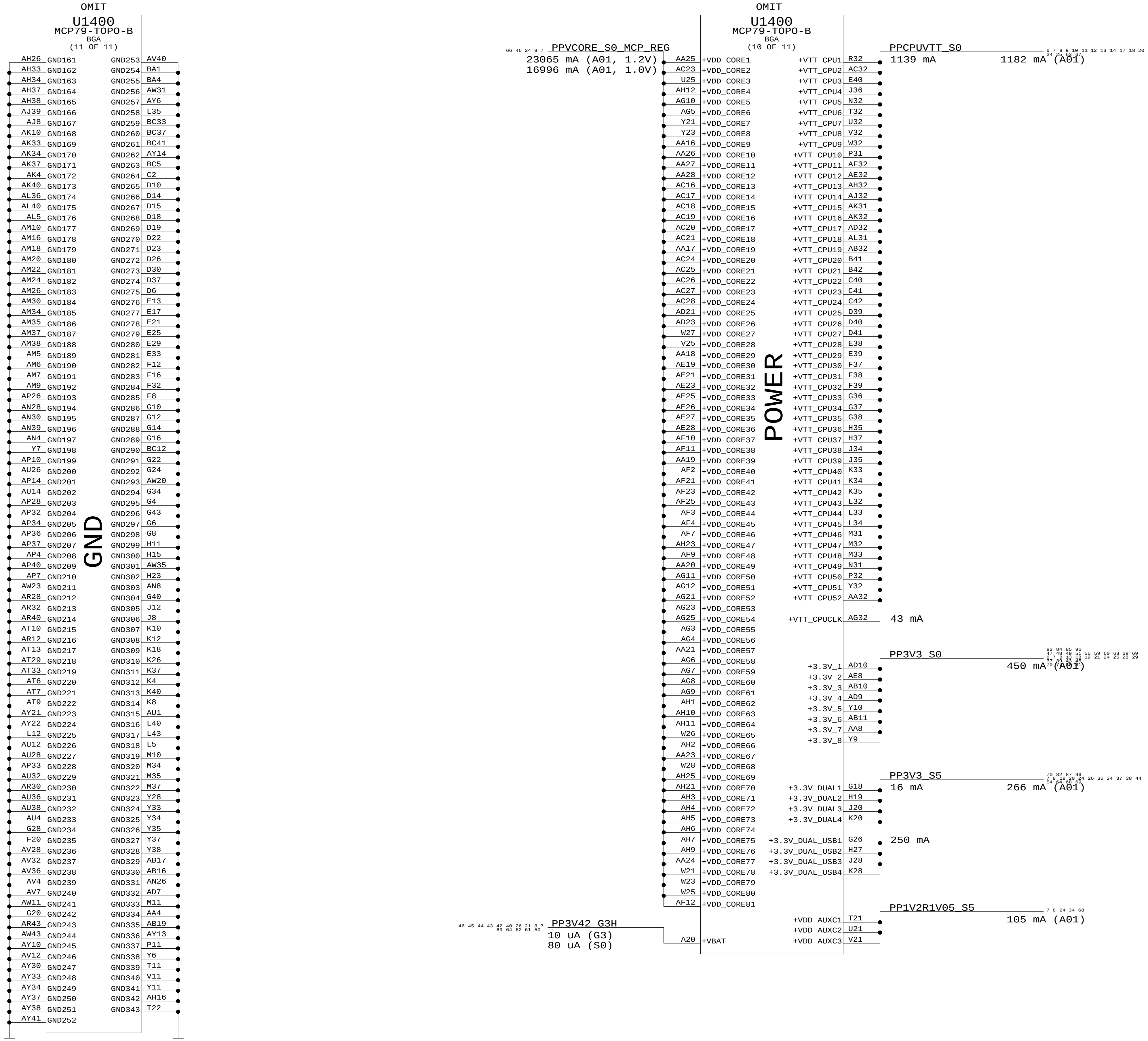
MCP HDA & MISC

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 21	OF 97

8 7 6 5 4 3 2 1



Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

MCP Power & Ground

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

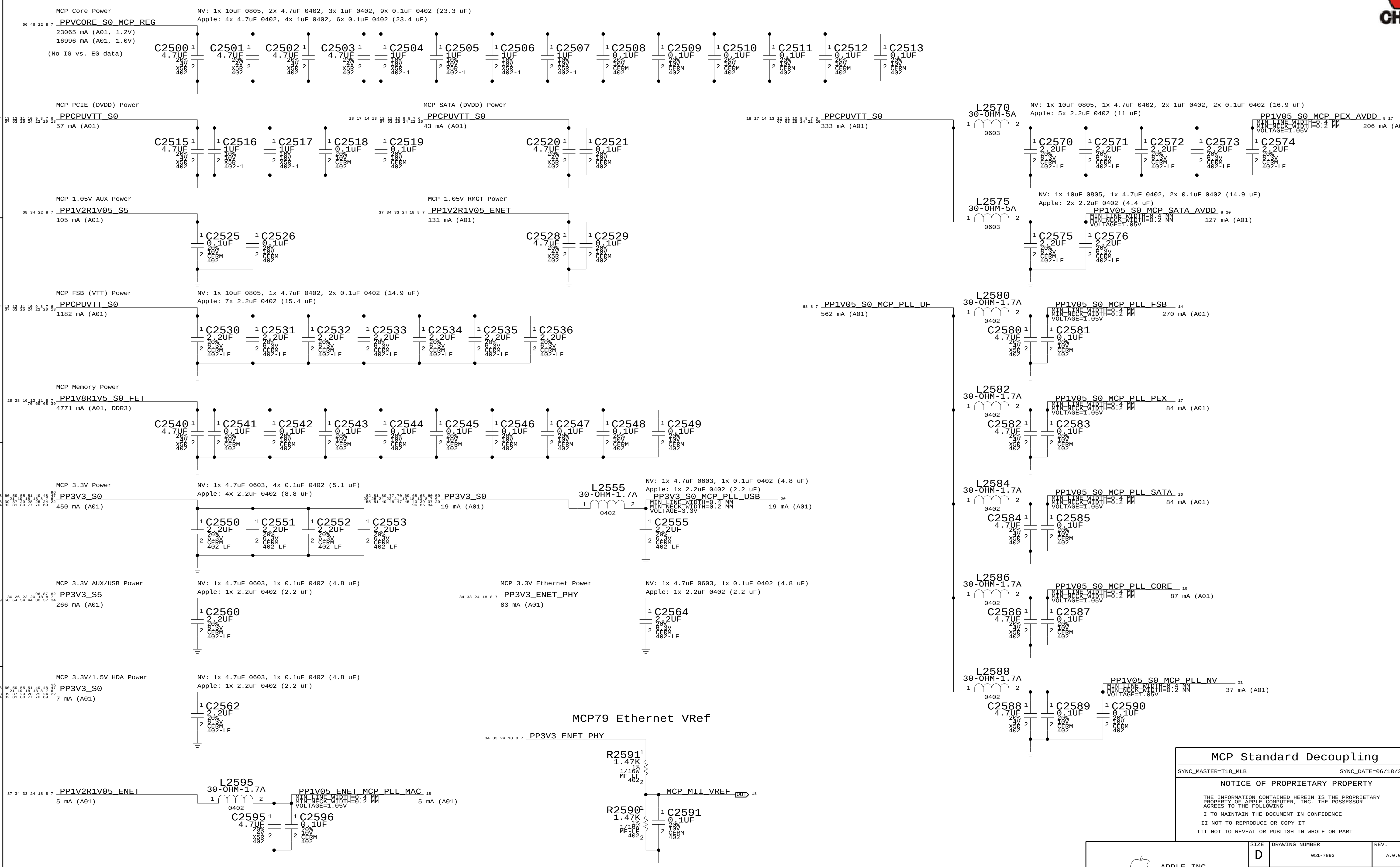
SCALE	SHT	OF
NONE	22	97

MCP79 A01 Silicon Support

SYNC_MASTER=T18_MLB SYNC_DATE=03/31/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 23	OF 97



Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

MCP Standard Decoupling

SYNC_MASTER=T18_MLB SYNC_DATE=06/18/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

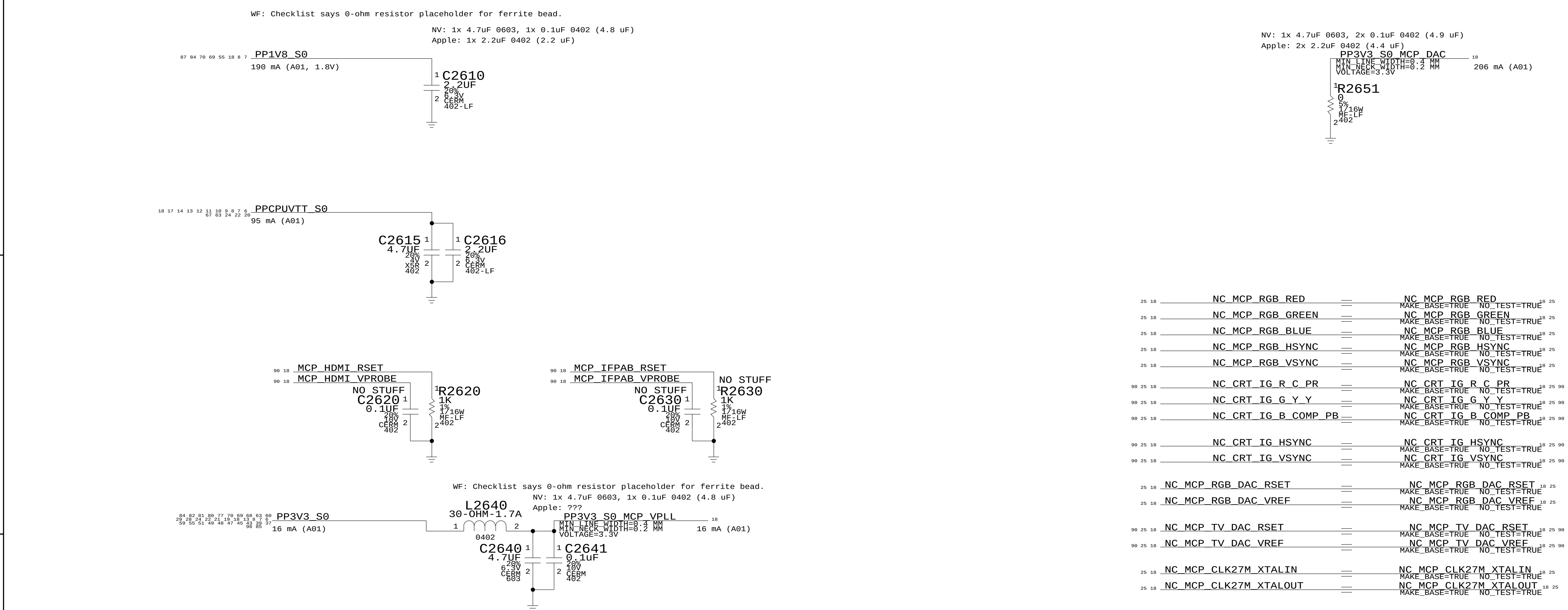
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 24	OF 97



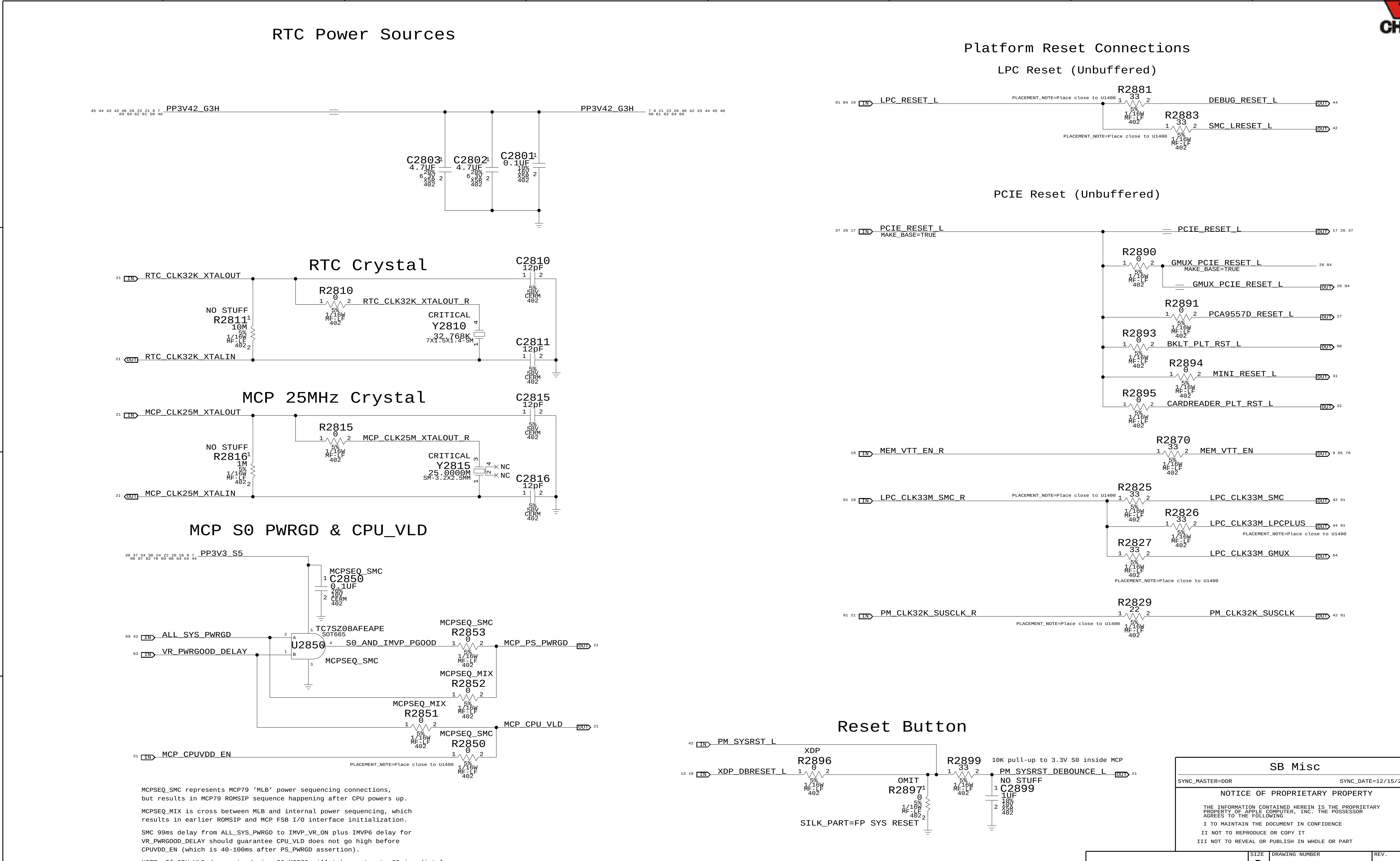
Current numbers from email Xiaowei Lin provided 11/12/2007 3:22pm (no official document number).

MCP Graphics Support

SYNC_MASTER=AMASON_M98_MLB SYNC_DATE=06/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 25	OF 97



SB Misc

SYNC_MASTER=DDR SYNC_DATE=12/15/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE NONE SHT 26 OF 97

Page Notes

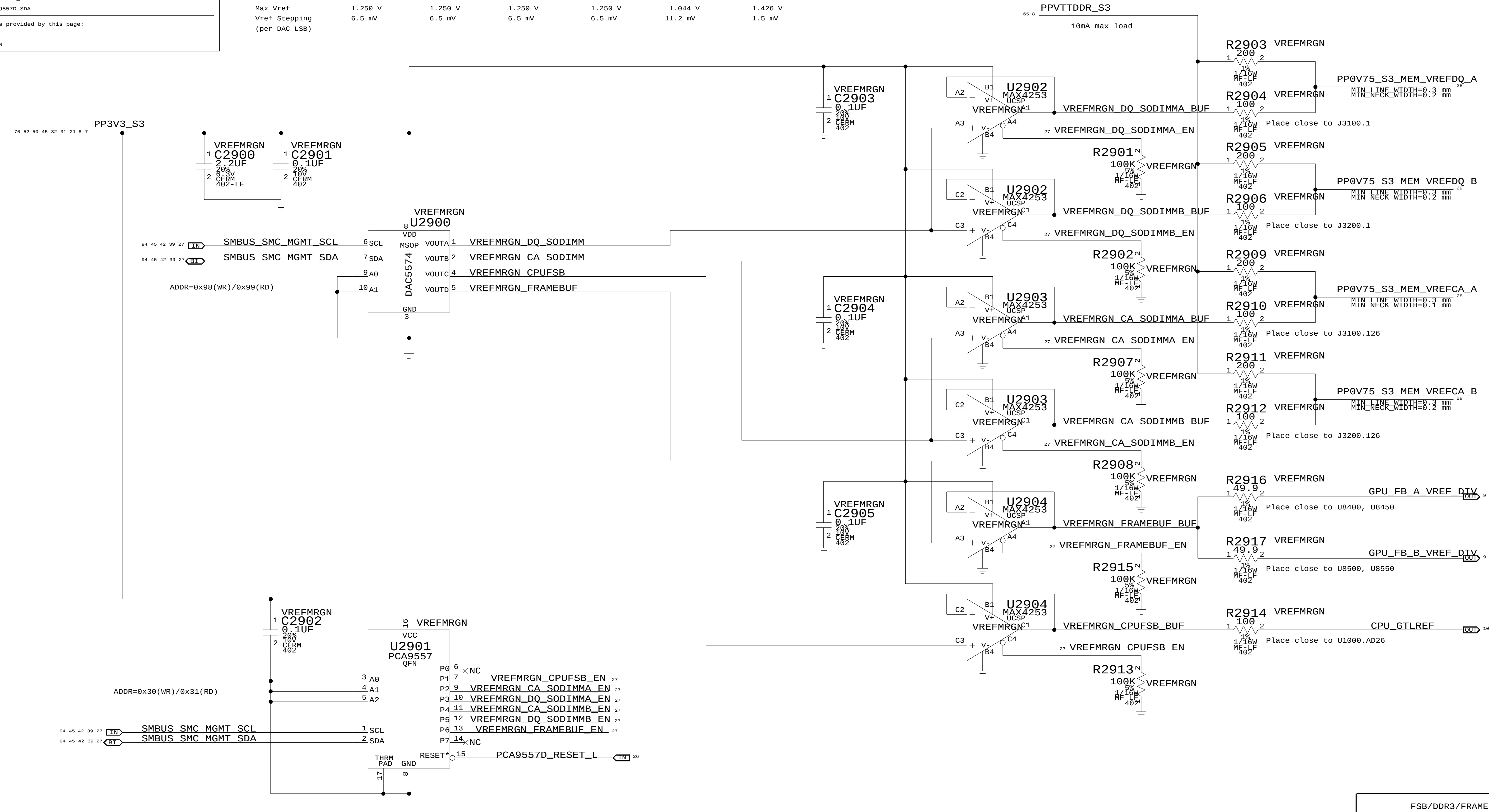
Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN
NO_VREFMRGN

	MEM A VREF DQ	MEM A VREF CA	MEM B VREF DQ	MEM B VREF CA	CPU FSB VREF	FRAME BUFFER VREF
DAC channel	A	B	A	B	C	D
Min DAC code	0x00	0x00	0x00	0x00	0x00	0x00
Max DAC code	0x87	0x87	0x87	0x87	0x55	0xFF
Max sink I	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-0.91 mA	-59.04 mA
Max source I	5 mA	5 mA	5 mA	5 mA	0.52 mA	51.15 mA
Nominal Vref	0.75 V	0.75 V	0.75 V	0.75 V	0.70 V	1.248 V
Min Vref	0.375 V	0.375 V	0.375 V	0.375 V	0.091 V	1.042 V
Max Vref	1.250 V	1.250 V	1.250 V	1.250 V	1.044 V	1.426 V
Vref Stepping (per DAC LSB)	6.5 mV	6.5 mV	6.5 mV	6.5 mV	11.2 mV	1.5 mV

SO-DIMM A and SO-DIMM B Vref settings should be margined separately
(i.e. not simultaneously) due to current limitation of TPS51116 regulator.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2903	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2905	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2909	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2911	CRITICAL	NO_VREFMRGN

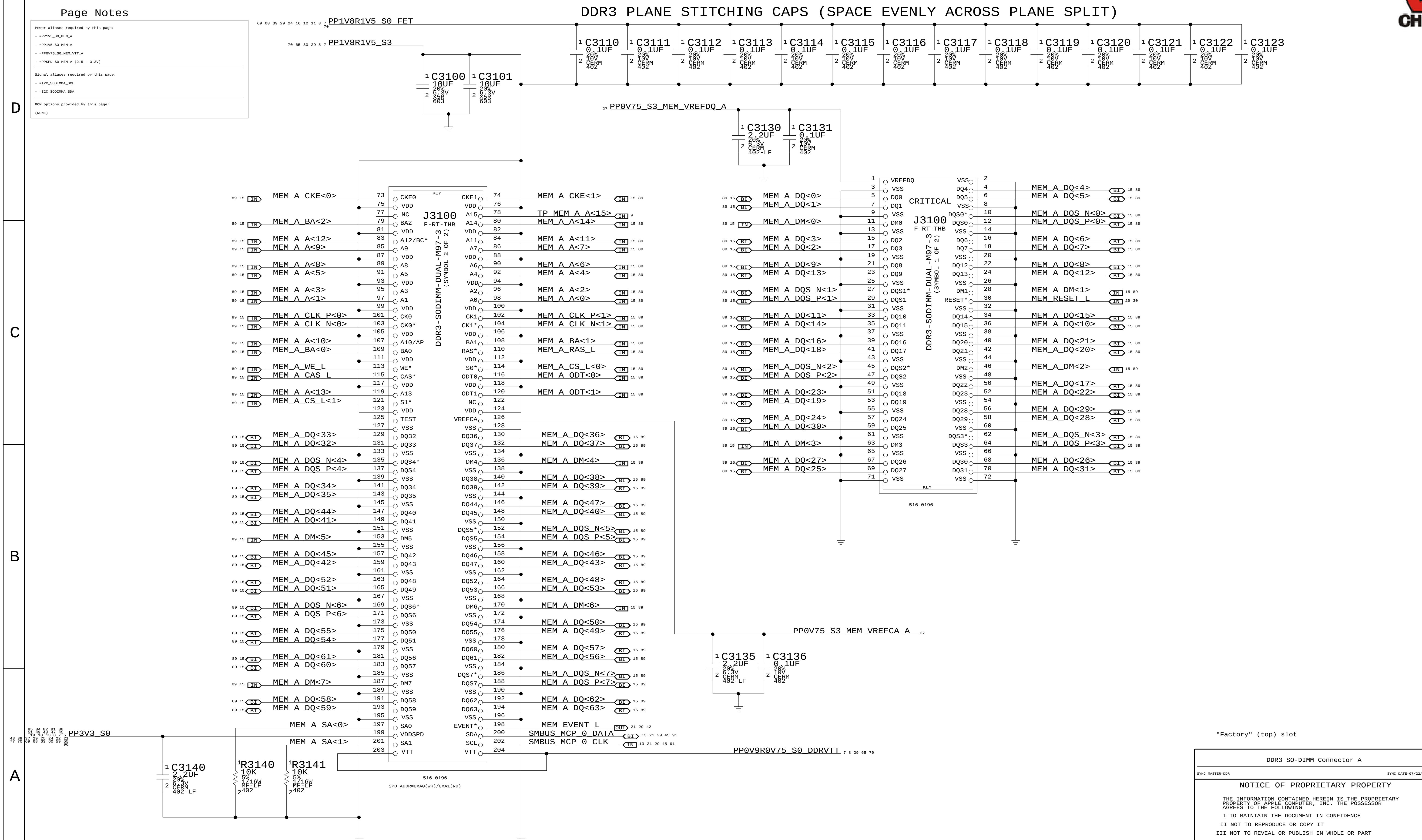
FSB/DDR3/FRAMEBUF Vref Margining

SYNC_MASTER=DDR SYNC_DATE=12/05/2008

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE | SHT 27 OF 97



DDR3 SO-DIMM Connector A

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

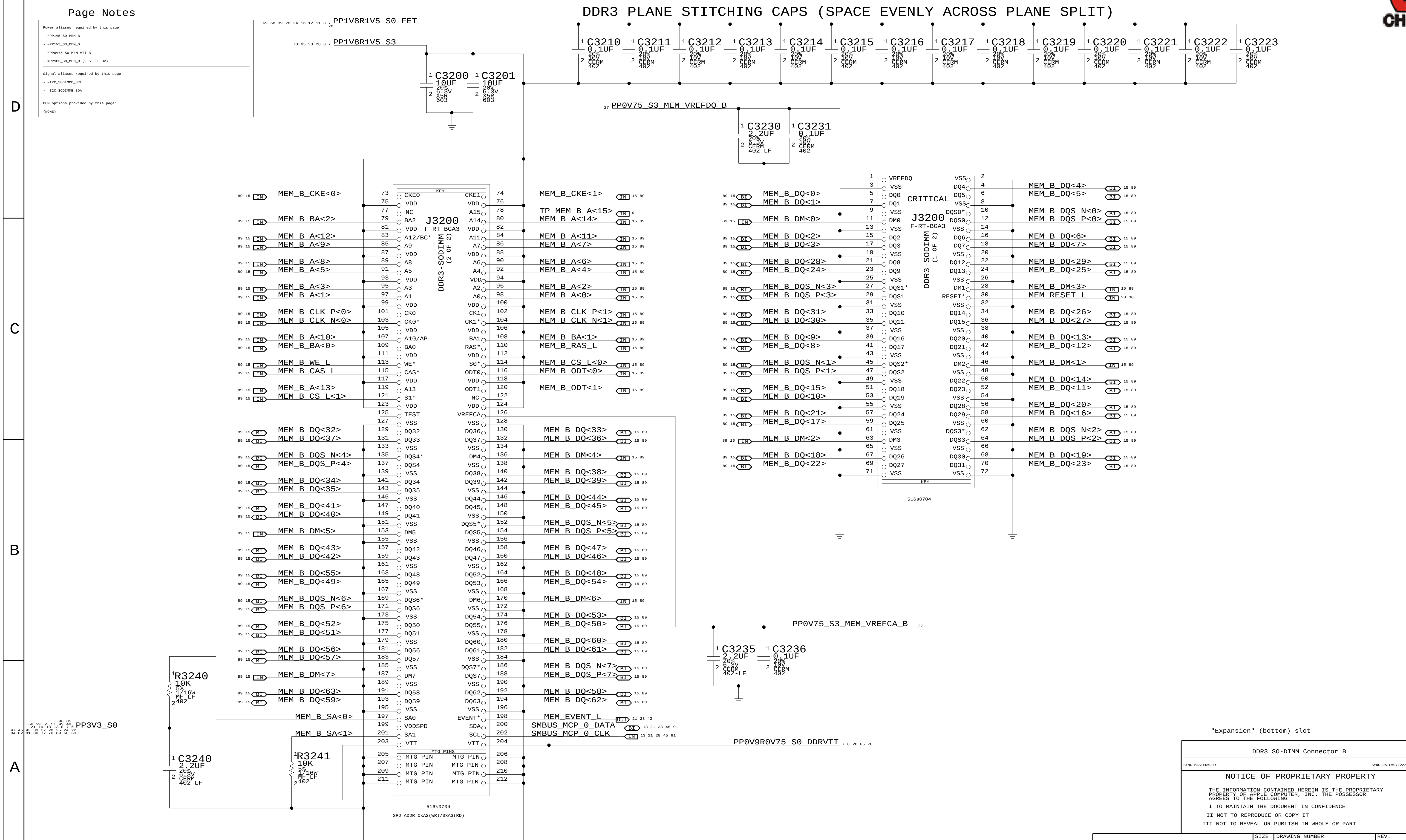
II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE NONE | SHT 28 | OF 97



"Expansion" (bottom) slot

DDR3 SO-DIMM Connector B

SYNC_MASTER=DDR SYNC_DATE=07/22/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

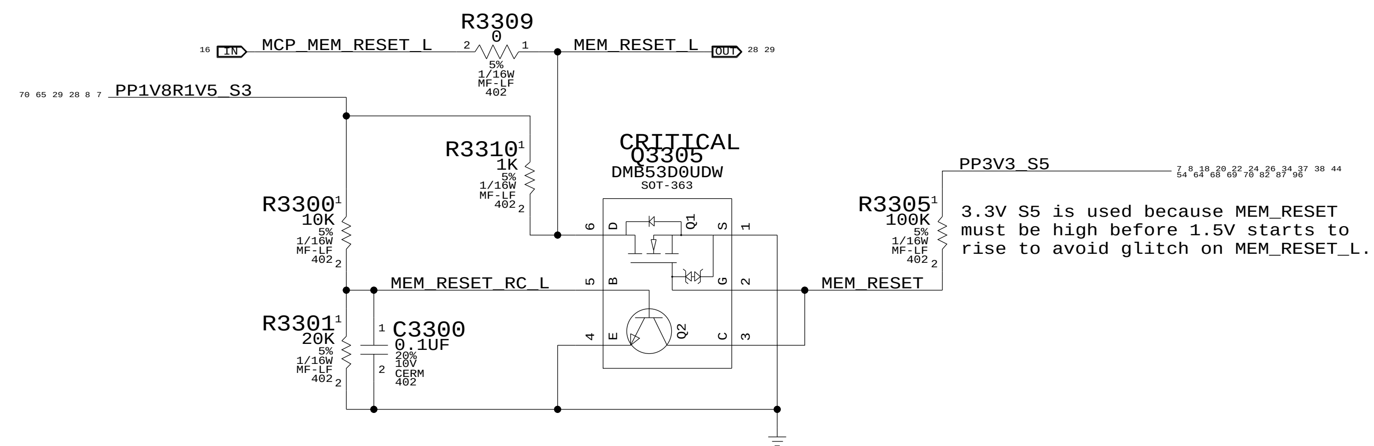
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 29	OF 97

DDR3 RESET Support

Required becaues MCP79 does not meet DDR3 spec power-up reset timing requirement.



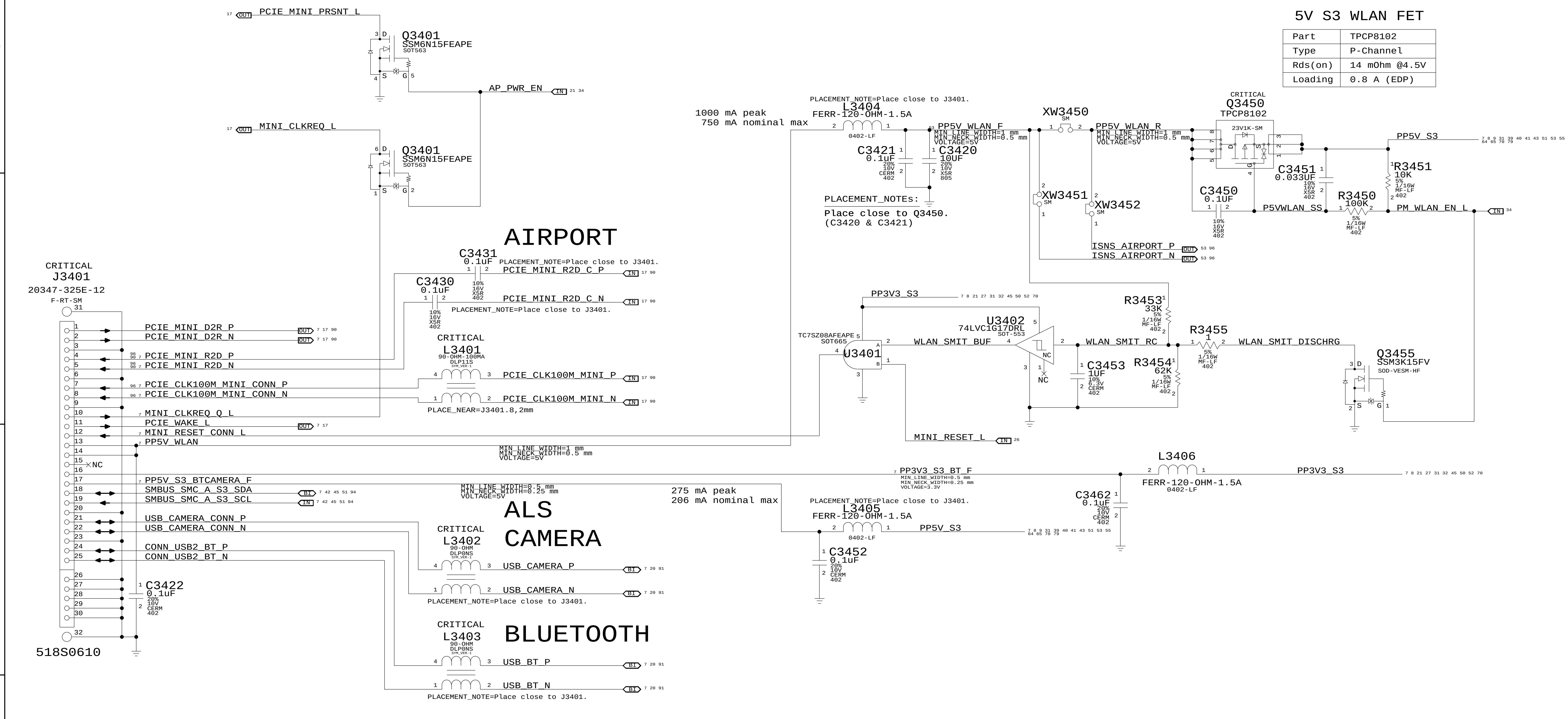
DDR3 Support

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	30	97



Part	TPCP8102
Type	P-Channel
Rds(on)	14 mOhm @4.5V
Loading	0.8 A (EDP)

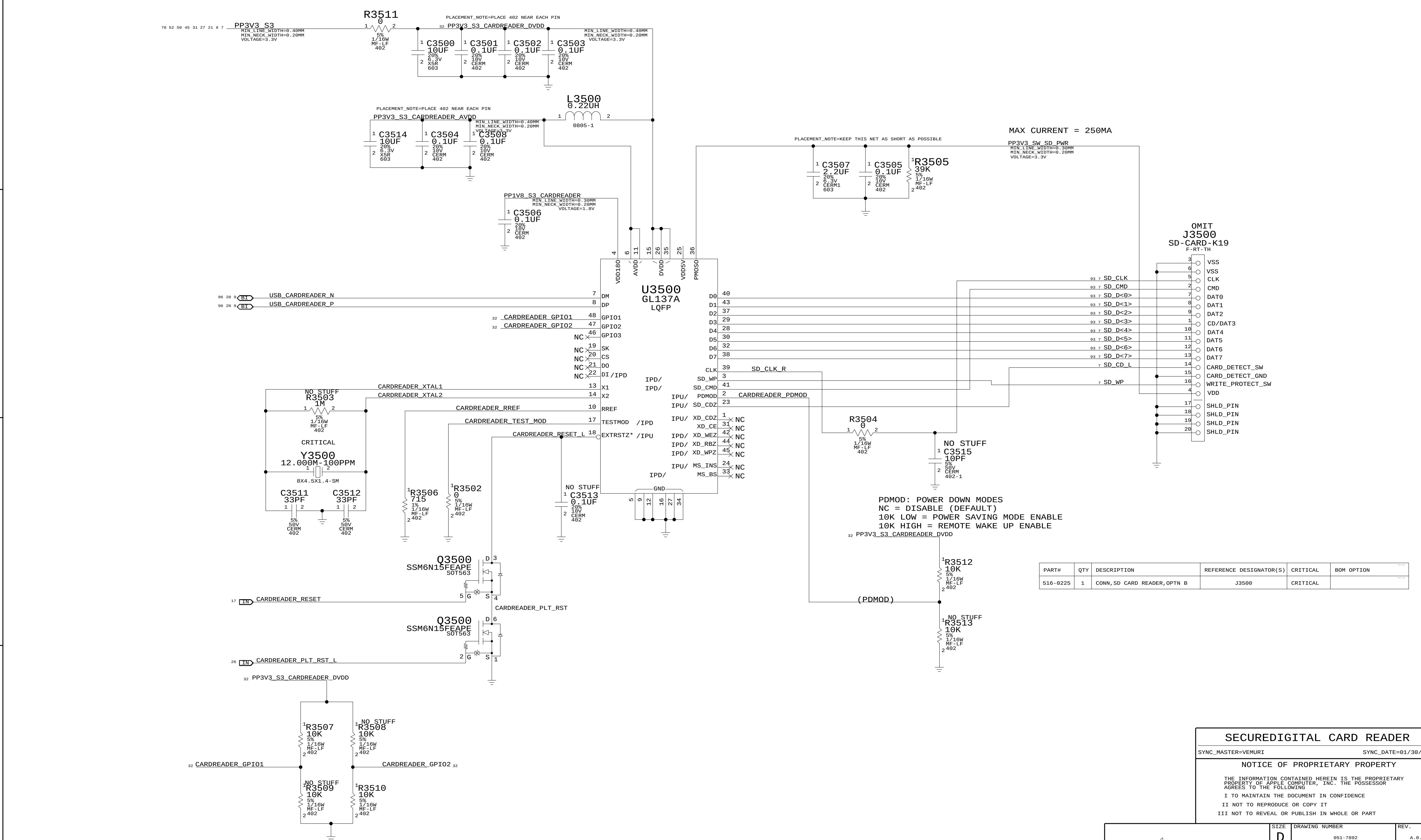
Right Clutch Connector

SYNC_MASTER=MUXGFX SYNC_DATE=12/08/2008

APPLE INC.

SIZE D DRAWING NUMBER 051-7892 REV. A.0.0

SCALE NONE SHT 31 OF 97



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
516-0225	1	CONN,SD CARD READER,OPTN B	J3500	CRITICAL	

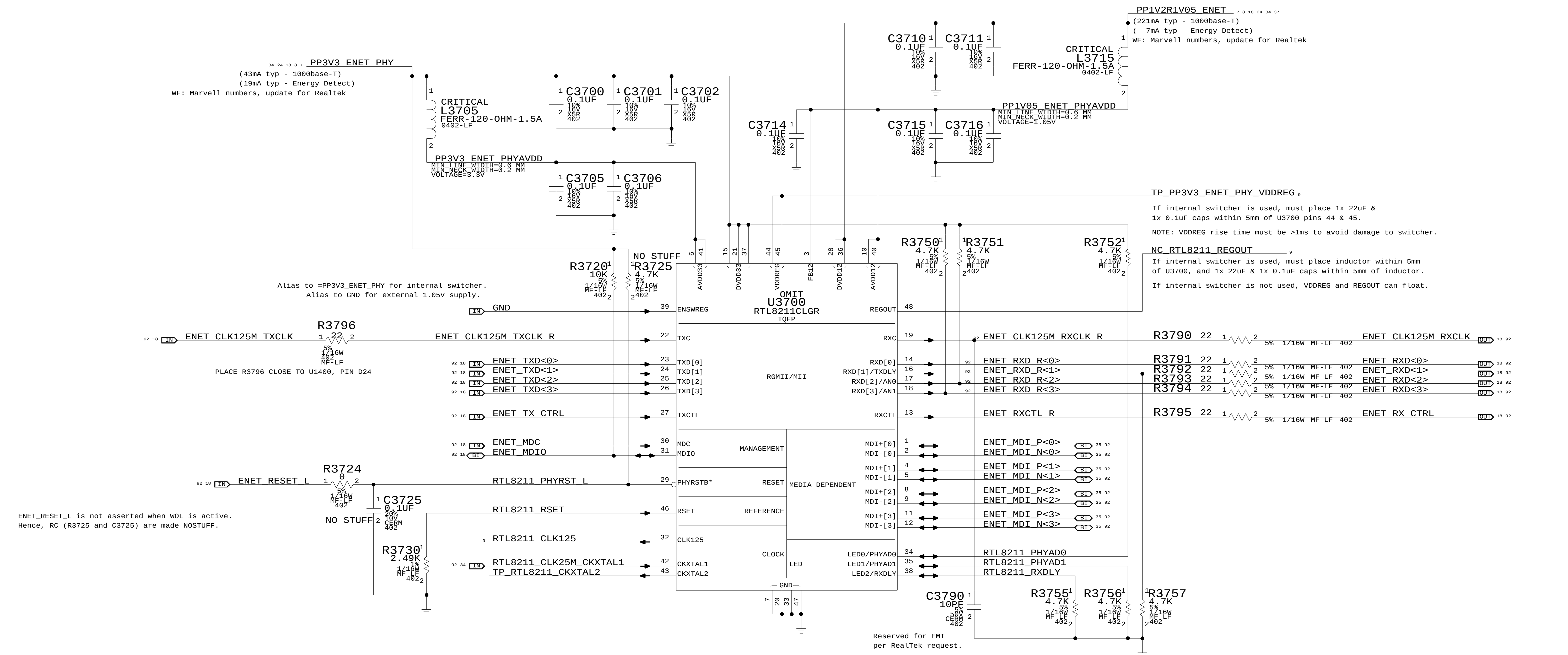
SECUREDIGITAL CARD READER

SYNC_MASTER=VEMURI SYNC_DATE=01/30/2009

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE NONE SHT 32 OF 97



Configuration Settings:

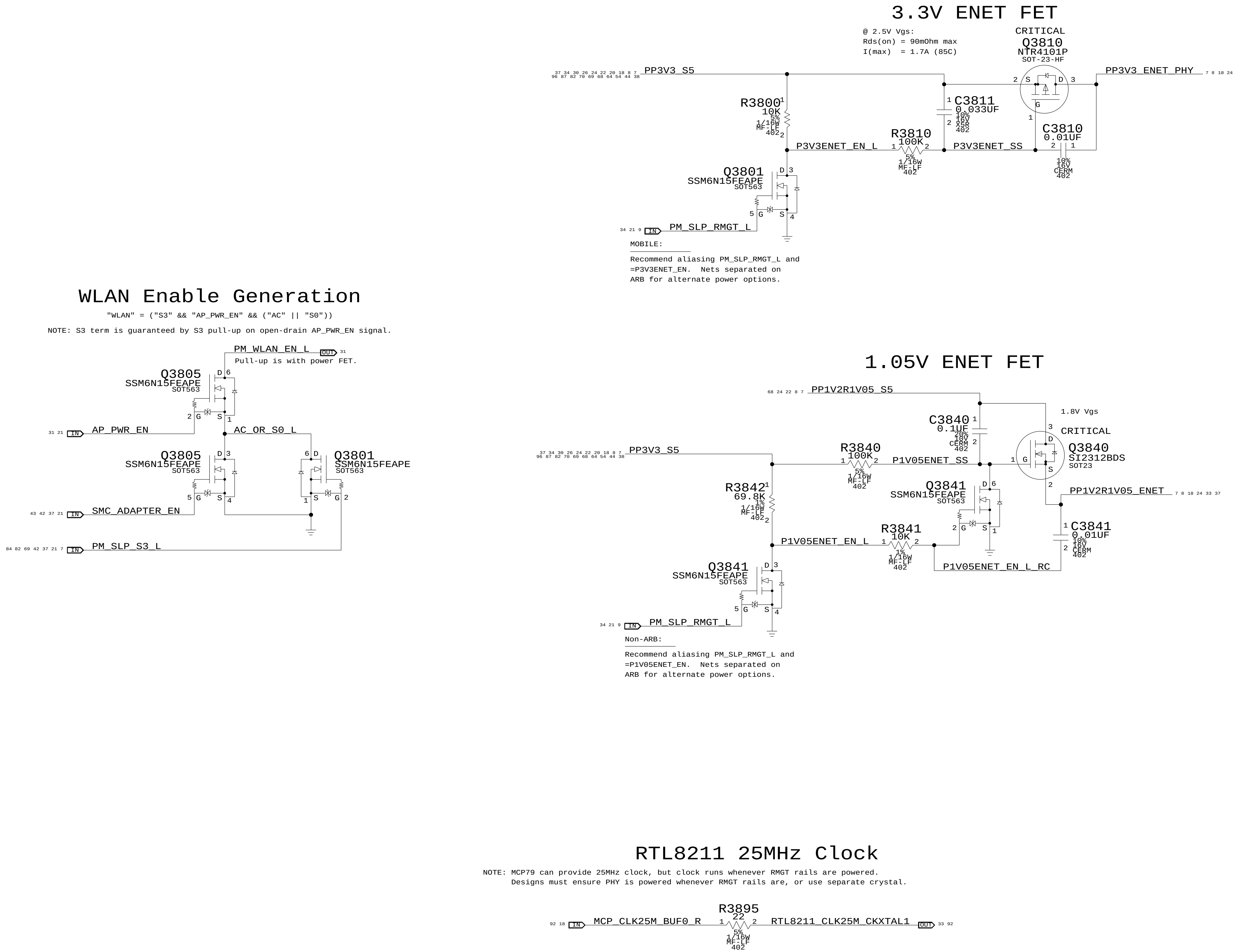
PHYAD = 01 (PHY Address 00001)
AN[1:0] = 11 (Full auto-negotiation)
RXDLY = 0 (RXCLK transitions with data)
TXDLY = 0 (No TXCLK Delay)

Ethernet PHY (RTL8211CL)

SYNC_MASTER=SUMA_M98_MLB SYNC_DATE=07/01/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 33	OF 97



Ethernet & AirPort Support

SYNC_MASTER=SUMA_M98_MLB SYNC_DATE=07/01/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

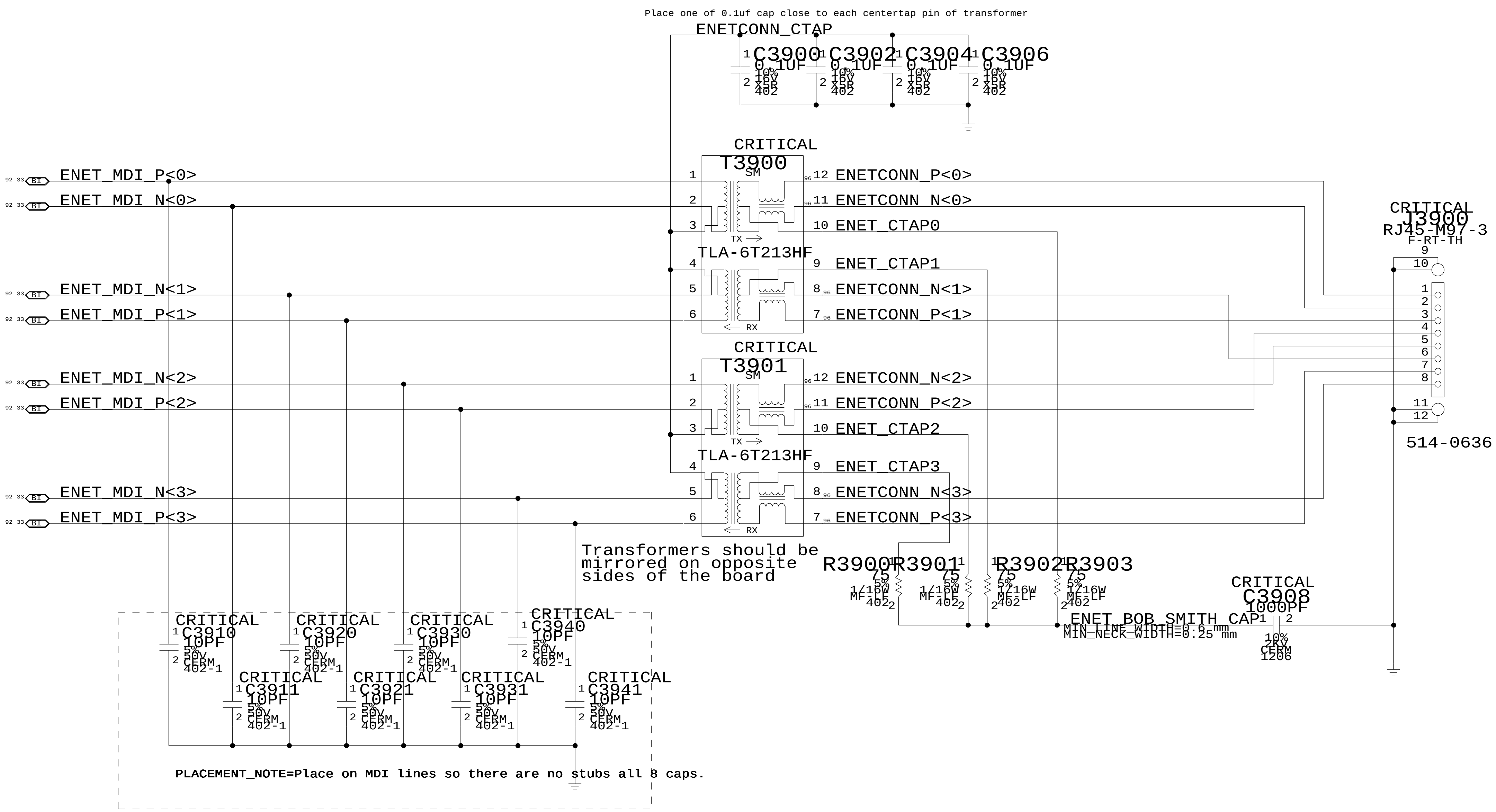
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 34	OF 97

Page Notes

Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



Ethernet Connector

SYNC_MASTER=AMASON_M98_MLB SYNC_DATE=12/16/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

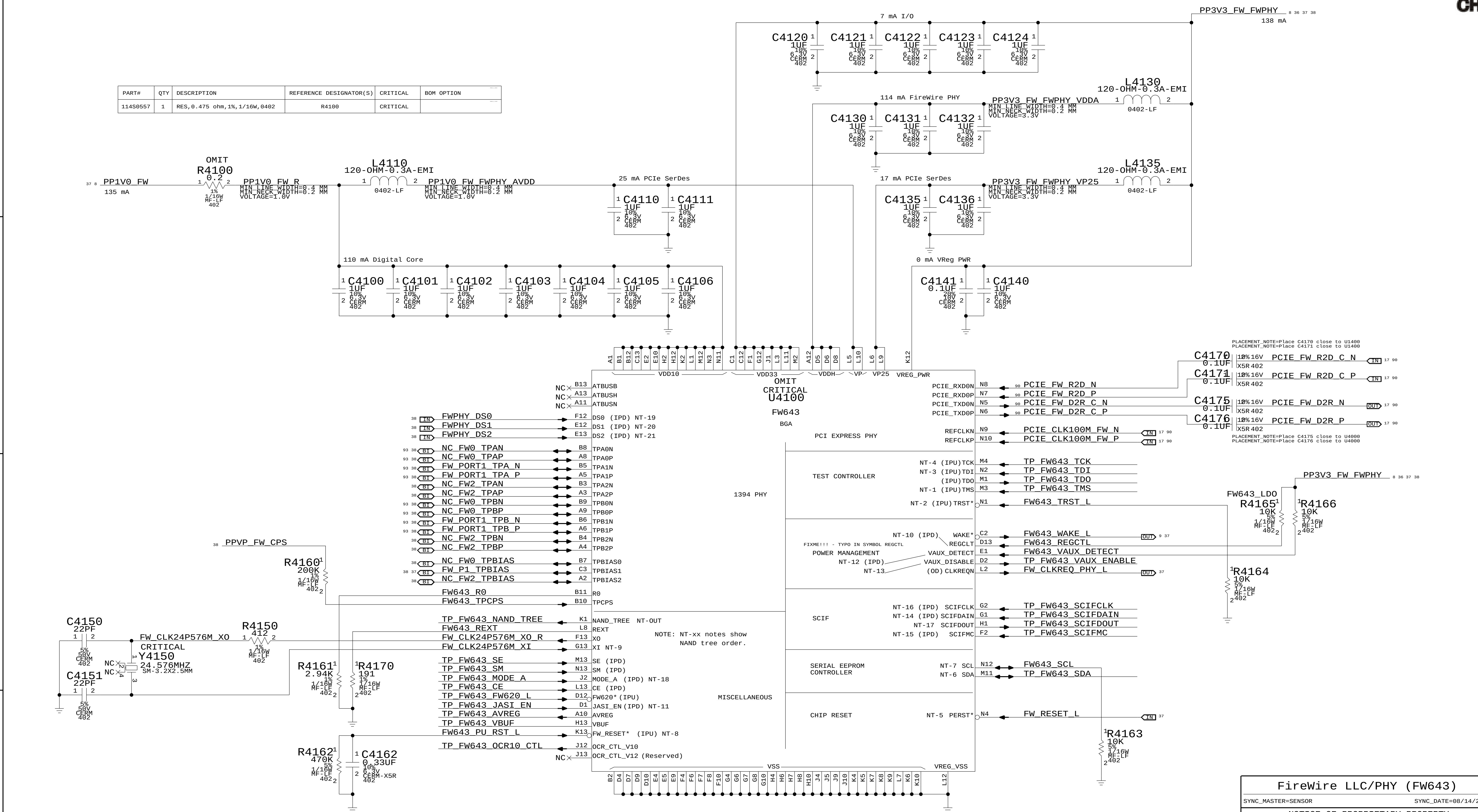
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	35	97

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
114S0557	1	RES,0.475 ohm,1%,1/16W,0402	R4100	CRITICAL	



FireWire LLC/PHY (FW643)

SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

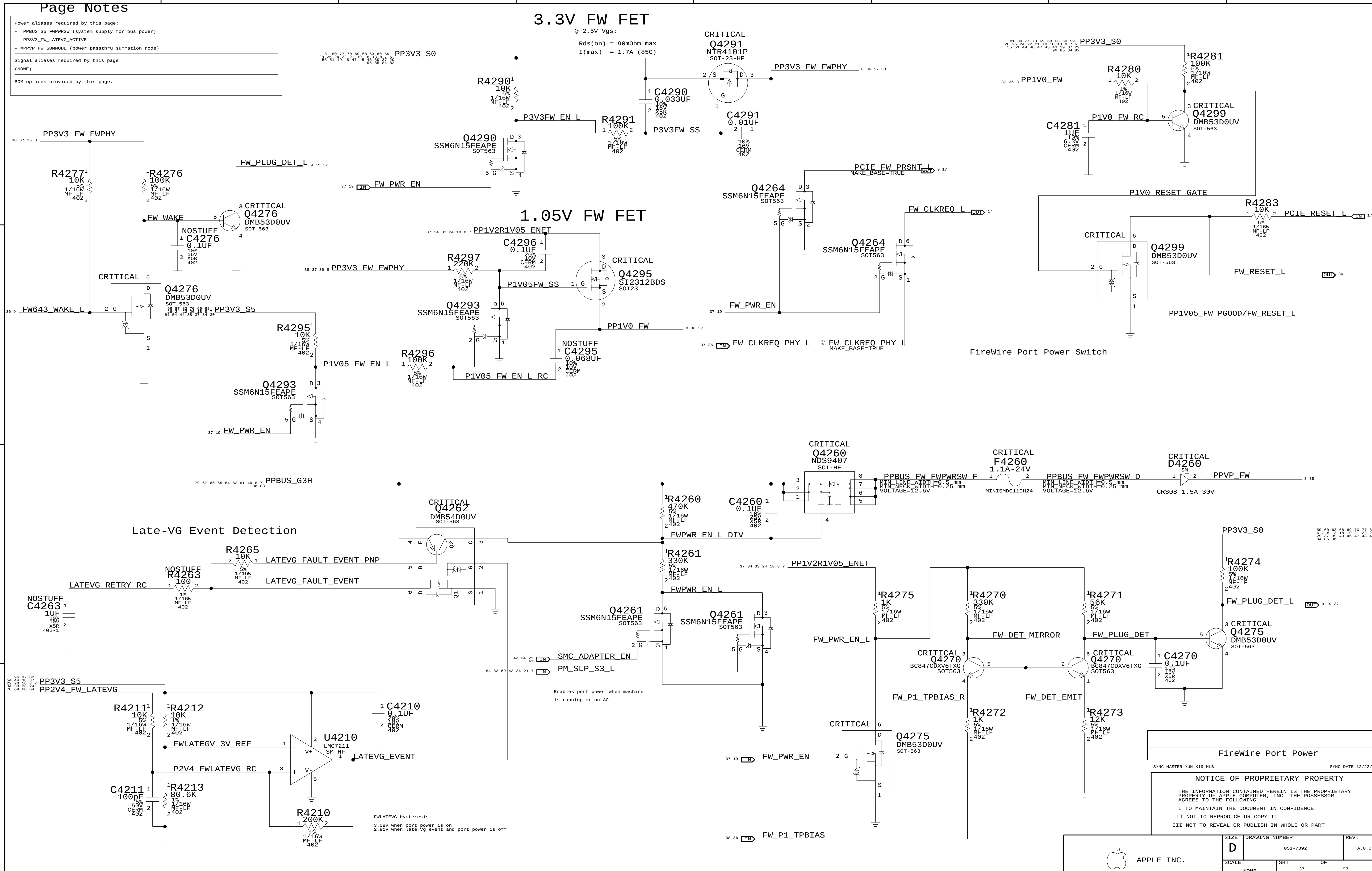
II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	4.12.0

SCALE	SHT	OF
NONE	36	97



FireWire Port Power

SYNC_MASTER=YUN_K19_MLB SYNC_DATE=12/22/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 37	OF 97

Page Notes

Power aliases required by this page:

- =PPVP_FW_PORT1
- =PP3V3_FW_LATEVG

- =GND_CHASSIS_FW_PORT1
- =GND_CHASSIS_FW_EMI_R

Signal aliases required by this page:
(NONE)

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

BOM options provided by this page:
(NONE)

NOTE: FireWire TPA/TPB pairs are NOT constrained on this page. It is assumed that FireWire PHY page will provide the appropriate constraints to apply to entire TPA/TPB XNets.

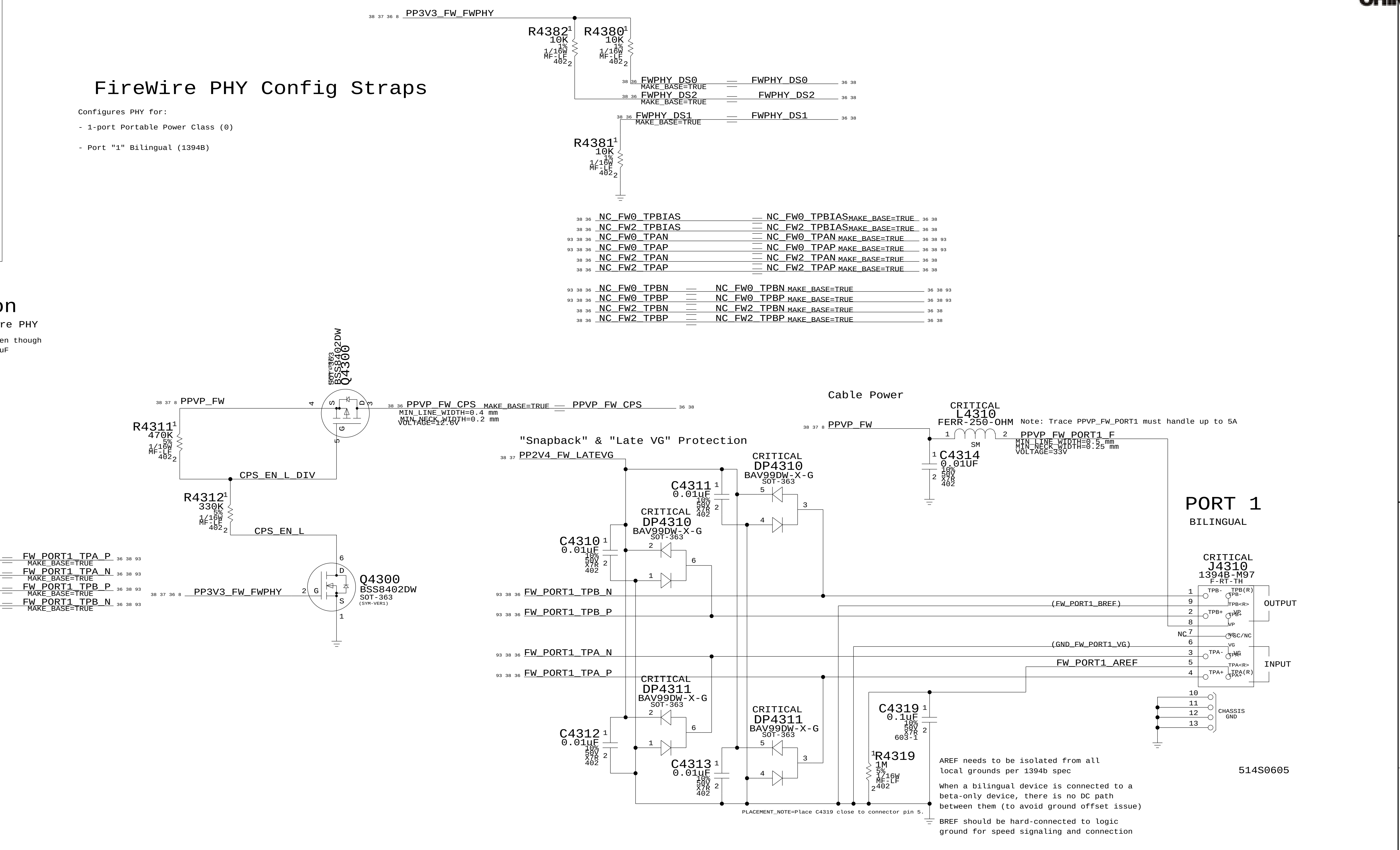
1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

FireWire PHY Config Straps

Configures PHY for:

- 1-port Portable Power Class (0)

- Port "1" Bilingual (1394B)



AREF needs to be isolated from all local grounds per 1394b spec

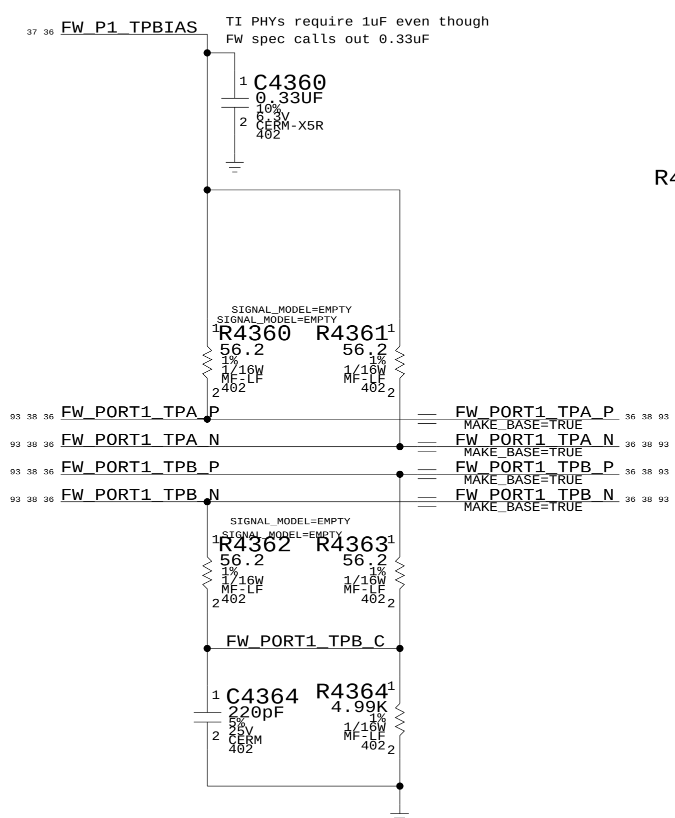
When a bilingual device is connected to a beta-only device, there is no DC path between them (to avoid ground offset issue)

BREF should be hard-connected to logic ground for speed signaling and connection

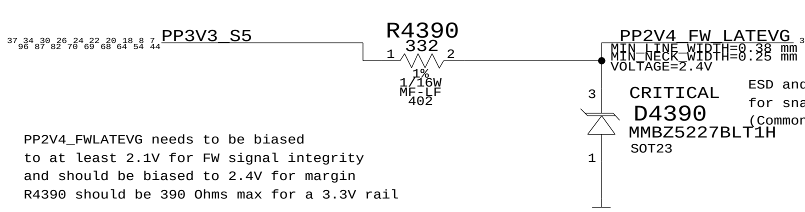
Termination

Place close to FireWire PHY

TI PHYs require 1uF even though FW spec calls out 0.33uF



Late-VG Protection Power



ESD and late-VG rail for snap-back diodes (Common to all ports)

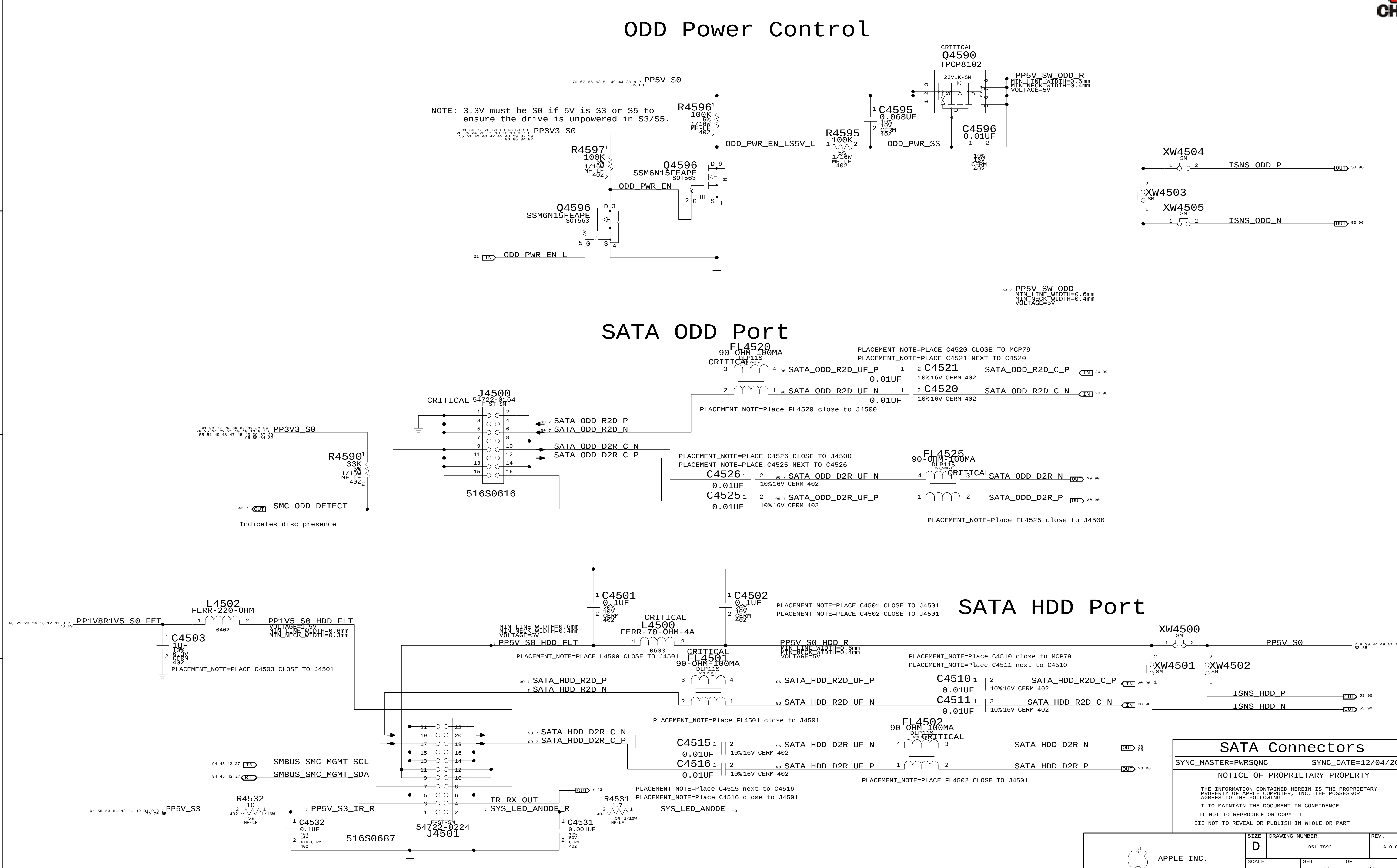
PP2V4_FWLATEVG needs to be biased to at least 2.1V for FW signal integrity and should be biased to 2.4V for margin
R4390 should be 390 Ohms max for a 3.3V rail

FireWire Ports

SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 38	OF 97

APPLE INC.



SATA Connectors

SYNC_MASTER=PWRSQNC SYNC_DATE=12/04/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

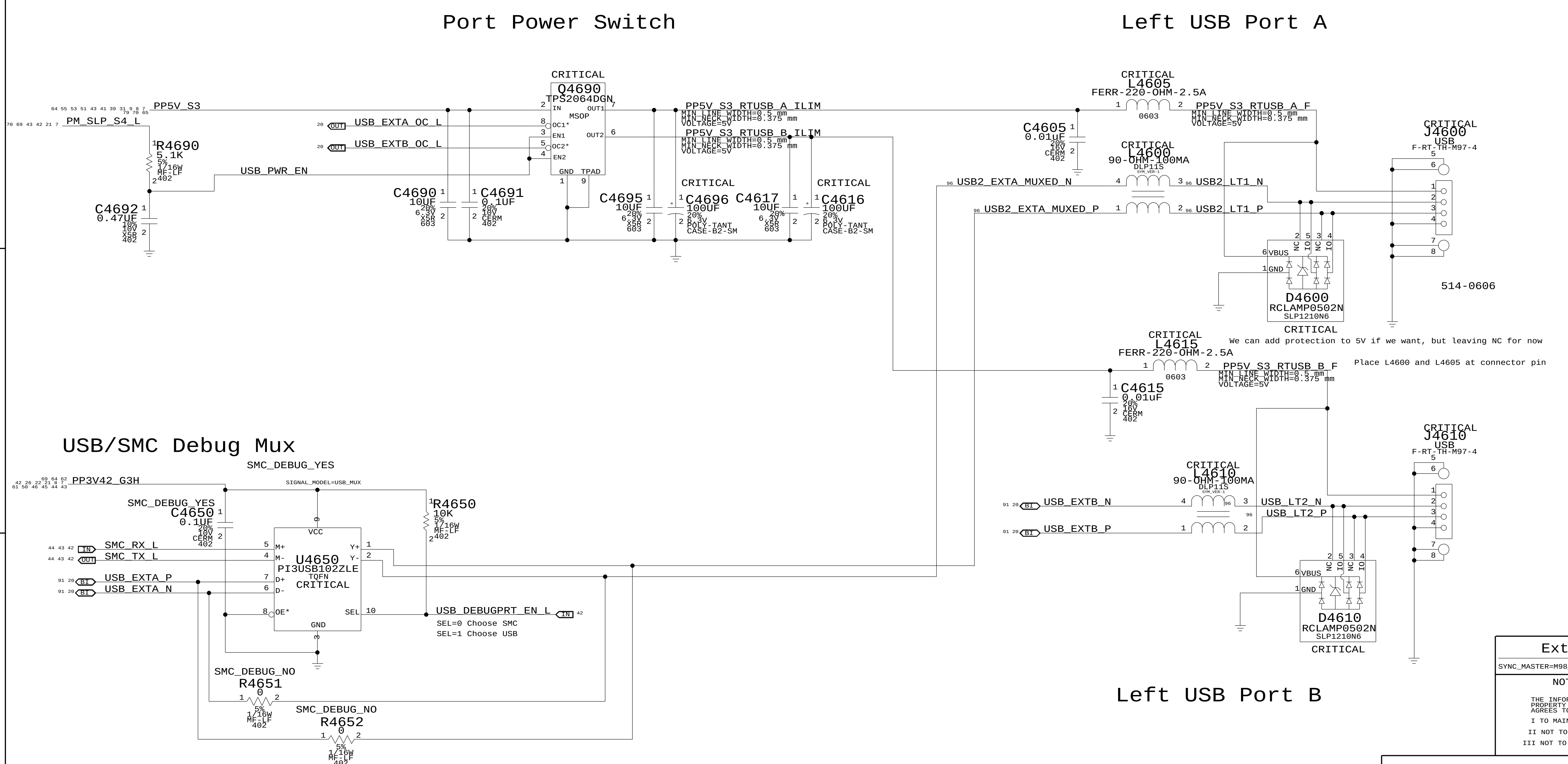
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 39	OF 97



External USB Connectors

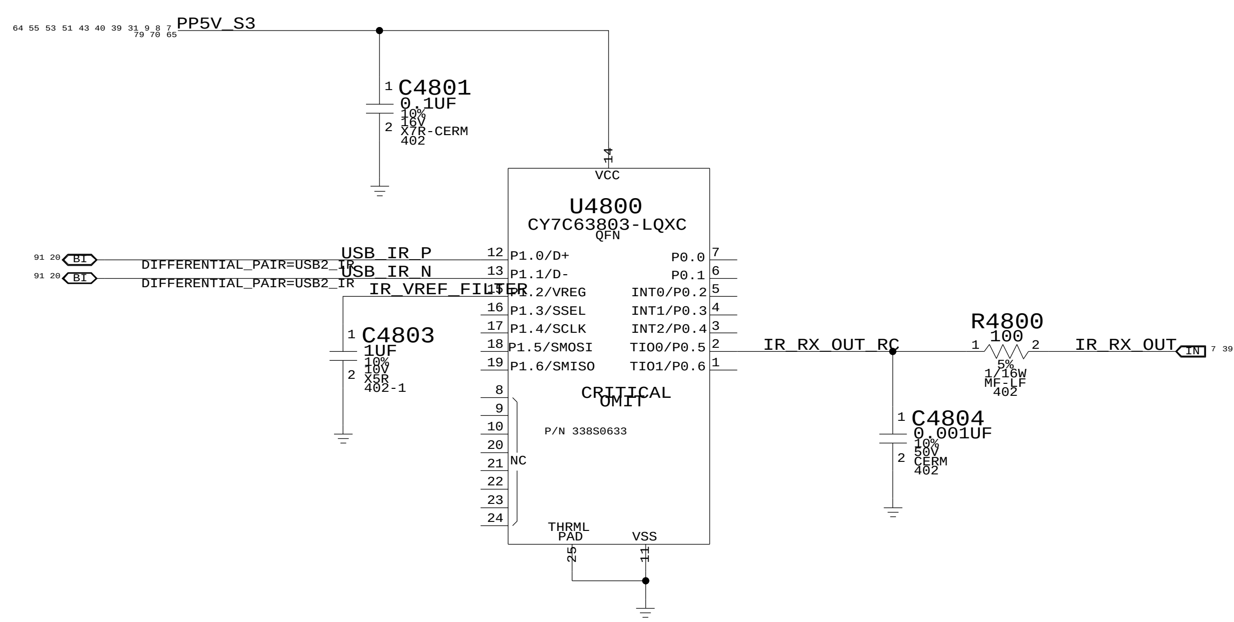
SYNC_MASTER=M98_MLB SYNC_DATE=11/14/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 40	OF 97

IR SUPPORT



Front Flex Support

SYNC_MASTER=PWRSQNC SYNC_DATE=12/04/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE: NONE	SHT 41	OF 97

NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SMC_PB3:

SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay (was ISENSE_CAL_EN)

SMC

SYNC_MASTER=T18_MLB SYNC_DATE=12/12/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

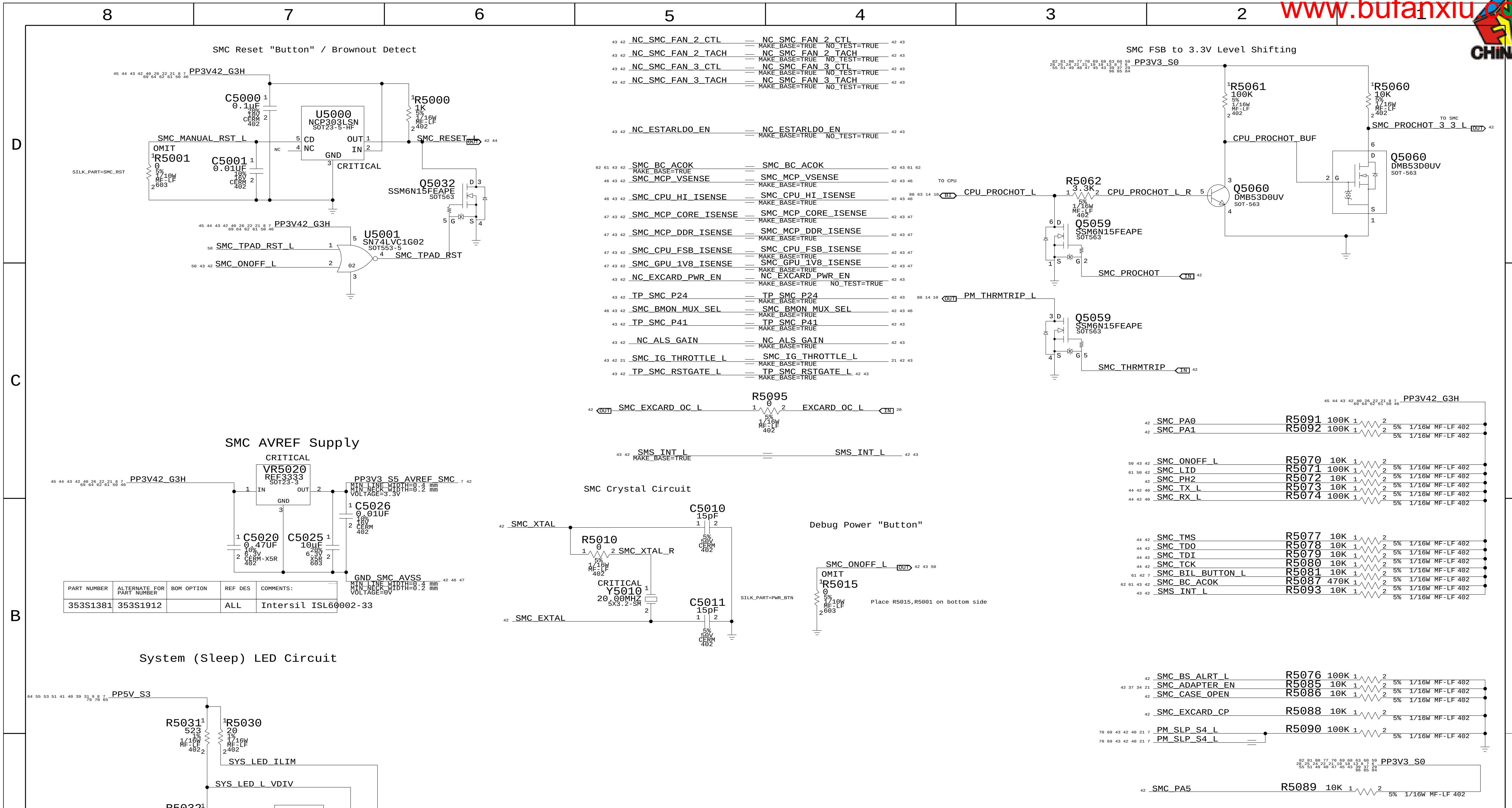
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 42	OF 97



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1912		ALL	Intersil ISL60002-33

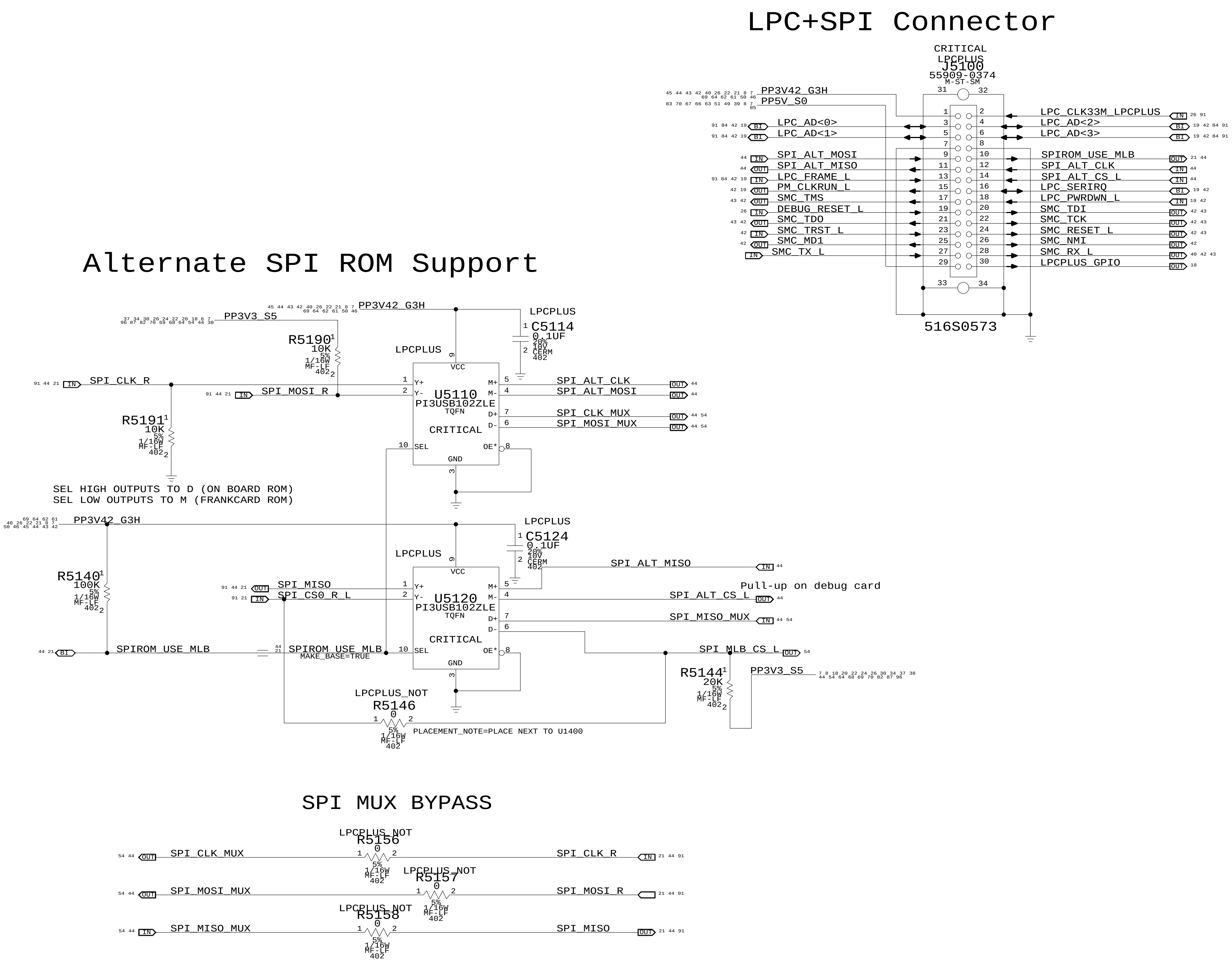
SMC Support

SYNC_MASTER=DDR SYNC_DATE=12/19/2006

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 43	OF 97



LPC+SPI Debug Connector

SYNC_MASTER=CHANGZHANG SYNC_DATE=05/09/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

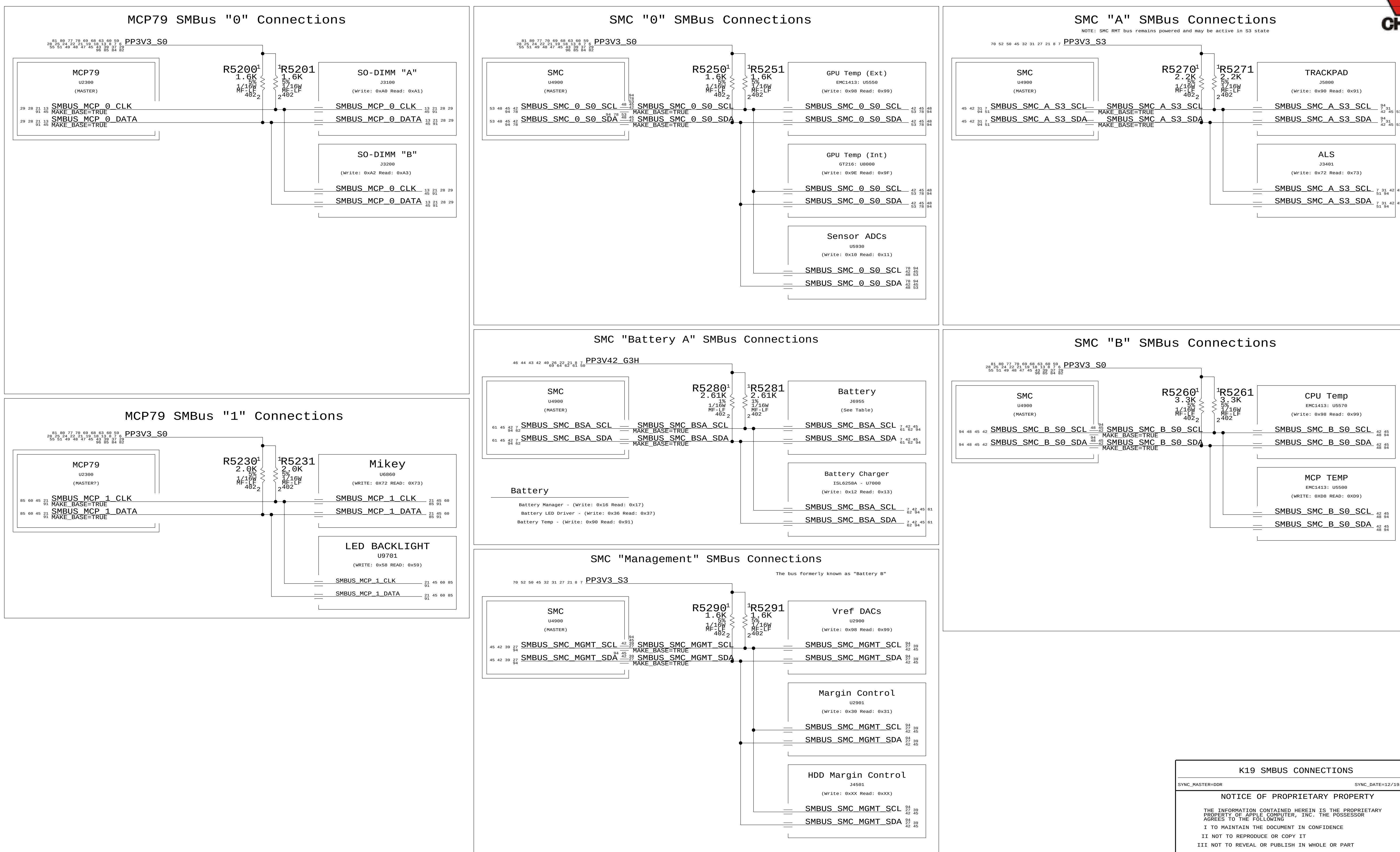
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 44	OF 97



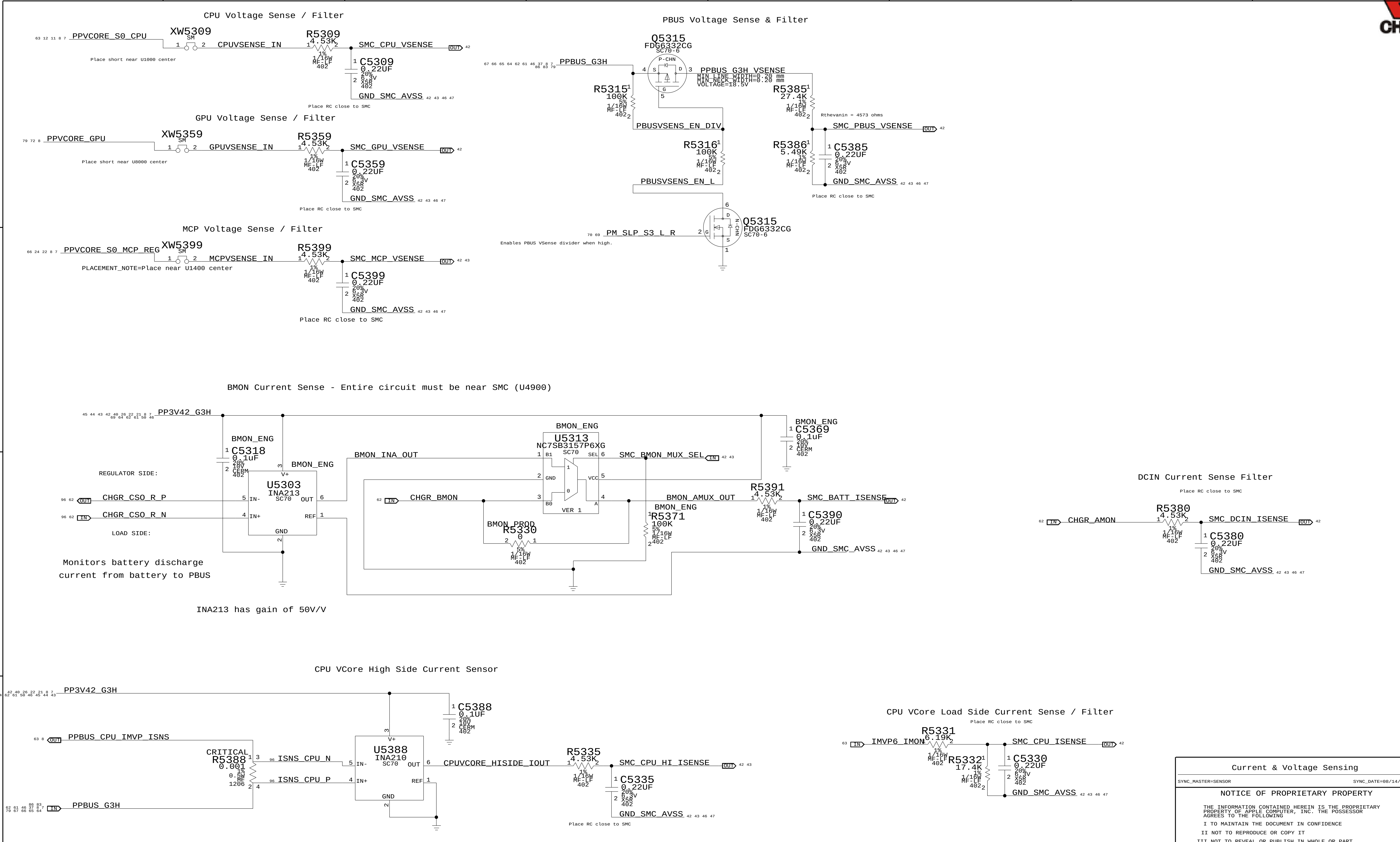
K19 SMBUS CONNECTIONS

SYNC_MASTER=DDR SYNC_DATE=12/19/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 45	OF 97



Current & Voltage Sensing

SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

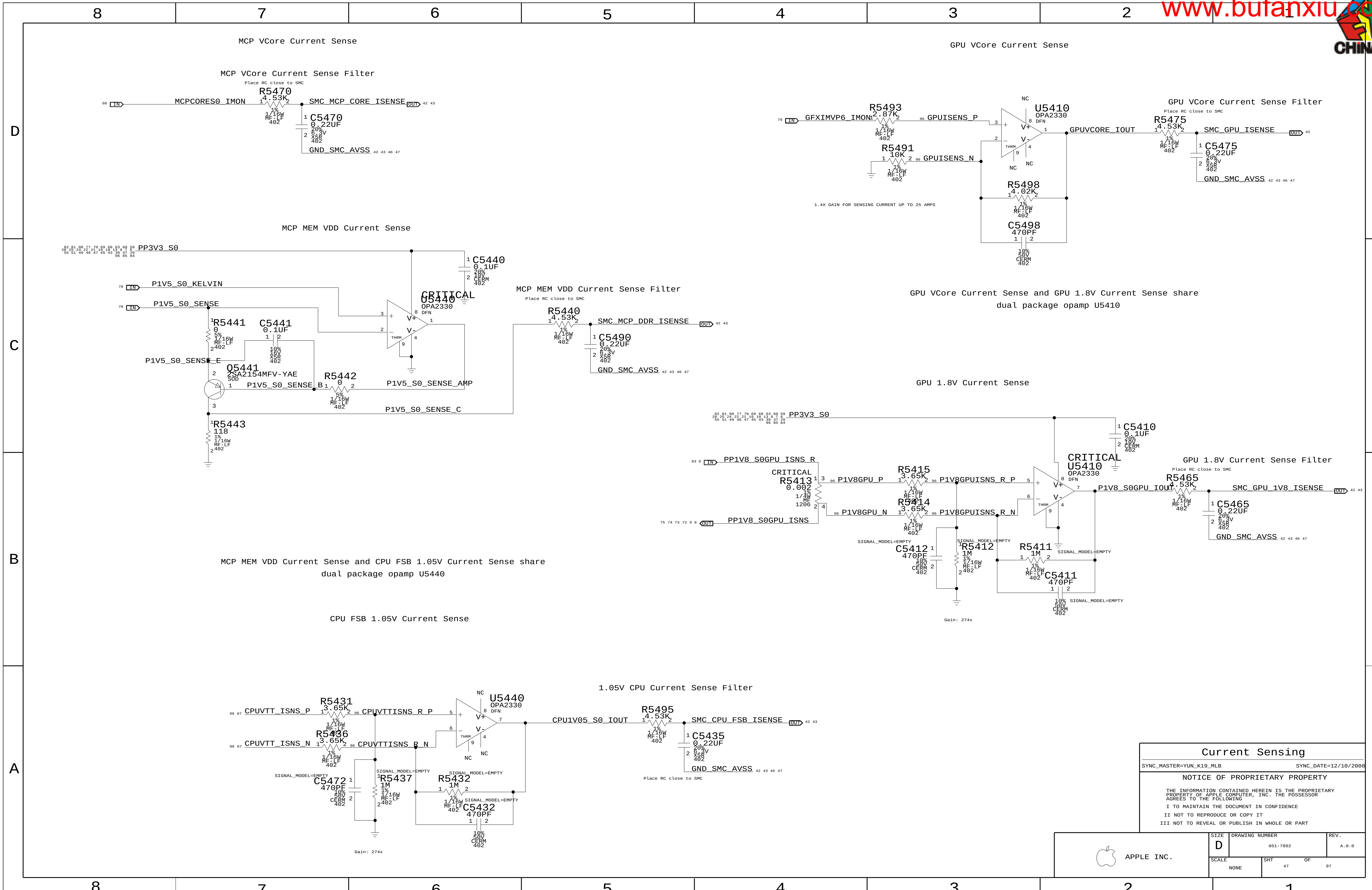
NOTICE OF PROPRIETARY PROPERTY

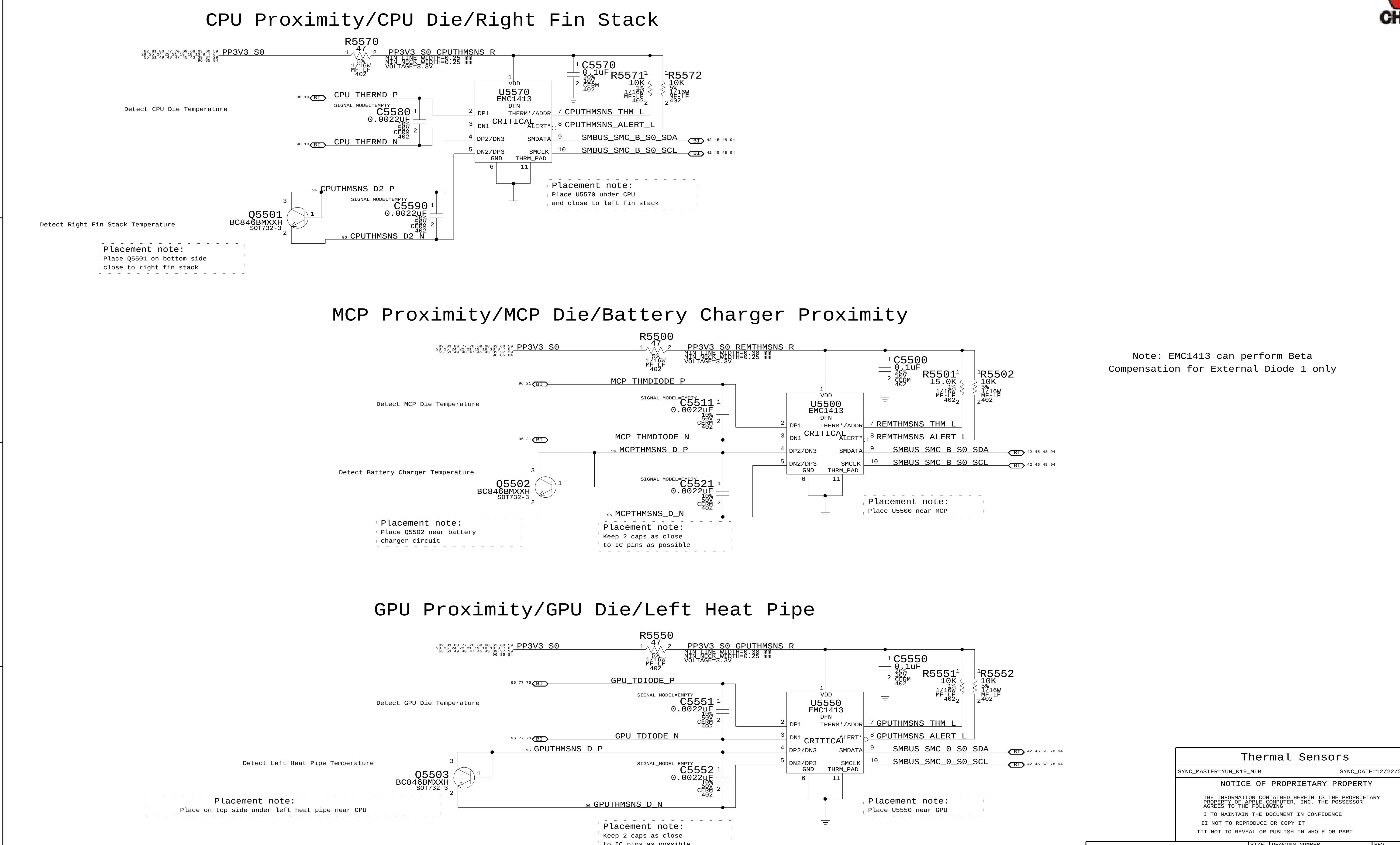
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 46	OF 97


MCP VCore Current Sense
MCP VCore Current Sense Filter
Place RC close to SMC
R5470 4.53K 1% 1/16W MF-LF 402
MCPCORES0_IMON
SMC_MCP_CORE_ISENSE
C5470 0.22UF 20% 6.3V X5R 402
GND_SMC_AVSS
GPU VCore Current Sense
U5410 OPA2330 DFN
R5493 2.87K
GFXIMVP6_IMON
GPUISENS_P
R5491 10K
GPUISENS_N
GPUVCORE_IOUT
R5498 4.02K
C5498 470PF 10% 50V CERM 402
1.4X GAIN FOR SENSING CURRENT UP TO 25 AMPS
GPU VCore Current Sense Filter
Place RC close to SMC
R5475 4.53K
SMC_GPU_ISENSE
C5475 0.22UF
GND_SMC_AVSS
MCP MEM VDD Current Sense
PP3V3_S0
C5440 0.1UF 20% 10V CERM 402
P1V5_S0_KELVIN
P1V5_S0_SENSE
CRITICAL U5440 OPA2330 DFN
R5441 0 5% 1/16W MF-LF 402
C5441 0.1UF 10% 16V X5R 402
P1V5_S0_SENSE_E
Q5441 2SA2154MFV-YAE SOD
P1V5_S0_SENSE_B
R5442 0
P1V5_S0_SENSE_AMP
P1V5_S0_SENSE_C
R5443 118 1% 1/16W MF-LF 402
MCP MEM VDD Current Sense Filter
Place RC close to SMC
R5440 4.53K
SMC_MCP_DDR_ISENSE
C5490 0.22UF
GND_SMC_AVSS
GPU VCore Current Sense and GPU 1.8V Current Sense share dual package opamp U5410
GPU 1.8V Current Sense
PP3V3_S0
C5410 0.1UF 20% 10V CERM 402
PP1V8_S0GPU_ISNS_R
CRITICAL R5413 0.002 1% 1/4W MF 1206
P1V8GPU_P
R5415 3.65K
P1V8GPUISNS_R_P
CRITICAL U5410 OPA2330 DFN
P1V8_S0GPU_IOUT
R5414 3.65K
P1V8GPU_N
P1V8GPUISNS_R_N
PP1V8_S0GPU_ISNS
SIGNAL_MODEL=EMPTY
C5412 470PF 10% 50V CERM 402
R5412 1M
R5411 1M
C5411 470PF
Gain: 274x
GPU 1.8V Current Sense Filter
Place RC close to SMC
R5465 4.53K
SMC_GPU_1V8_ISENSE
C5465 0.22UF
GND_SMC_AVSS
MCP MEM VDD Current Sense and CPU FSB 1.05V Current Sense share dual package opamp U5440
CPU FSB 1.05V Current Sense
CPUVTT_ISNS_P
R5431 3.65K
CPUVTTISNS_R_P
U5440 OPA2330 DFN
CPUVTT_ISNS_N
R5436 3.65K
CPUVTTISNS_R_N
C5472 470PF
R5437 1M
R5432 1M
C5432 470PF
Gain: 274x
1.05V CPU Current Sense Filter
CPU1V05_S0_IOUT
R5495 4.53K
SMC_CPU_FSB_ISENSE
C5435 0.22UF
GND_SMC_AVSS
Place RC close to SMC
Current Sensing
SYNC_MASTER=YUN_K19_MLB
SYNC_DATE=12/10/2008
NOTICE OF PROPRIETARY PROPERTY
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART
APPLE INC.
SIZE D
DRAWING NUMBER 051-7892
REV. A.0.0
SCALE NONE
SHT 47 OF 97



Thermal Sensors

SYNC_MASTER=YUN_K19_MLB SYNC_DATE=12/22/2008

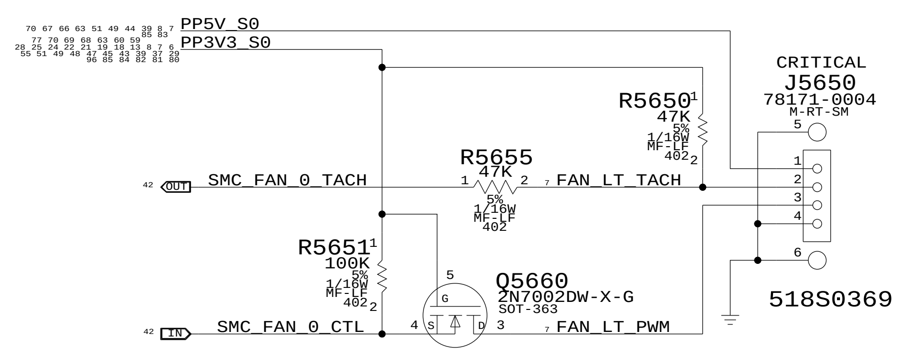
NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

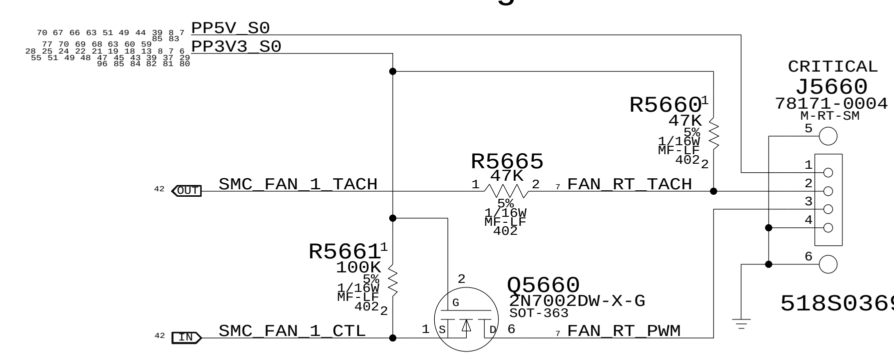
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE NONE SHT 48 OF 97

Left Fan



Right Fan

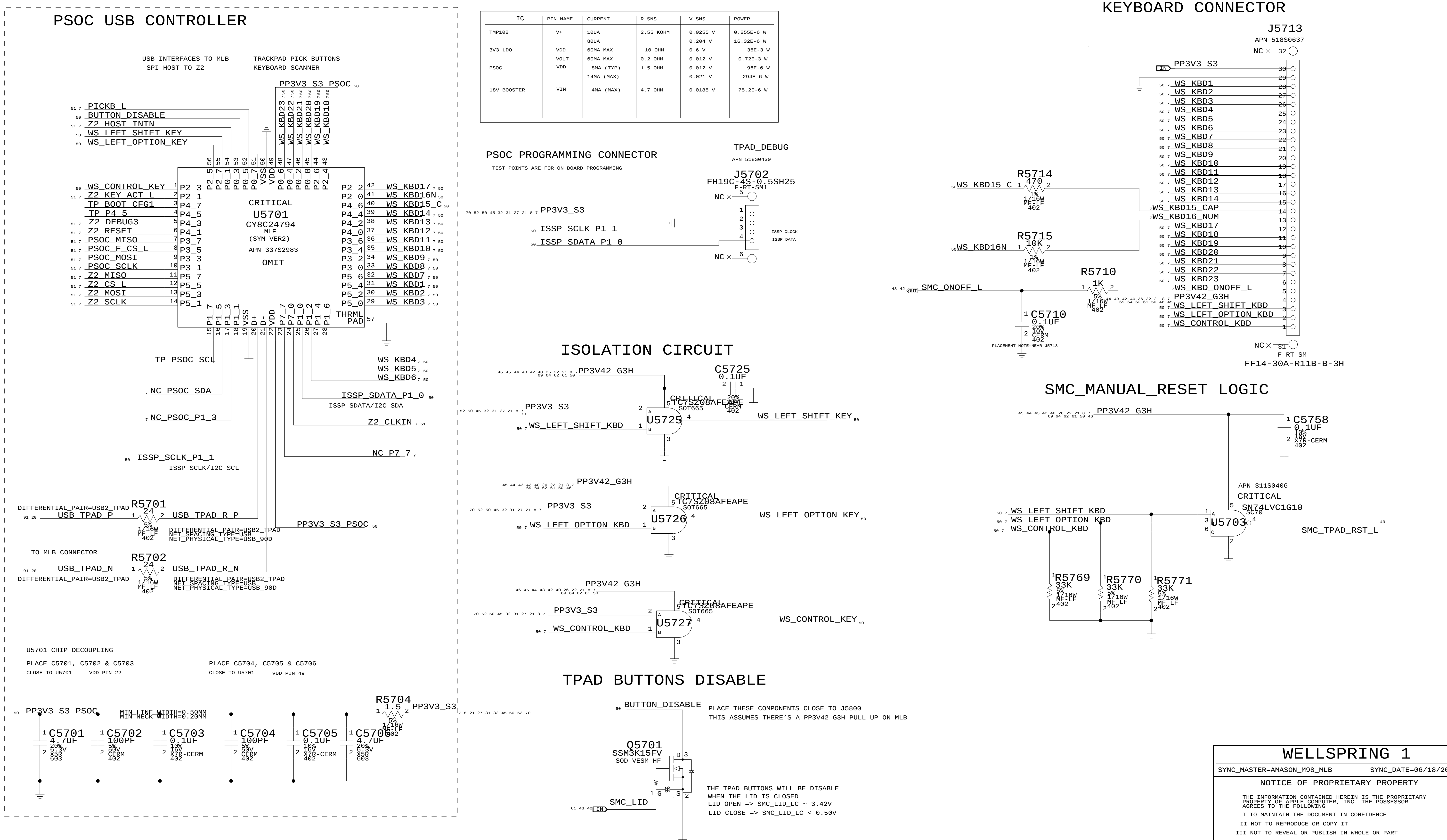


Fan Connectors

SYNC_MASTER=M87_MLB SYNC_DATE=10/17/2007

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 49	OF 97



IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA	2.55 KOHM	0.0255 V	0.255E-6 W
		80UA		0.204 V	16.32E-6 W
3V3 LDO	VDD	60MA MAX	10 OHM	0.6 V	36E-3 W
	VOUT	60MA MAX	0.2 OHM	0.012 V	0.72E-3 W
PSOC	VDD	8MA (TYP)	1.5 OHM	0.012 V	96E-6 W
		14MA (MAX)		0.021 V	294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

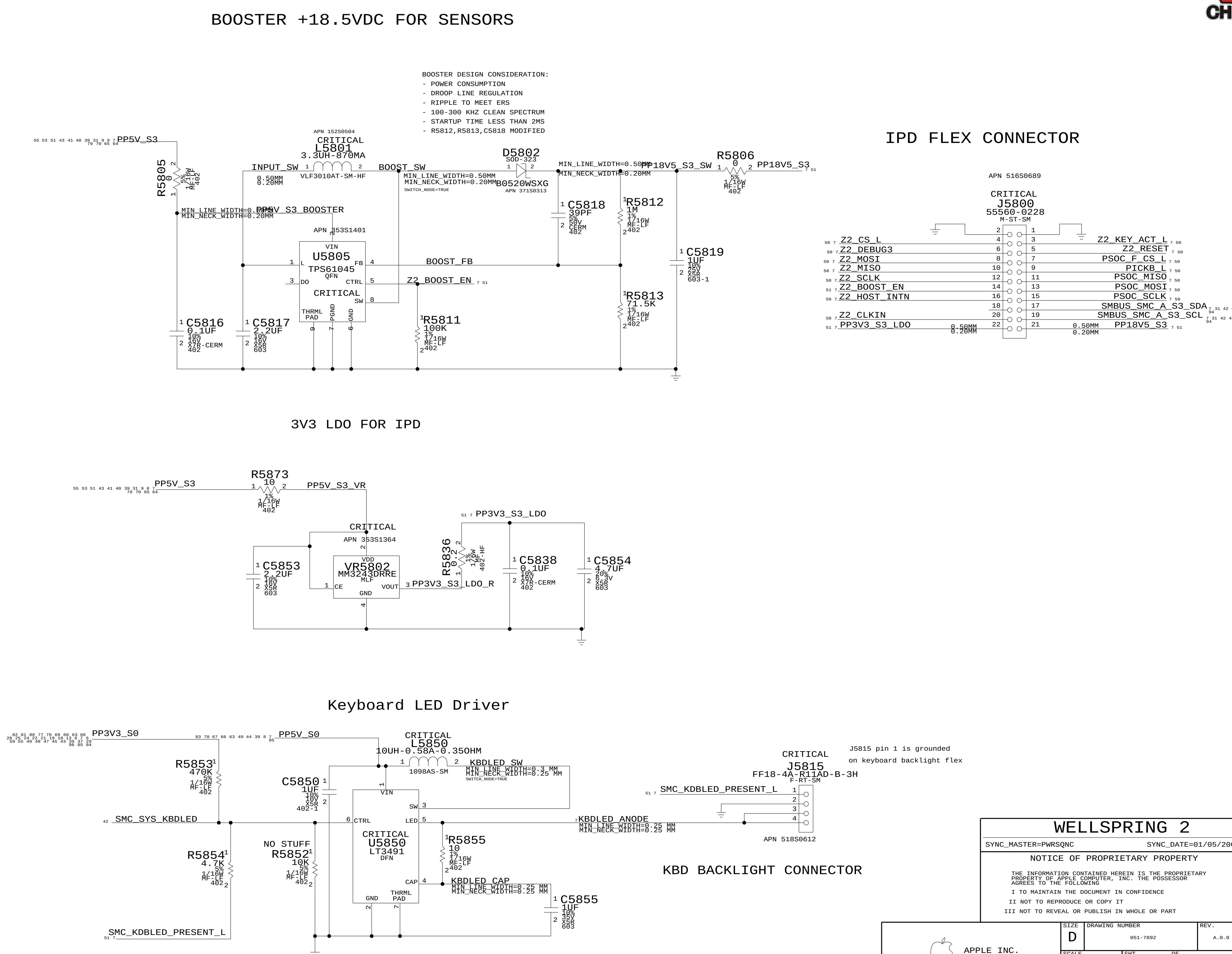
WELLSPRING 1

SYNC_MASTER=AMASON_M98_MLB SYNC_DATE=06/18/2008

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE | SHT 50 OF 97



To detect Keyboard backlight, SMC will tristate SMC_SYS_KBDLED:

LOW = keyboard backlight present

HIGH= keyboard backlight not present

BOM OPTION: KBDLED_YES

R5853 ALWAYS PRESENT

WELLSPRING 2

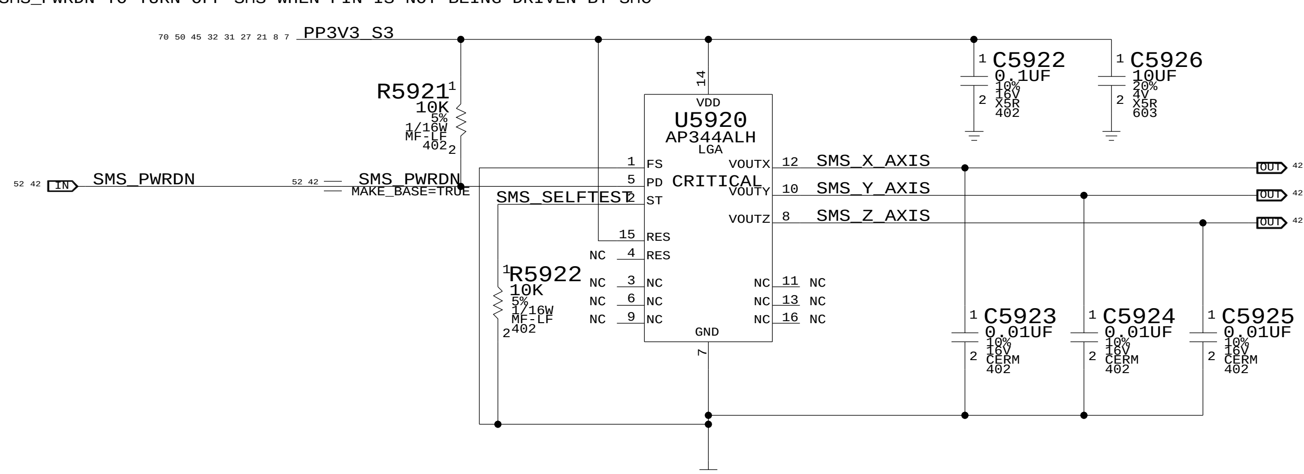
SYNC_MASTER=PWRSQNC SYNC_DATE=01/05/2009

APPLE INC.

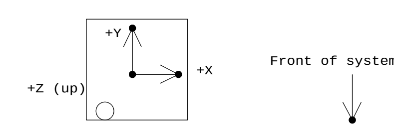
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 51	OF 97

Analog SMS

R5921 PULLS UP SMS_PWRDN TO TURN OFF SMS WHEN PIN IS NOT BEING DRIVEN BY SMC



Desired orientation when placed on board top-side:



Circle indicates pin 1 location when placed in correct orientation

Sudden Motion Sensor (SMS)

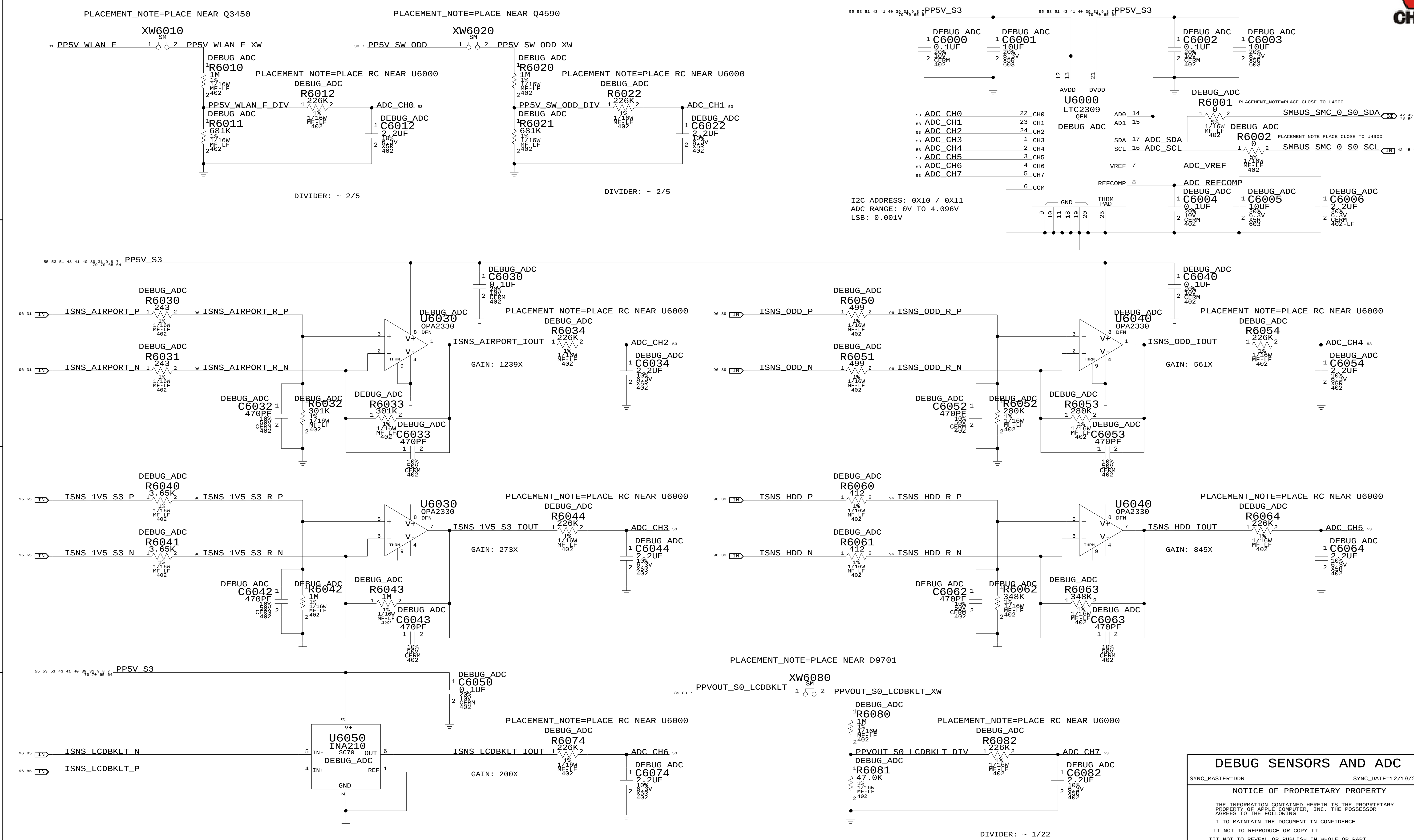
SYNC_MASTER=SENSOR SYNC_DATE=08/14/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	52	97



DEBUG SENSORS AND ADC

SYNC_MASTER=DDR
SYNC_DATE=12/19/2008

NOTICE OF PROPRIETARY PROPERTY

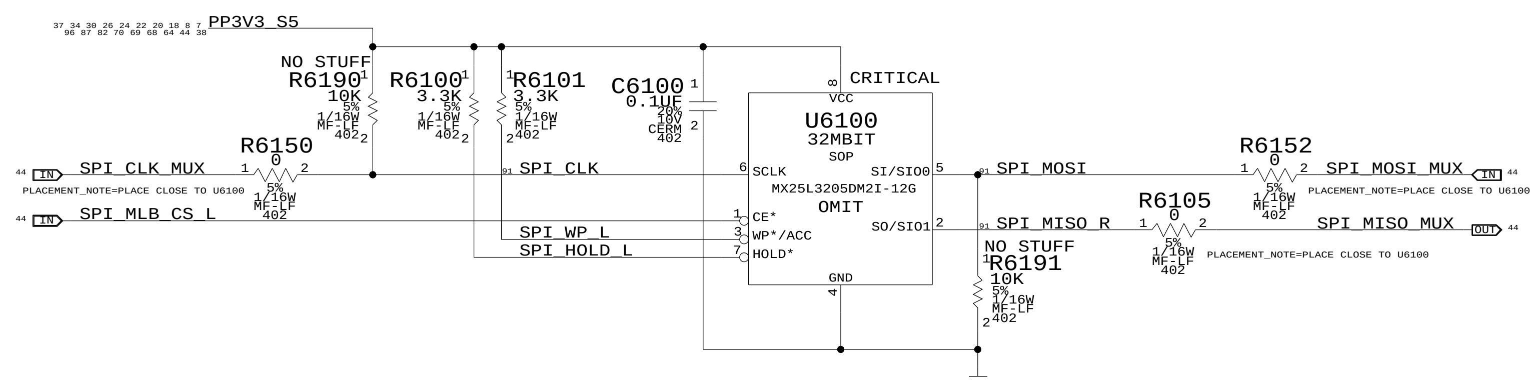
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	53	97



MCP79 SPI Frequency Select

Frequency	SPI_MOSI	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

25MHz is selected with R5190 and R5191
Any of the 4 frequencies can be selected
with R6190, R6191, R5190 and R5191

SPI ROM

SYNC_MASTER=CHANG_M98_MLB SYNC_DATE=07/01/2008

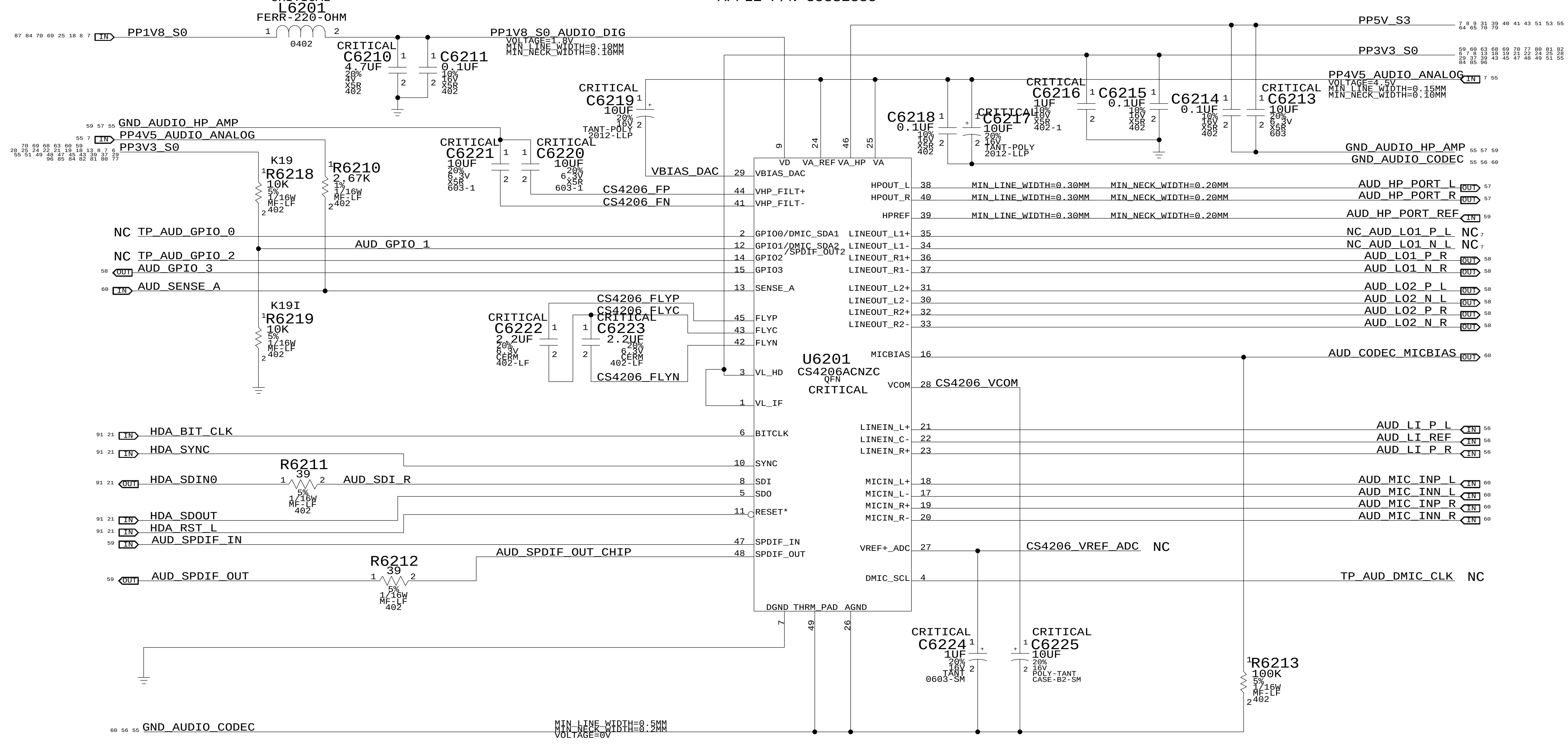
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE: NONE

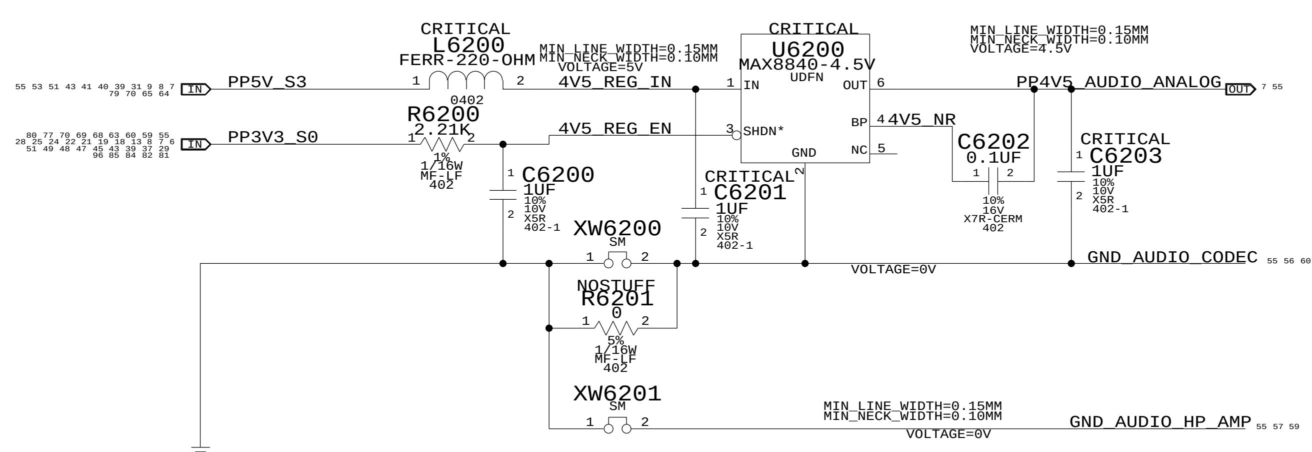
AUDIO CODEC

APPLE P/N 353S2355



4.5V POWER SUPPLY FOR CODEC

APPLE P/N 353S2234



NOTES ON CODEC I/O

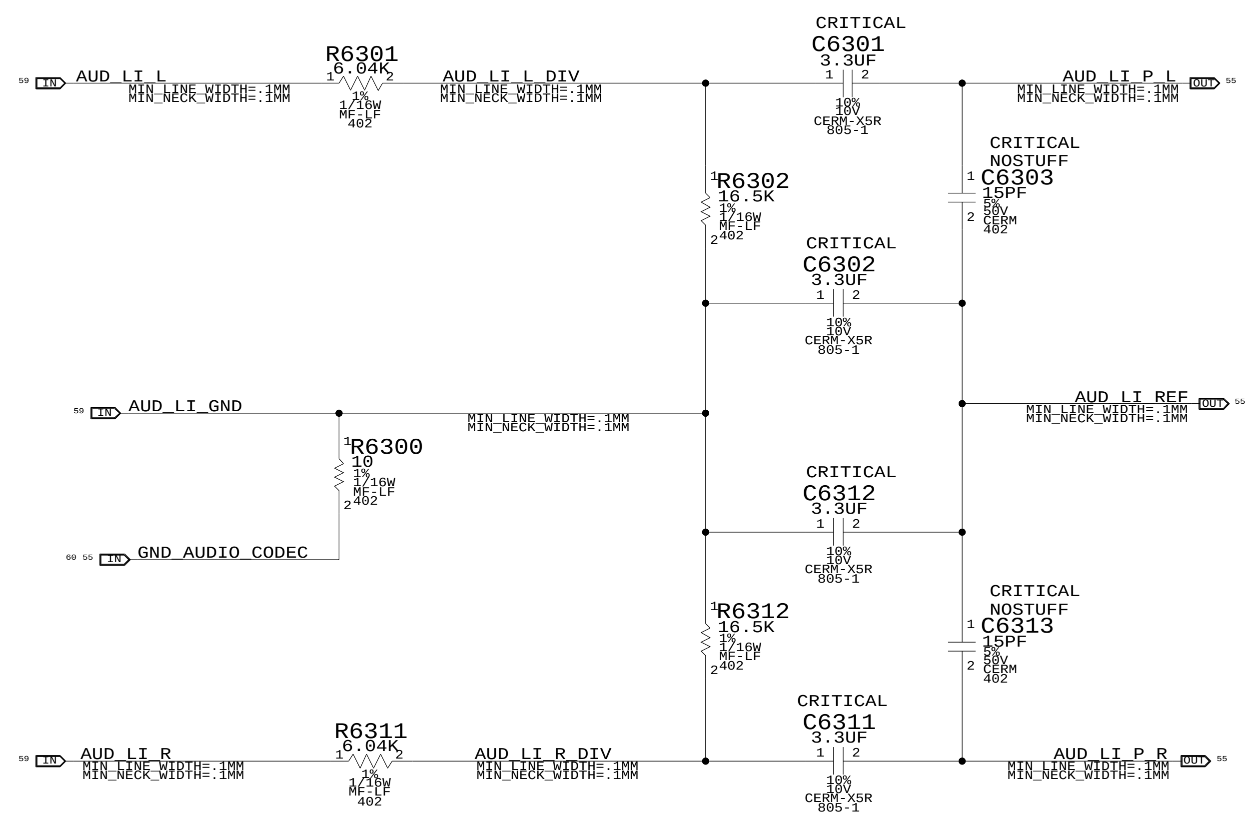
DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

AUDIO: CODEC/REGULATOR

APPLE INC.	SIZE D	DRAWING NUMBER 051-7892	REV. A.0.0
	SCALE NONE	SHT 55	OF 97

LINE INPUT VOLTAGE DIVIDER

CODEC RIN = 20K OHMS
NET RIN = 20K OHMS
FC = 8 HZ
VIN = 2VRMS, CODEC VIN = 1.21 VRMS



AUDIO: LINE INPUT FILTER

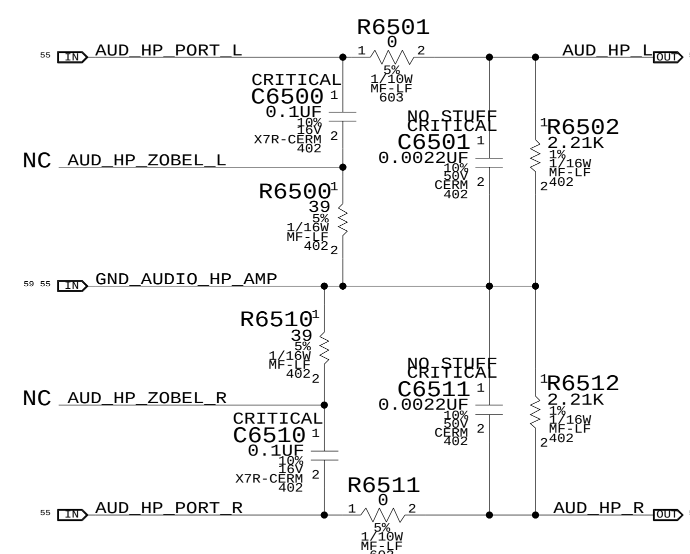
NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	56	97

ZOBEL NETWORK & 1ST ORDER DAC FILTER PLACEHOLDER



AUDIO: HEADPHONE FILTER

SYNC_MASTER=AUDIO SYNC_DATE=03/16/2009

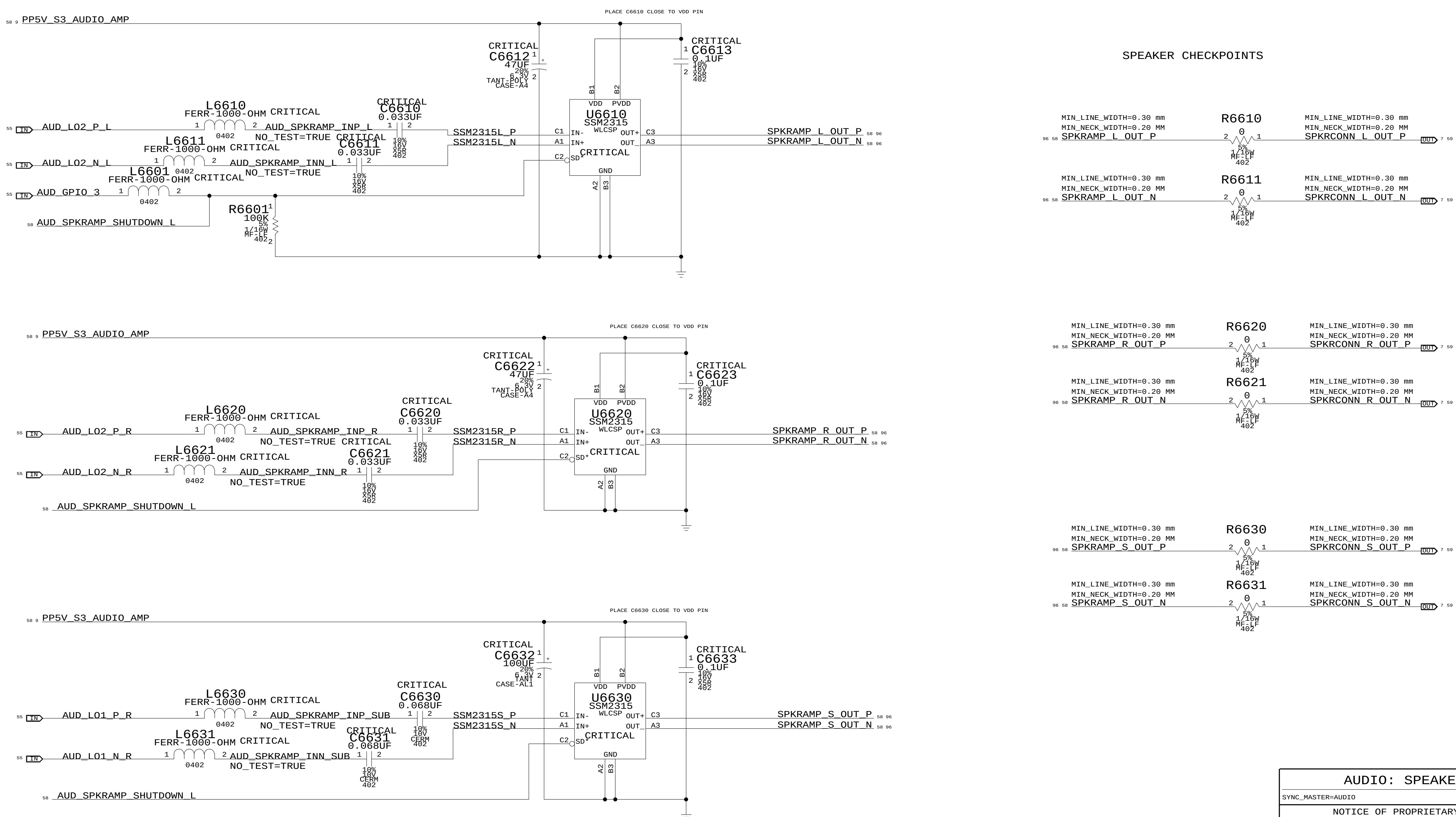
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	57	97

3X MONO SPEAKER AMPLIFIERS (SSM2315)
APN: 353S2500

GAIN = 6DB
1ST ORDER FC (L&R) = 120 HZ +/- 30%
1ST ORDER FC (SUB) = 58HZ +/- 30%



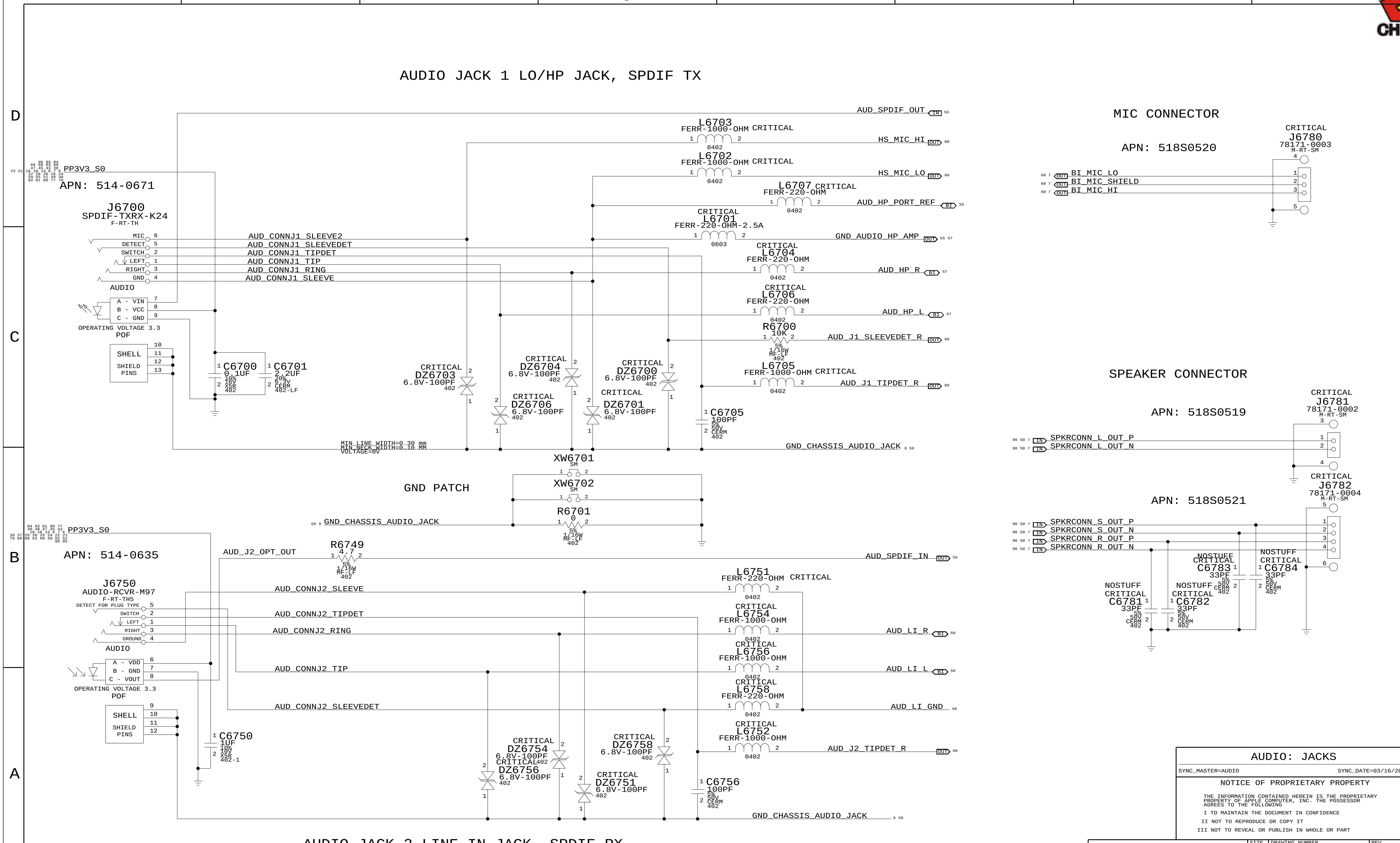
AUDIO: SPEAKER AMP

SYNC_MASTER=AUDIO SYNC_DATE=03/16/2009

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 58	OF 97



AUDIO: JACKS

SYNC_MASTER=AUDIO SYNC_DATE=03/16/2009

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

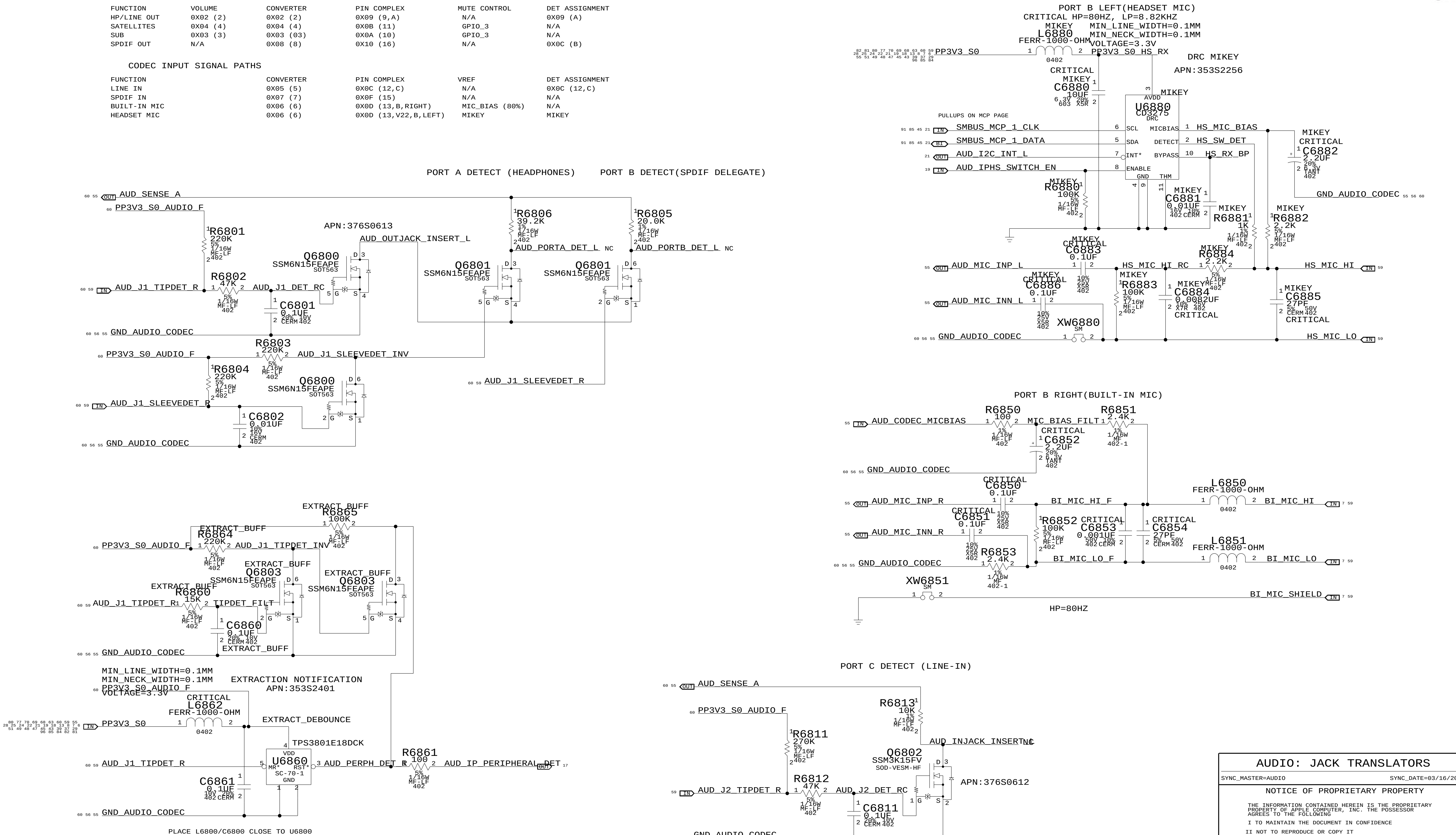
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 59	OF 97

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)	N/A	0X09 (A)
SATELLITES	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SUB	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (16)	N/A	0X0C (B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X0C (12,C)	N/A	0X0C (12,C)
SPDIF IN	0X07 (7)	0X0F (15)	N/A	N/A
BUILT-IN MIC	0X06 (6)	0X0D (13,B,RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY



AUDIO: JACK TRANSLATORS

SYNC_MASTER=AUDIO SYNC_DATE=03/16/2009

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

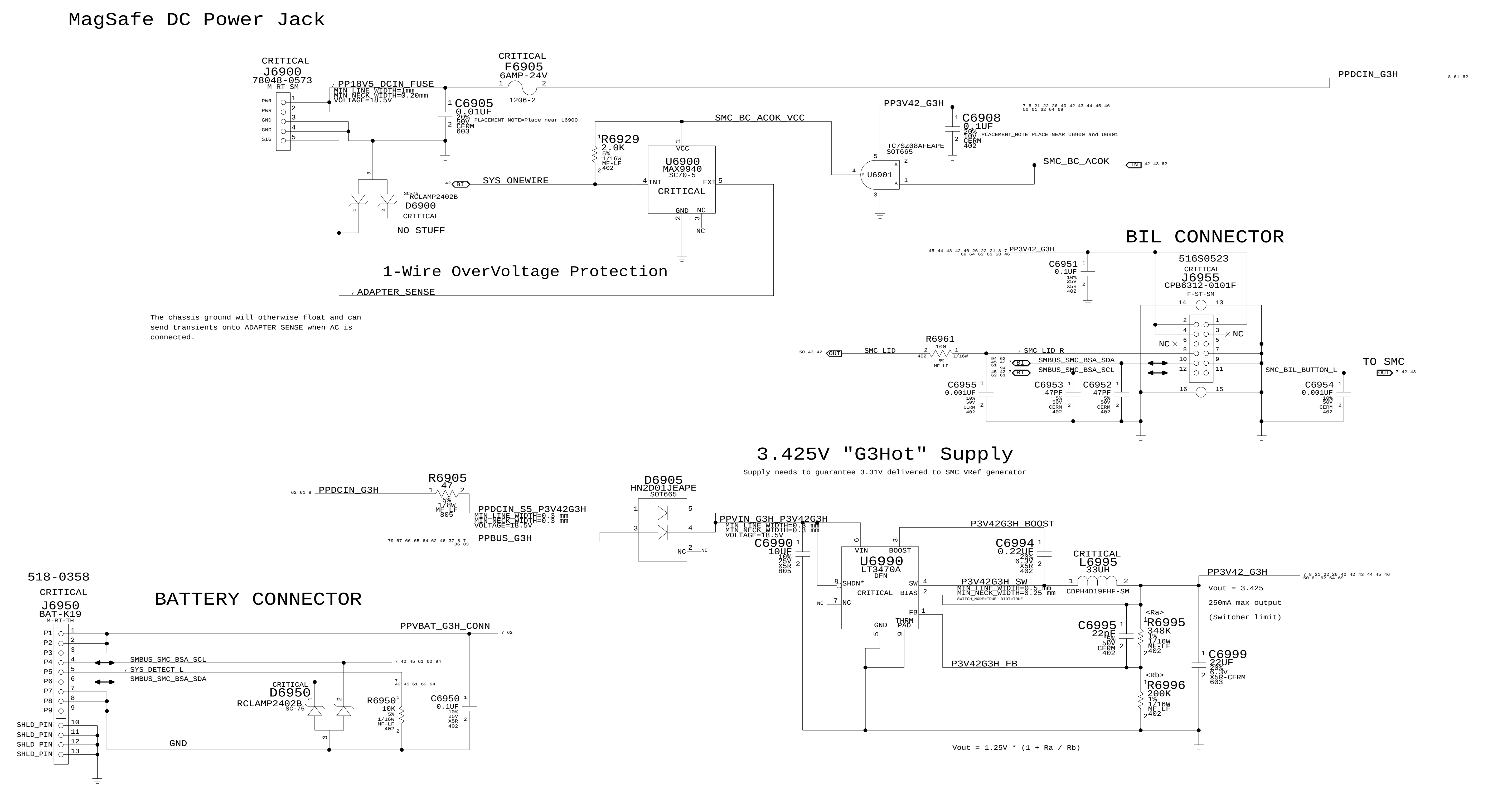
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 60	OF 97



DC-In & Battery Connectors	
SYNC_MASTER=YUN_K19_MLB	SYNC_DATE=12/16/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

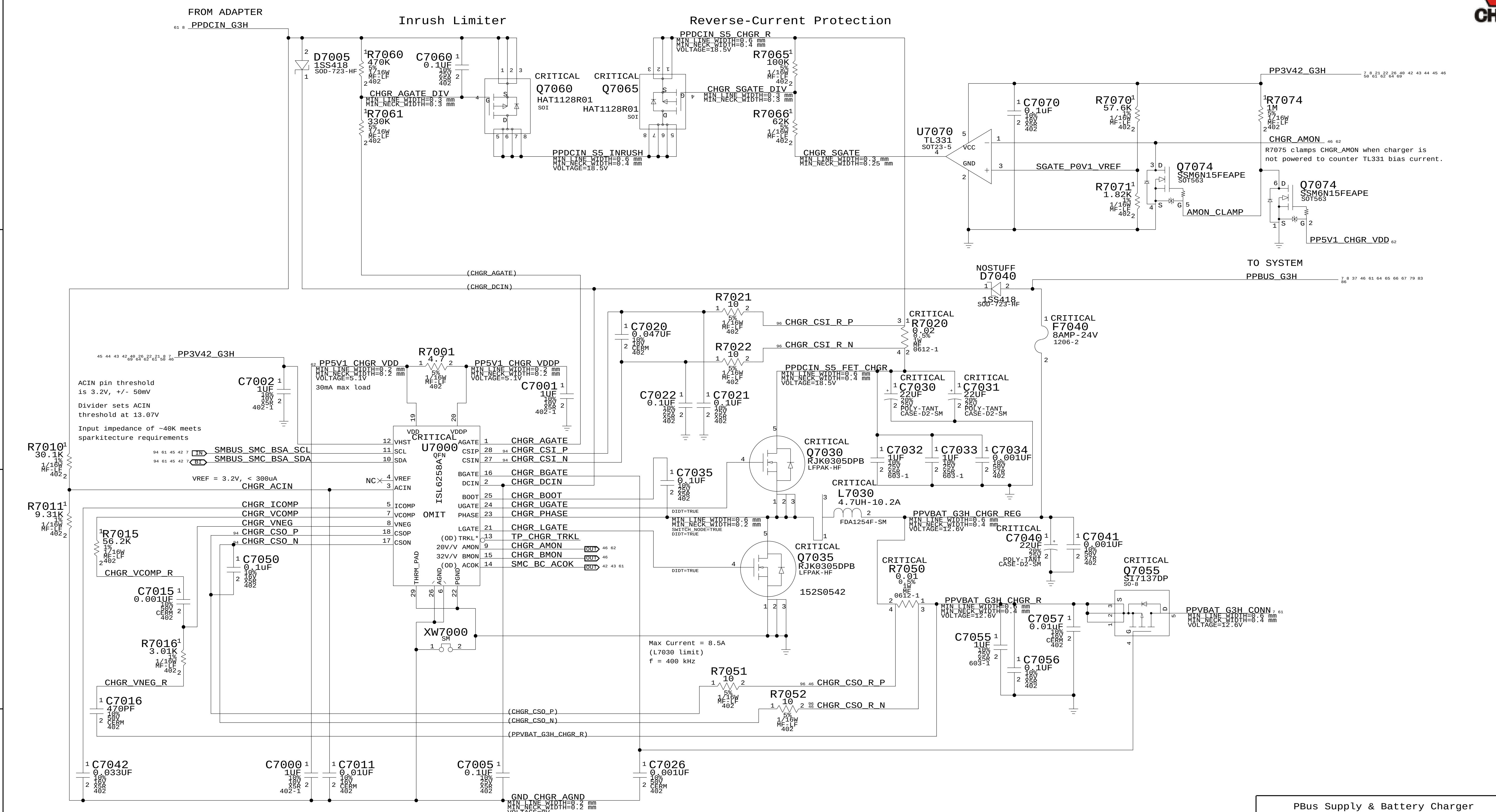
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 61	OF 97



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S1811	1	IC,ISL6258,BAT CHARGER,28P,4X4,QFN,L	U7000	CRITICAL	ISL6258
353S1832	1	IC,ISL6258A,BAT CHARGER,4X4MM,QFN28	U7000	CRITICAL	ISL6258A

2S Battery Default
3S Battery Default

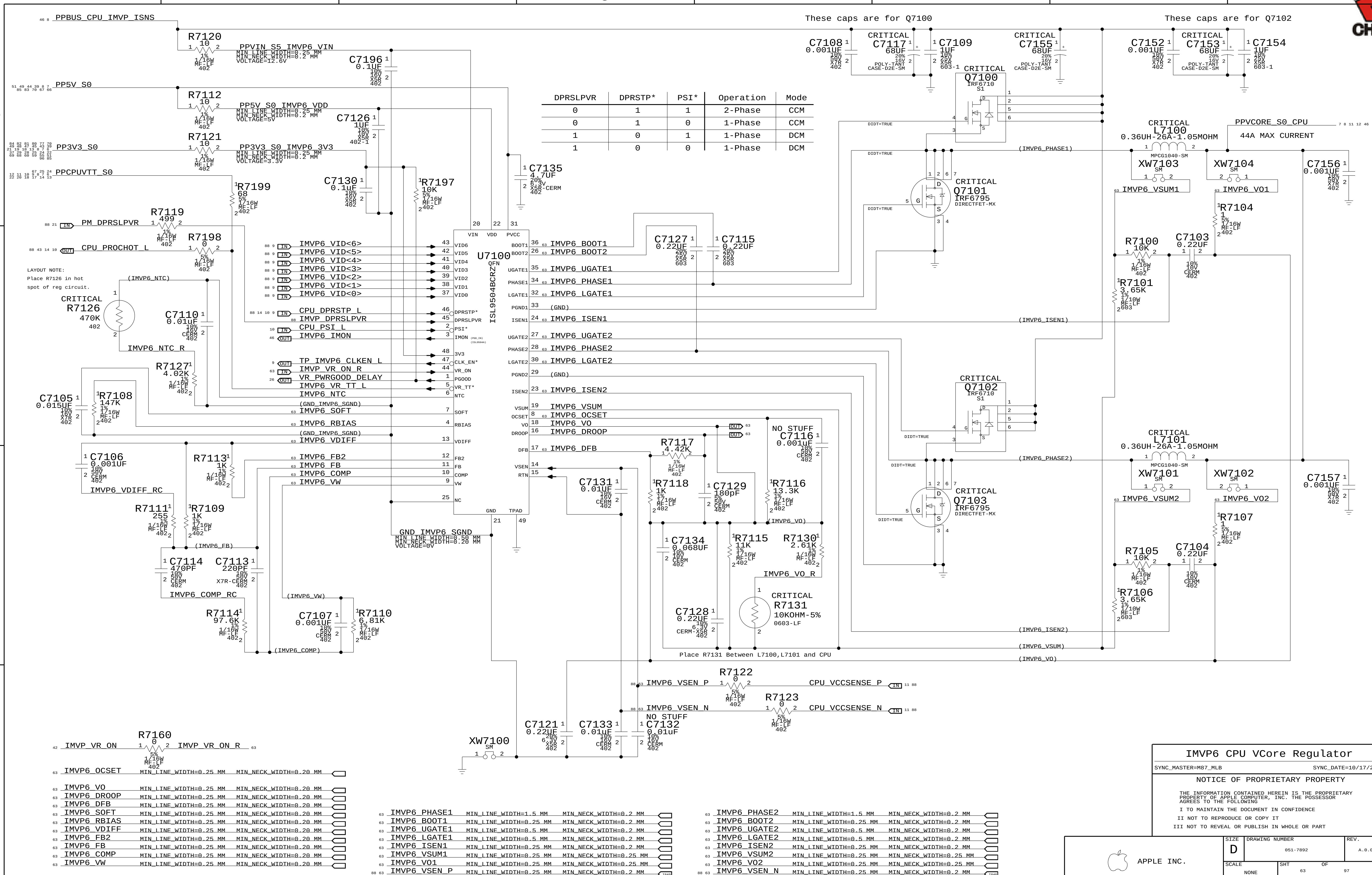
PBus Supply & Battery Charger

SYNC_MASTER=M99_MLB SYNC_DATE=12/10/2007

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE | SHT 62 OF 97



DPRSLPVR	DPRSTP*	PSI*	Operation	Mode
0	1	1	2-Phase	CCM
0	1	0	1-Phase	CCM
1	0	1	1-Phase	DCM
1	0	0	1-Phase	DCM

IMVP6 CPU VCore Regulator

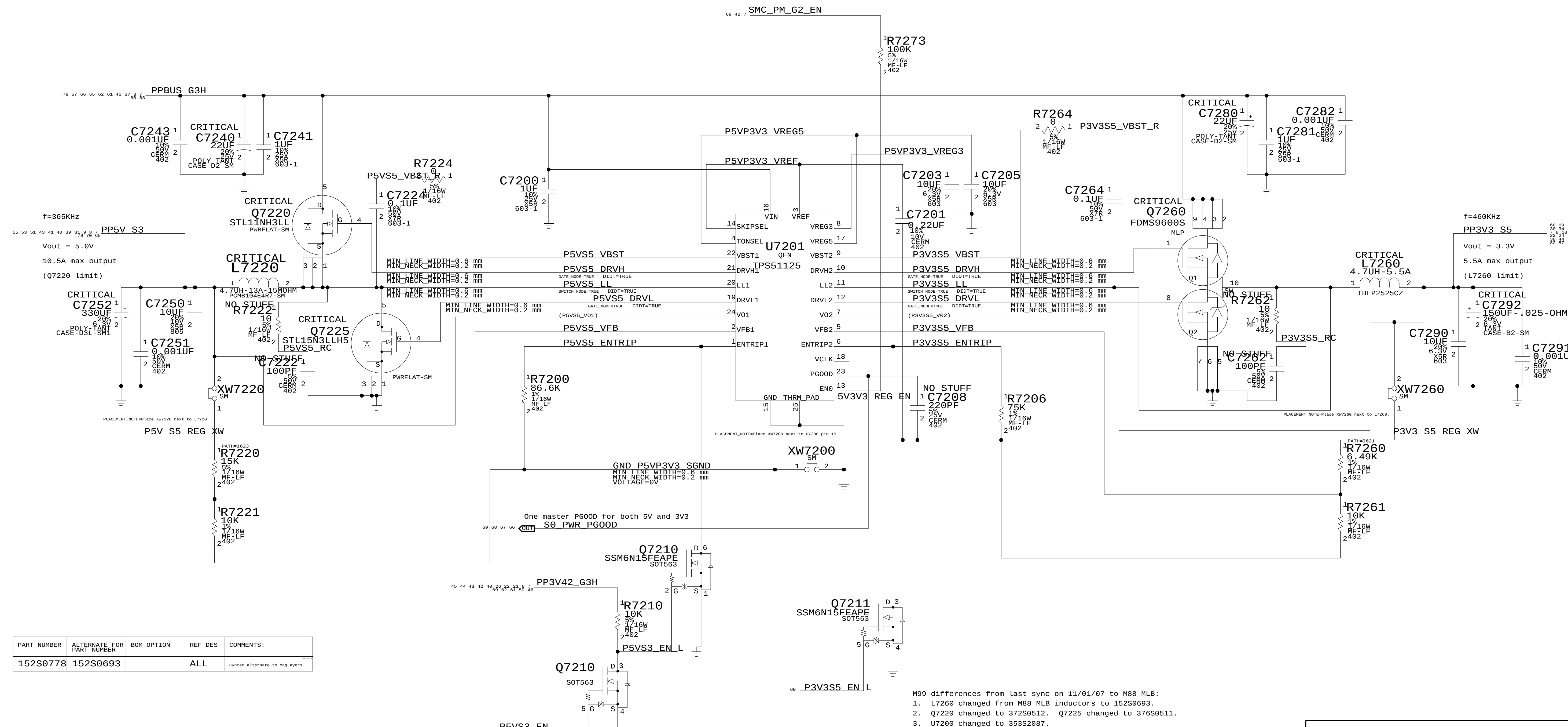
SYNC_MASTER=M87_MLB SYNC_DATE=10/17/2007

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	63	97



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S0778	152S0693		ALL	Cyntec alternate to MagLayers

M99 differences from last sync on 11/01/07 to M88 MLB:

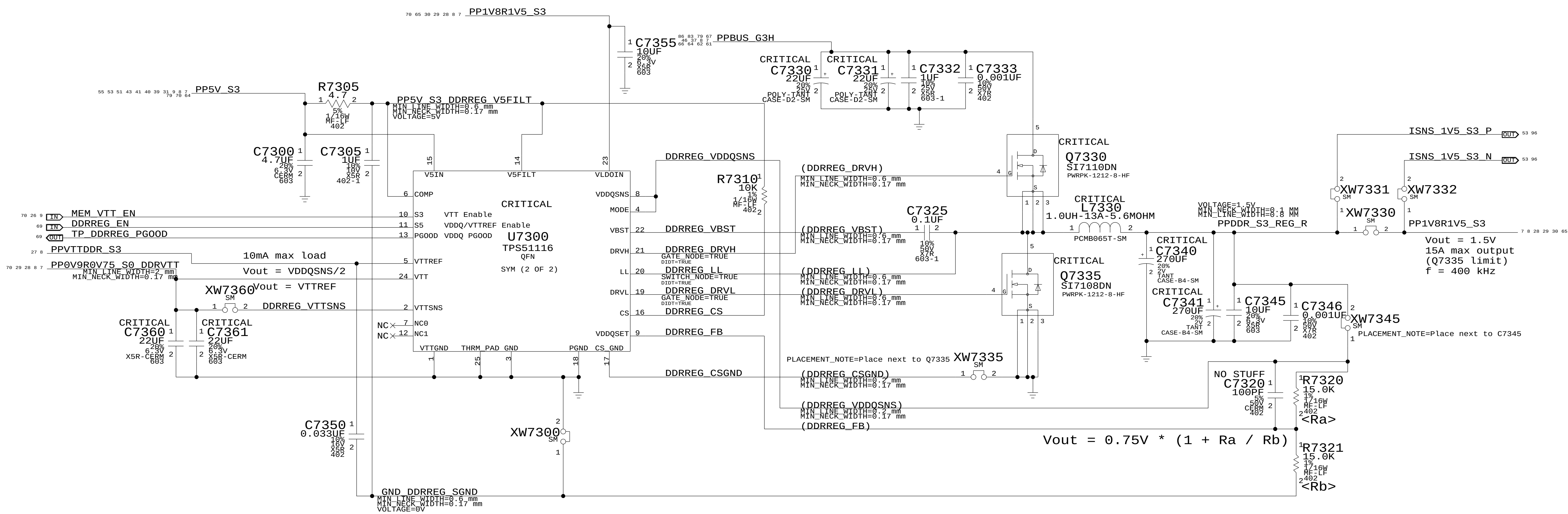
1. L7260 changed from M88 MLB inductors to 152S0693.
2. Q7220 changed to 372S0512. Q7225 changed to 376S0511.
3. U7200 changed to 353S2087.
4. Added R7200, R7220,R7221, R7260,R7261, C7201.

5V / 3.3V Power Supply

SYNC_MASTER=PWRSQNC SYNC_DATE=12/17/2008

APPLE INC.

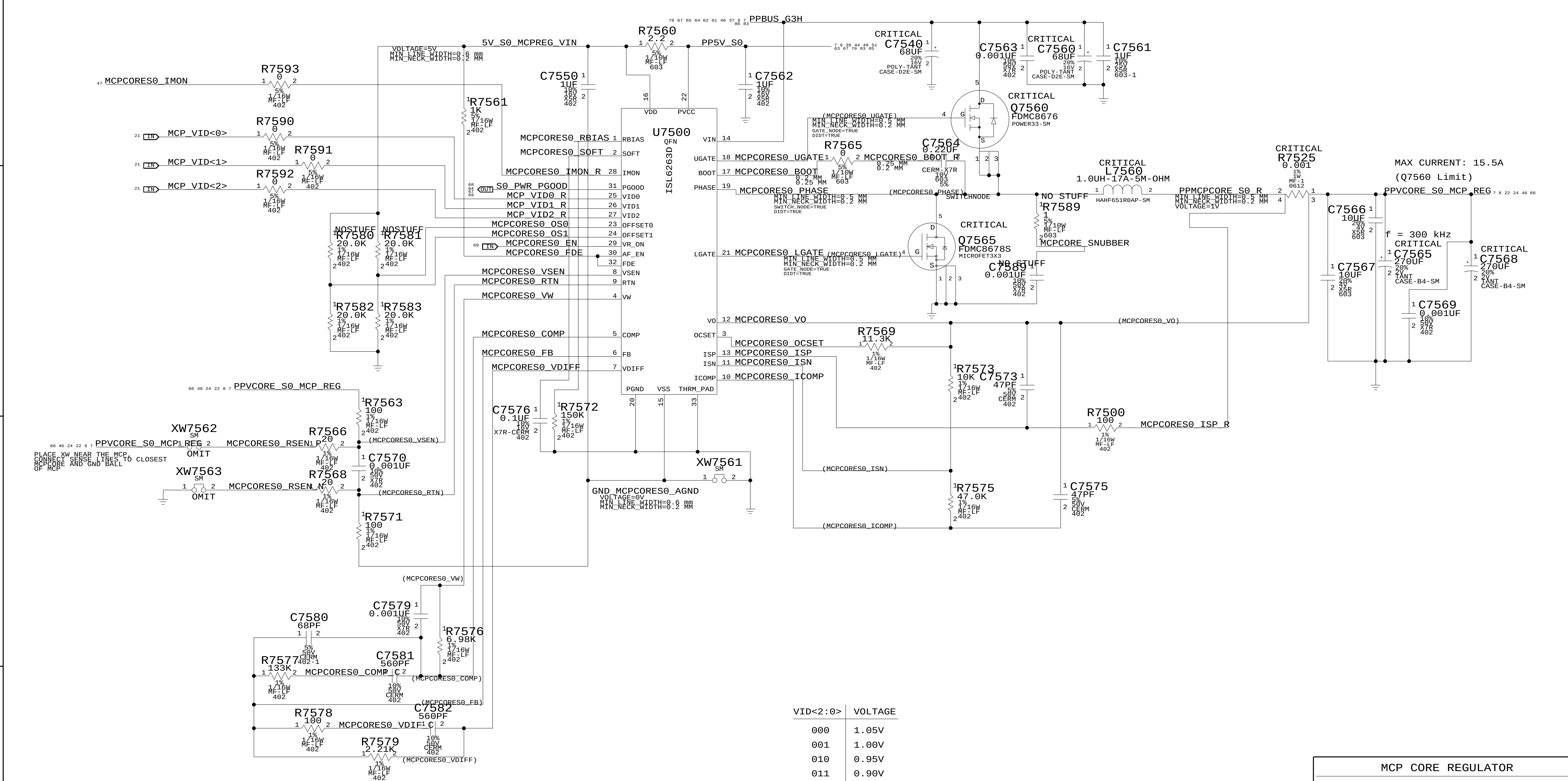
SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0
SCALE NONE | SHT 64 OF 97



1.5V DDR3 Supply

SYNC_MASTER=DDR SYNC_DATE=12/05/2008

APPLE INC.	SIZE D	DRAWING NUMBER 051-7892	REV. A.0.0
	SCALE NONE	SHT 65	OF 97



VID<2:0>	VOLTAGE
000	1.05V
001	1.00V
010	0.95V
011	0.90V
100	0.85V
101	0.80V
110	0.75V
111	0.70V

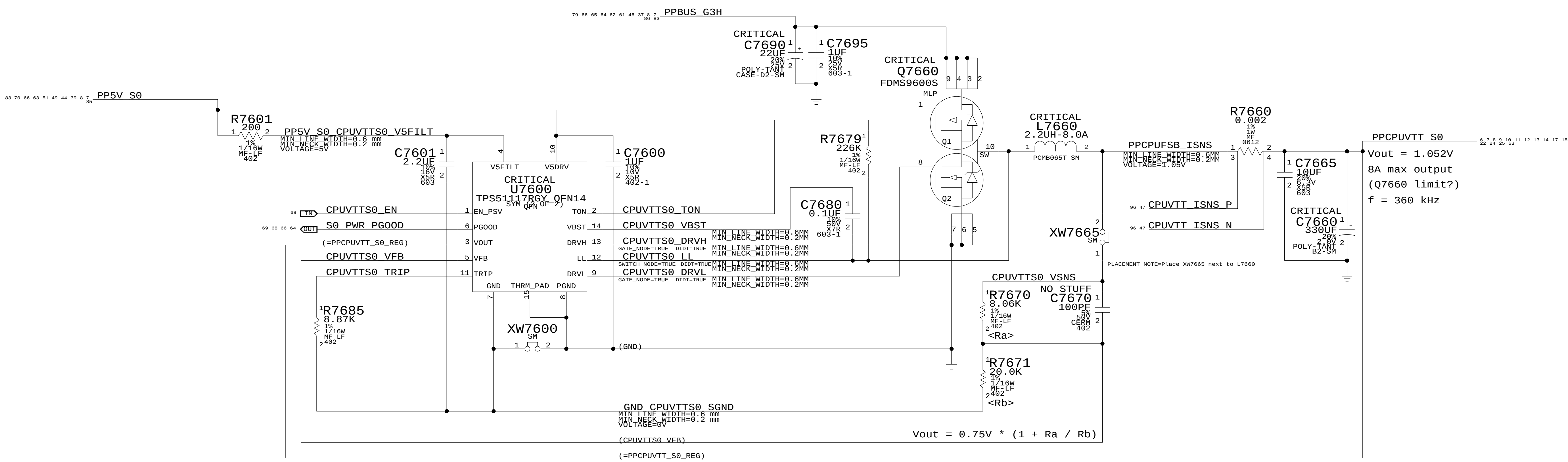
MCP CORE REGULATOR

SYNC_MASTER=M98_MLB SYNC_DATE=11/14/2008

APPLE INC.

SIZE D DRAWING NUMBER 051-7892 REV. A.0.0

SCALE NONE SHT 66 OF 97



M99 differences from last sync on 12/03/07 to T18 MLB:
1. Tied THERMAL_PAD to PGND. GND and THERMAL_PAD disconnected.

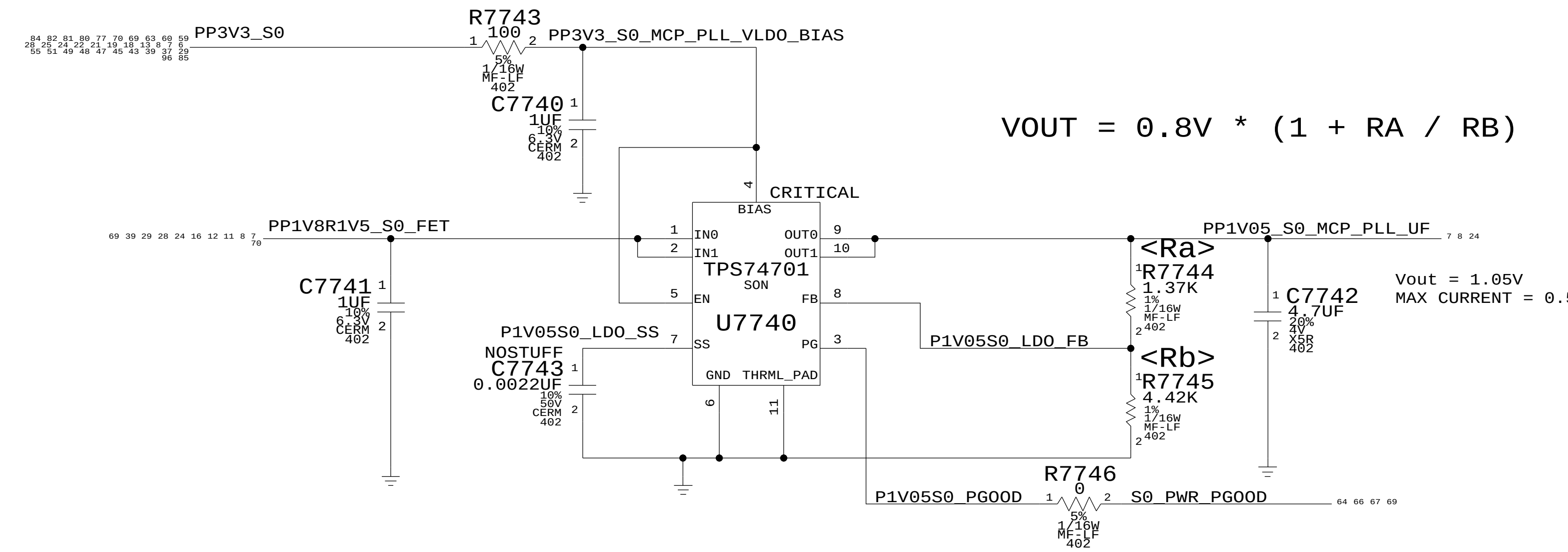
CPU VTT / 1V05 S0 Power Supply

SYNC_MASTER=M99_MLB SYNC_DATE=12/14/2007

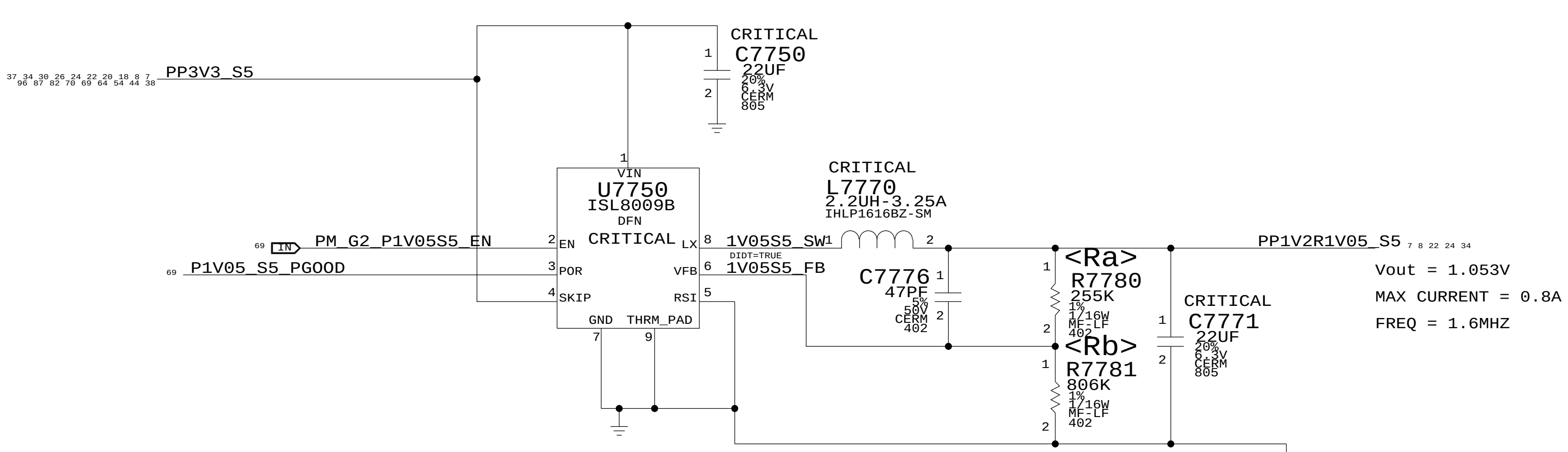
APPLE INC.	SIZE D	DRAWING NUMBER 051-7892	REV. A.0.0
	SCALE NONE	SHT 67	OF 97

1.05V S0 PLL LDO

VOUT = 0.8V * (1 + RA / RB)



MCP 1.05V S5 (AUXC) SUPPLY



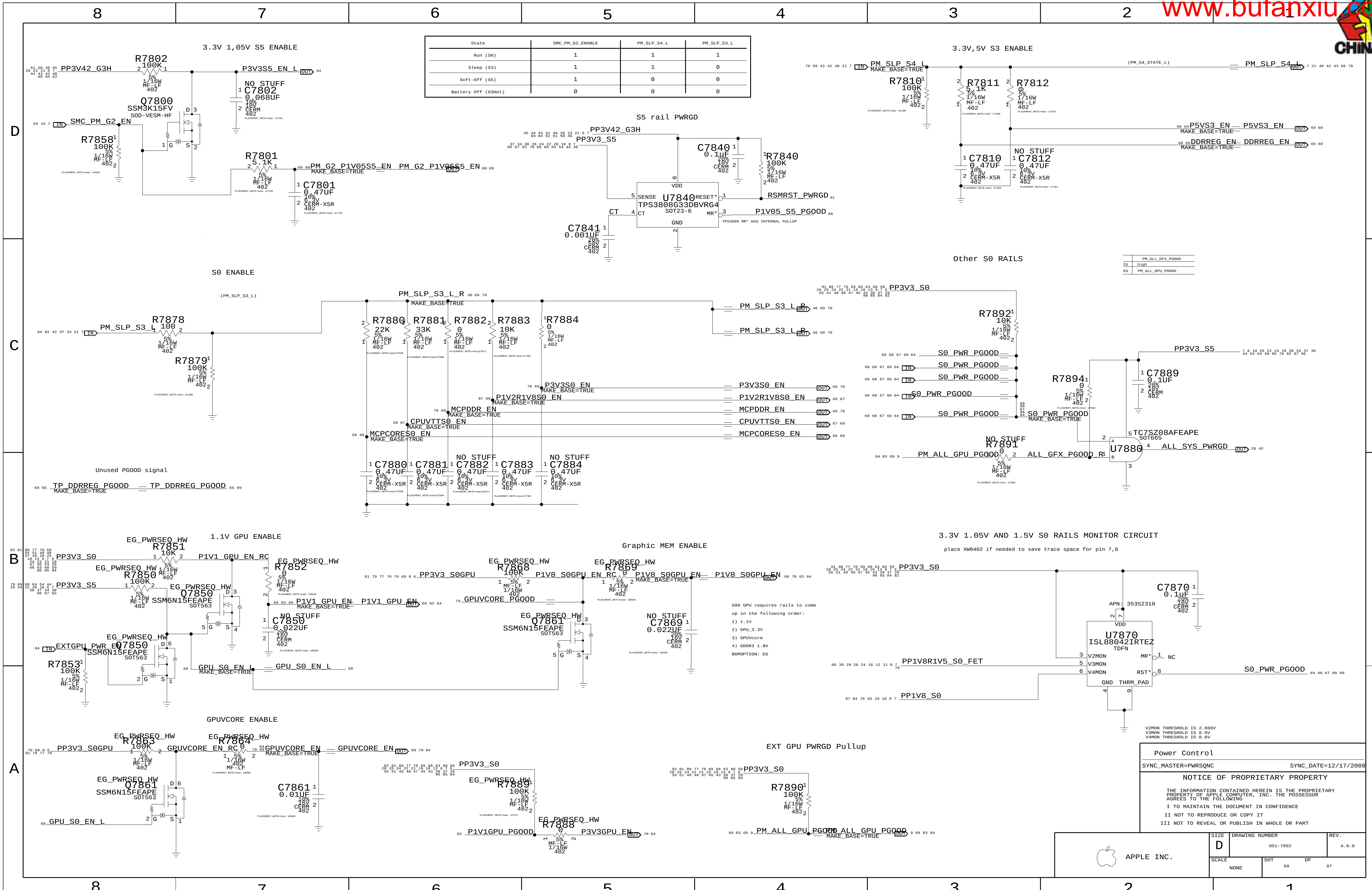
VOUT = 0.8V * (1 + RA / RB)

Misc Power Supplies

SYNC_MASTER=M99_MLB SYNC_DATE=12/14/2007

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 68	OF 97



State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0

	PM_ALL_GFX_PGOOD
IG	high
EG	PM_ALL_GPU_PGOOD

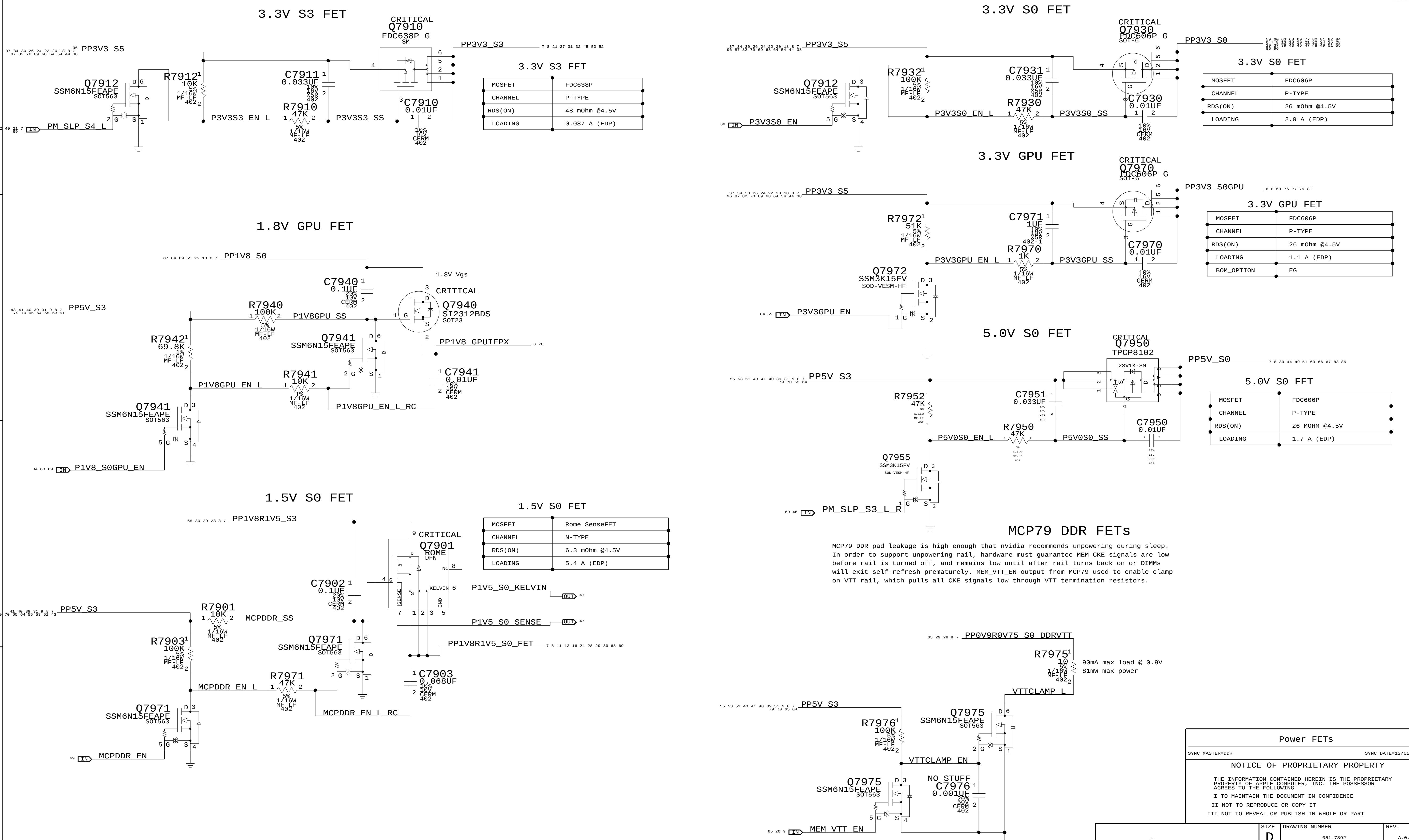
Power Control

SYNC_MASTER=PWRSQNC SYNC_DATE=12/17/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 69	OF 97



3.3V S3 FET

MOSFET	FDC638P
CHANNEL	P-TYPE
RDS(ON)	48 mOhm @4.5V
LOADING	0.087 A (EDP)

3.3V S0 FET

MOSFET	FDC606P
CHANNEL	P-TYPE
RDS(ON)	26 mOhm @4.5V
LOADING	2.9 A (EDP)

3.3V GPU FET

MOSFET	FDC606P
CHANNEL	P-TYPE
RDS(ON)	26 mOhm @4.5V
LOADING	1.1 A (EDP)
BOM_OPTION	EG

5.0V S0 FET

MOSFET	FDC606P
CHANNEL	P-TYPE
RDS(ON)	26 MOHM @4.5V
LOADING	1.7 A (EDP)

1.5V S0 FET

MOSFET	Rome SenseFET
CHANNEL	N-TYPE
RDS(ON)	6.3 mOhm @4.5V
LOADING	5.4 A (EDP)

MCP79 DDR FETs

MCP79 DDR pad leakage is high enough that nVidia recommends unpowering during sleep. In order to support unpowering rail, hardware must guarantee MEM_CKE signals are low before rail is turned off, and remains low until after rail turns back on or DIMMs will exit self-refresh prematurely. MEM_VTT_EN output from MCP79 used to enable clamp on VTT rail, which pulls all CKE signals low through VTT termination resistors.

Power FETs

SYNC_MASTER=DDR SYNC_DATE=12/05/2008

APPLE INC.

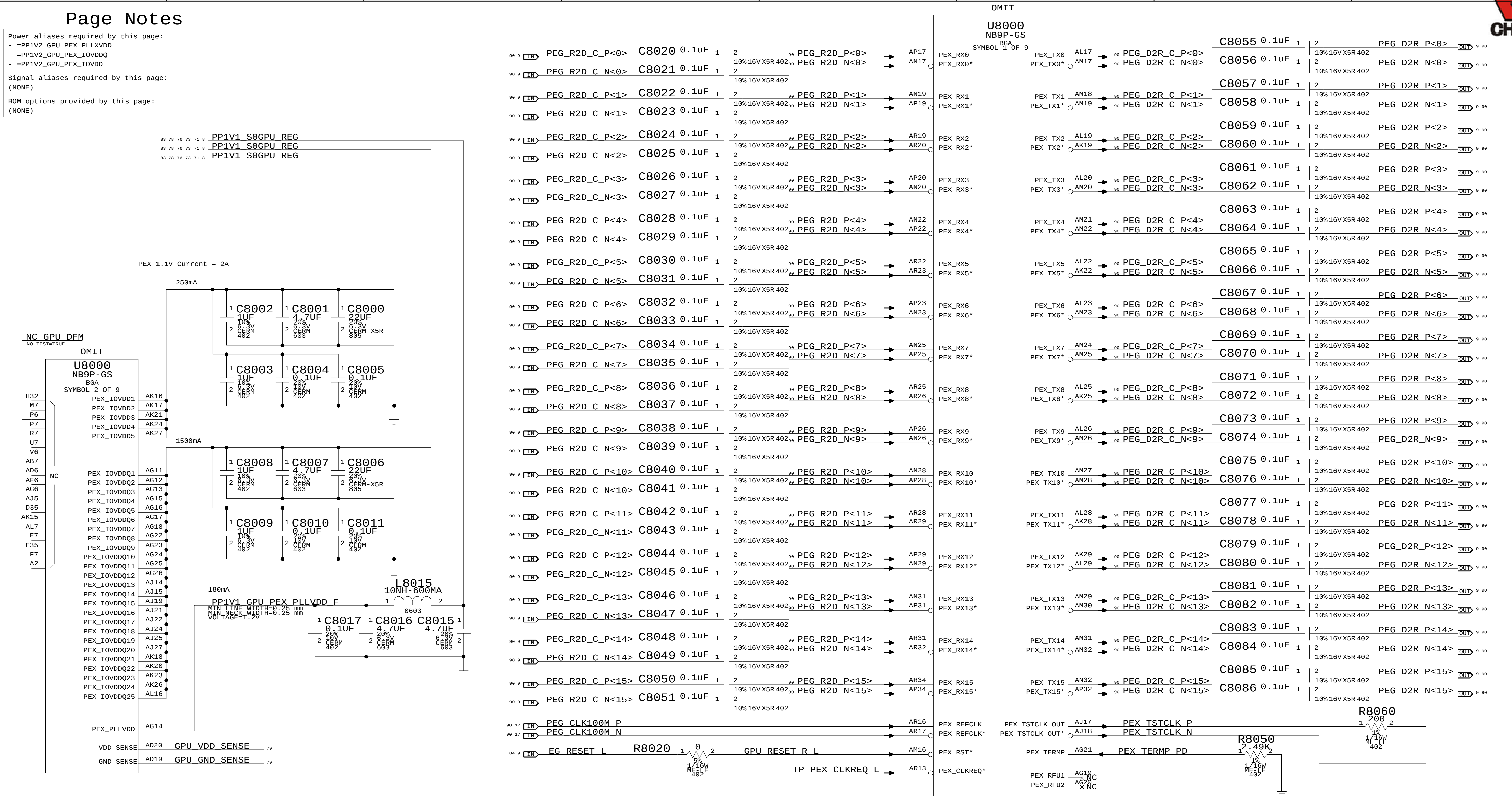
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 70	OF 97

Page Notes

Power aliases required by this page:
- =PP1V2_GPU_PEX_PLLXVDD
- =PP1V2_GPU_PEX_IOVDDQ
- =PP1V2_GPU_PEX_IOVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G96 PCI-E

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

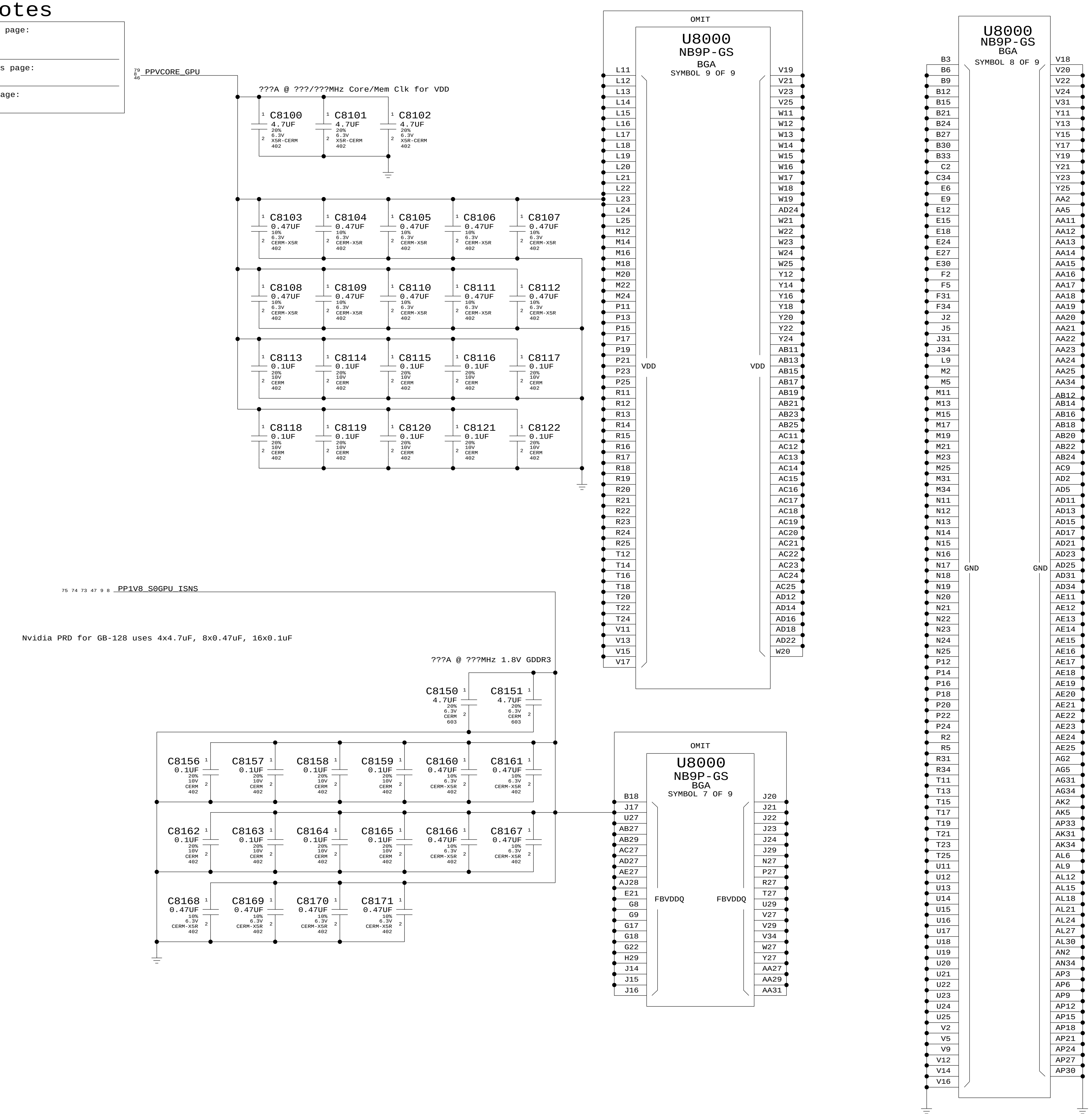
SCALE	SHT	OF
NONE	71	97

Page Notes

Power aliases required by this page:
- =PPVCORE_GPU
- =PP1V8_GPU_FBVDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G96 Core/FB Power

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

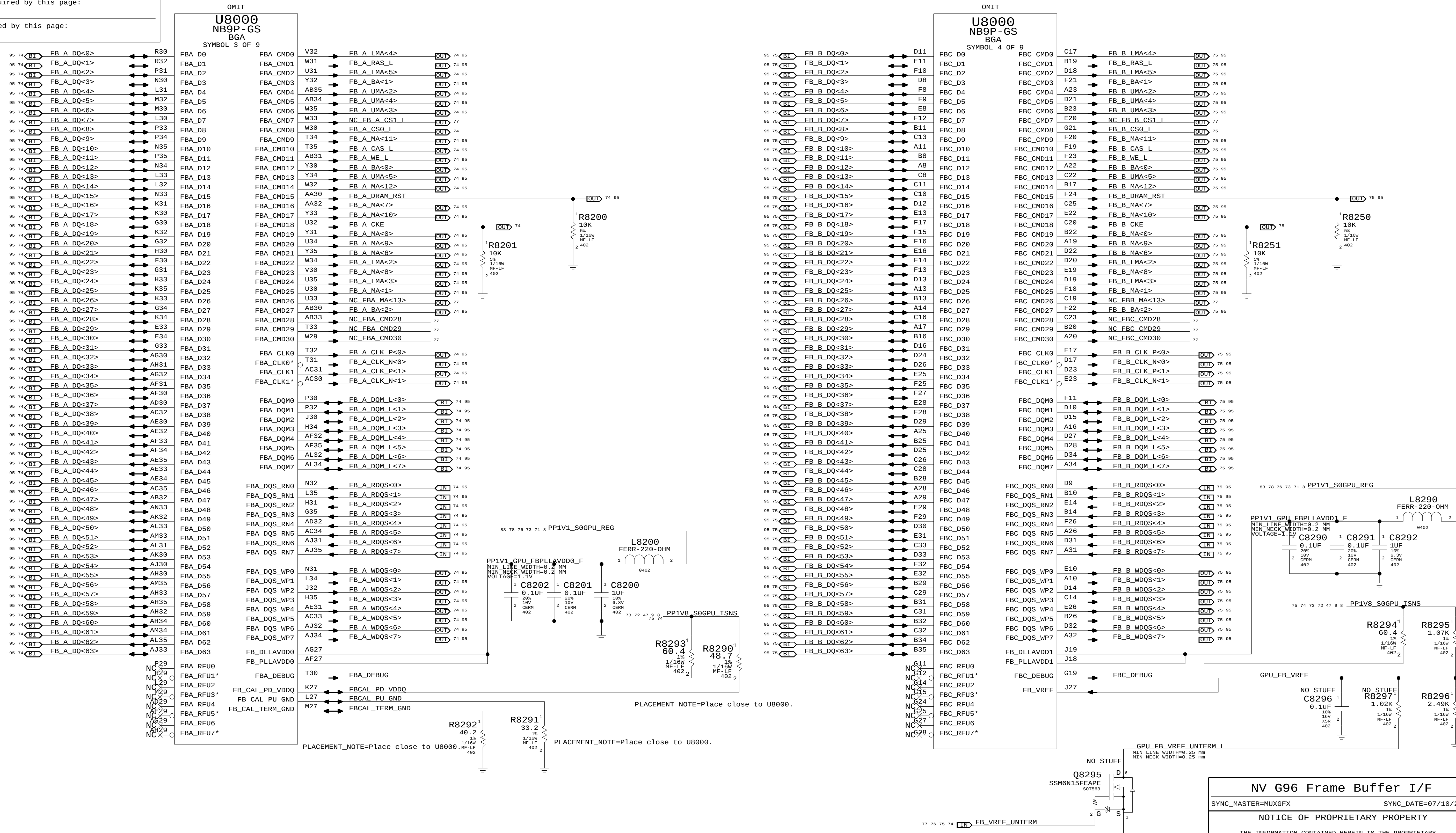
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 72	OF 97

Page Notes

Power aliases required by this page:
- =PP1V2_GPU_FBPLLAVDD
- =PP1V8_GPU_FBIO

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G96 Frame Buffer I/F

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

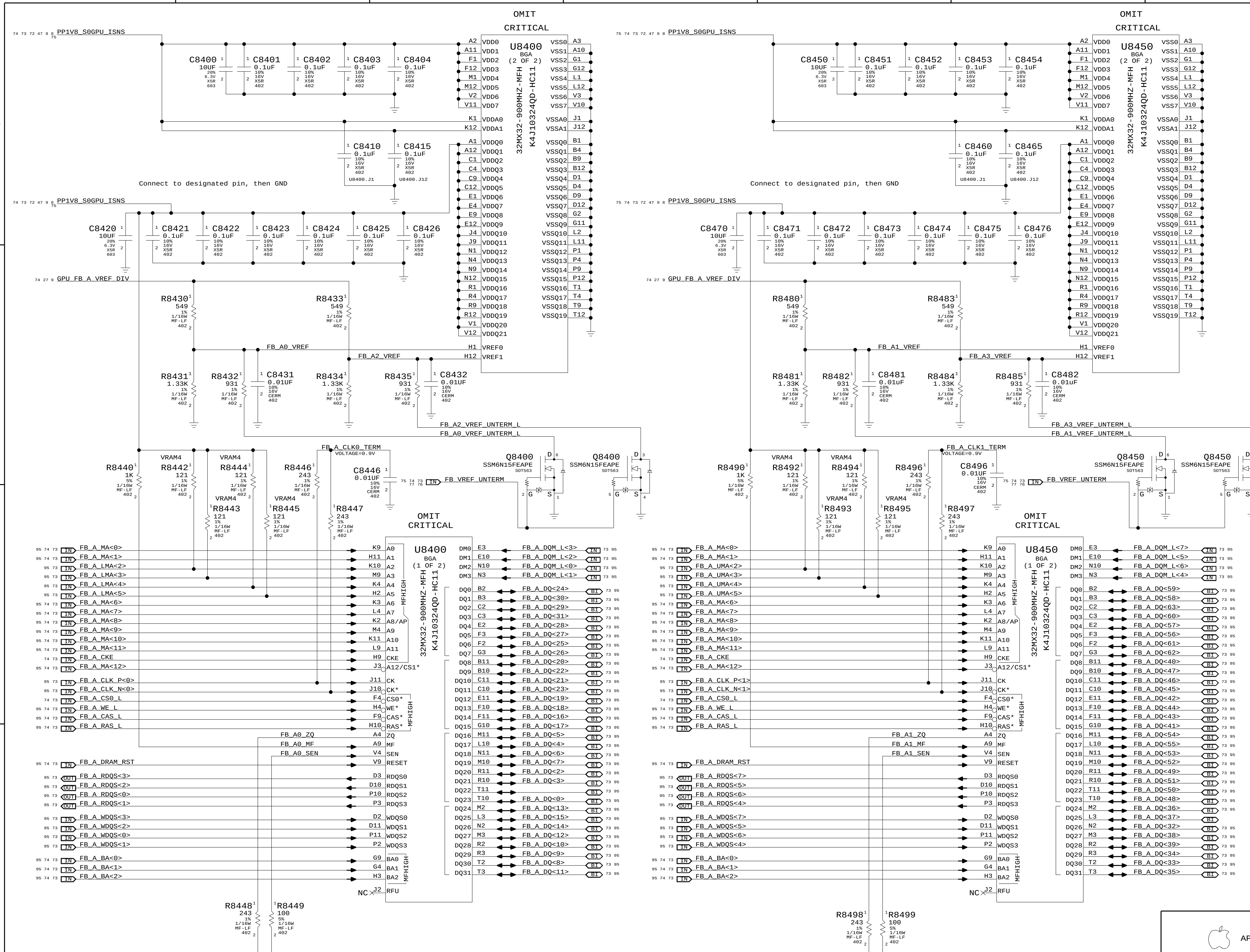
NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 73	OF 97



Page Notes

Power aliases required by this page:
- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VREFA

Signal aliases required by this page:
(NONE)

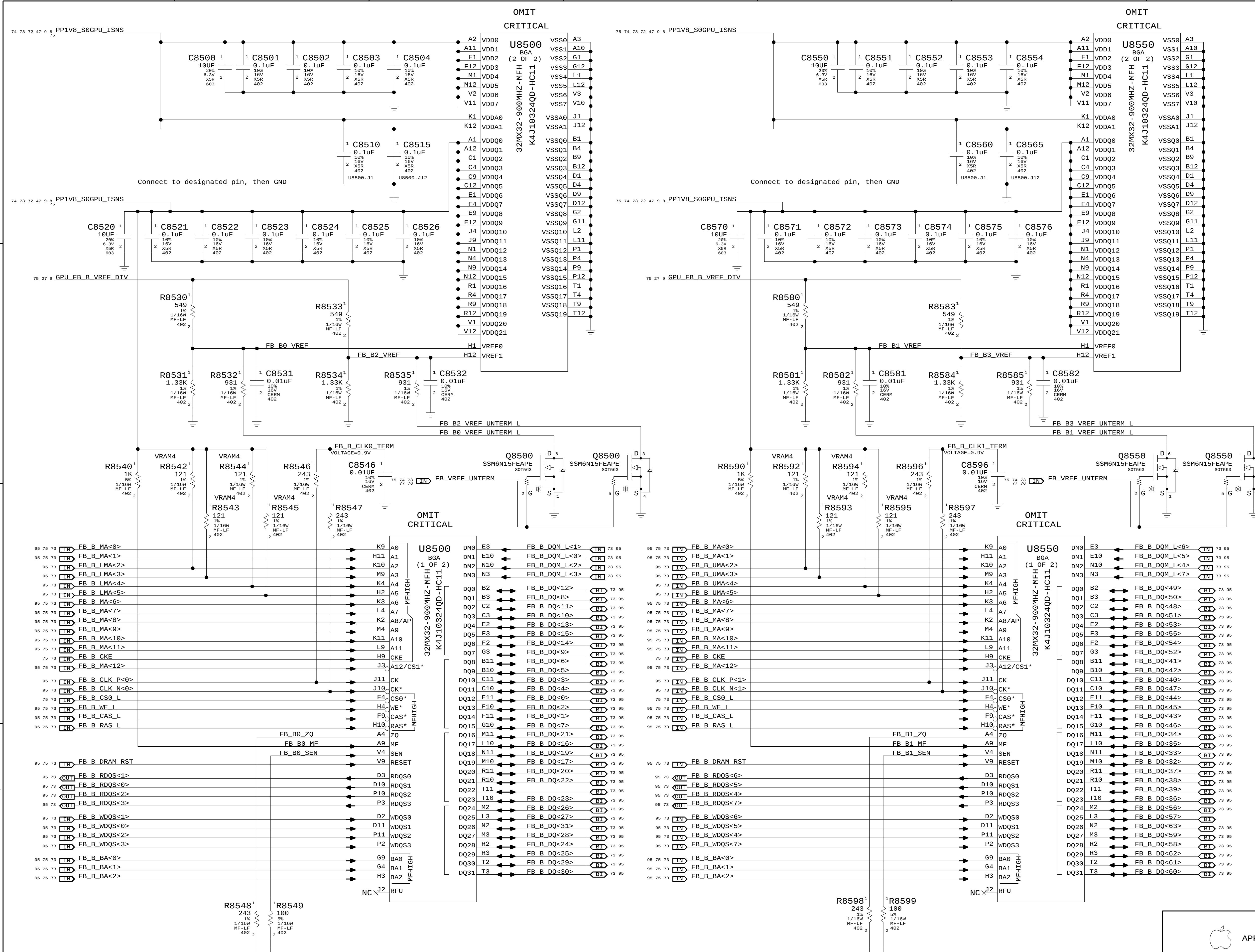
BOM options provided by this page:
VRAM4

GDDR3 Frame Buffer A (Top)

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 74 OF 97	



Page Notes

Power aliases required by this page:
- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VREF_B

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
VRAM4

GDDR3 Frame Buffer B (Top)

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

APPLE INC.

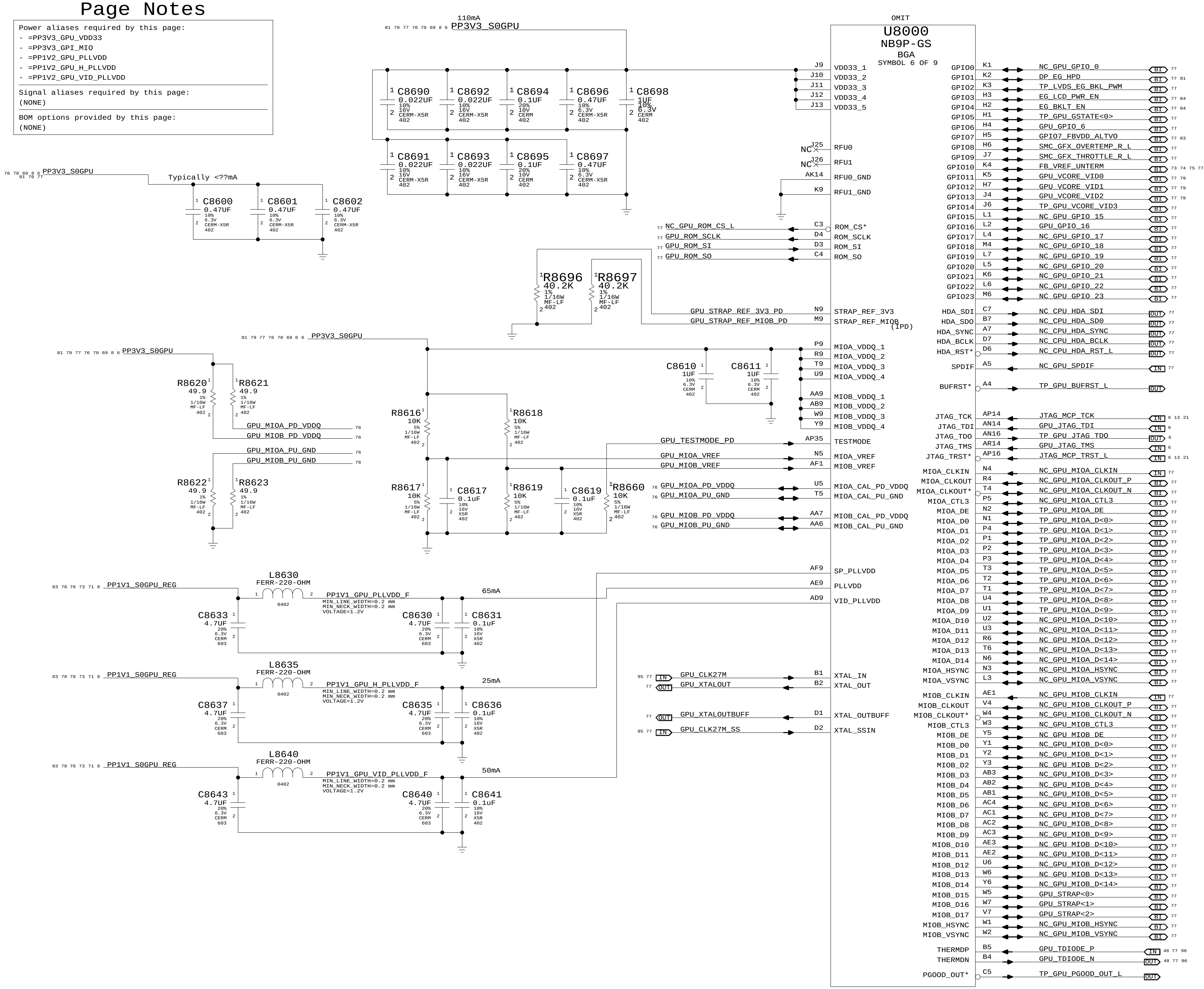
SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 75	OF 97

Page Notes

Power aliases required by this page:
- =PP3V3_GPU_VDD33
- =PP3V3_GPI_MIO
- =PP1V2_GPU_PLLVDD
- =PP1V2_GPU_H_PLLVDD
- =PP1V2_GPU_VID_PLLVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G96 GPIO/MIO/Misc

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

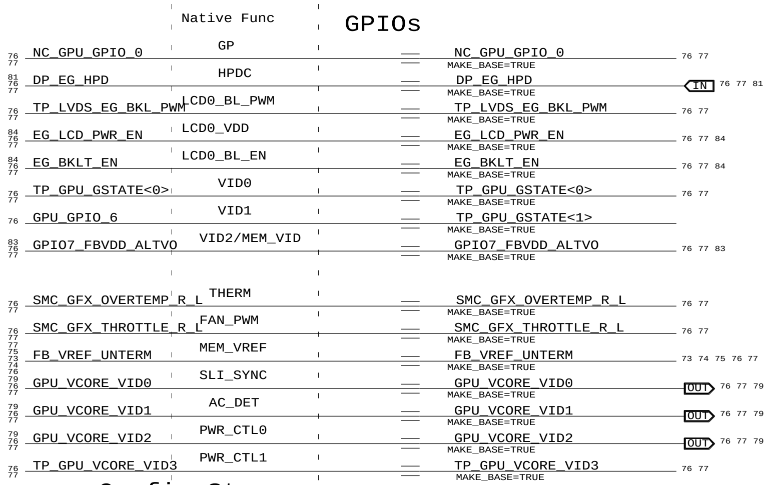
II NOT TO REPRODUCE OR COPY IT

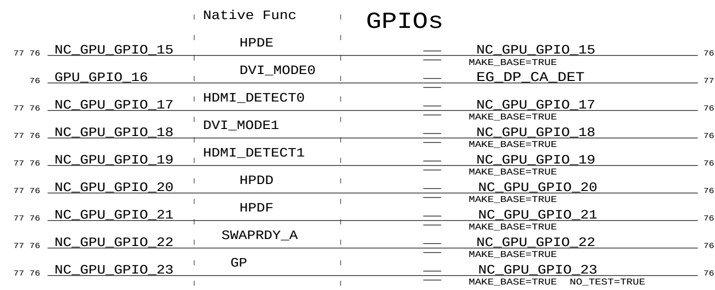
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 76	OF 97

GPIOs

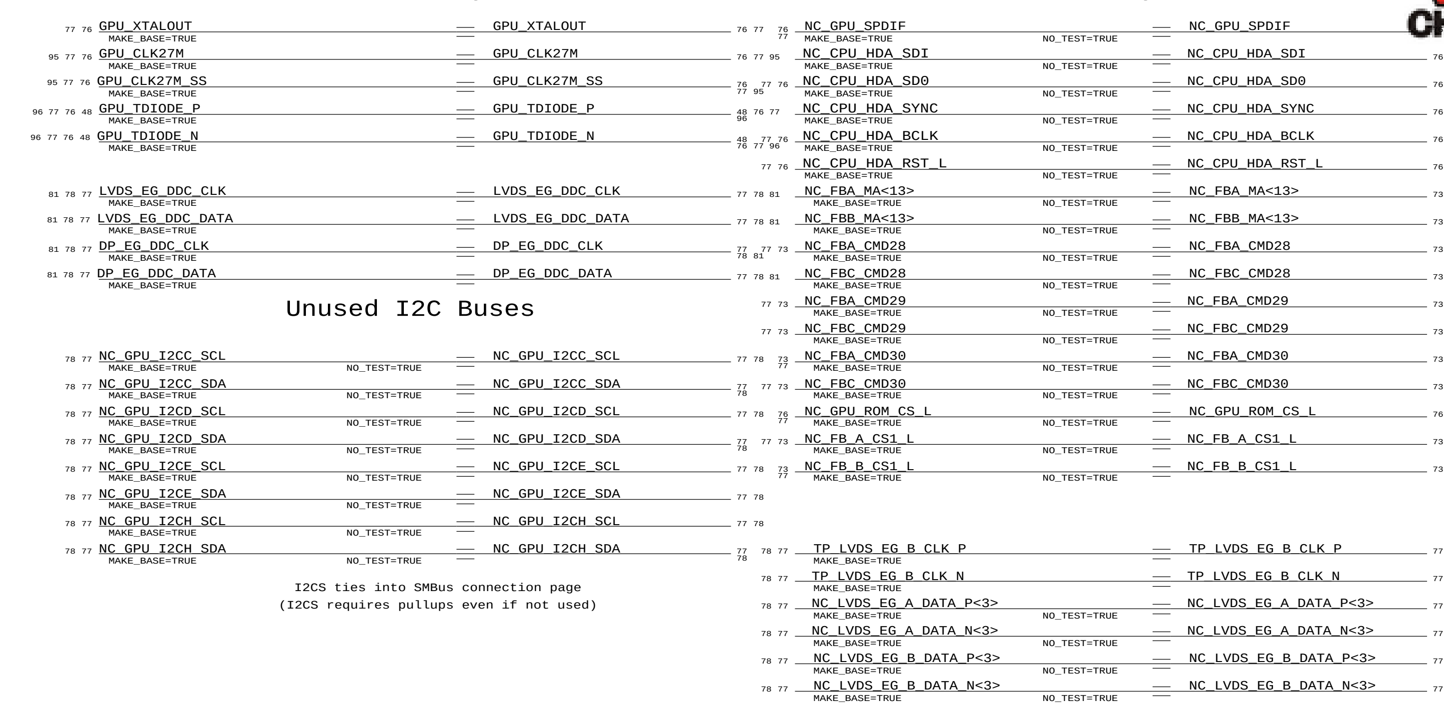




Renamed signals

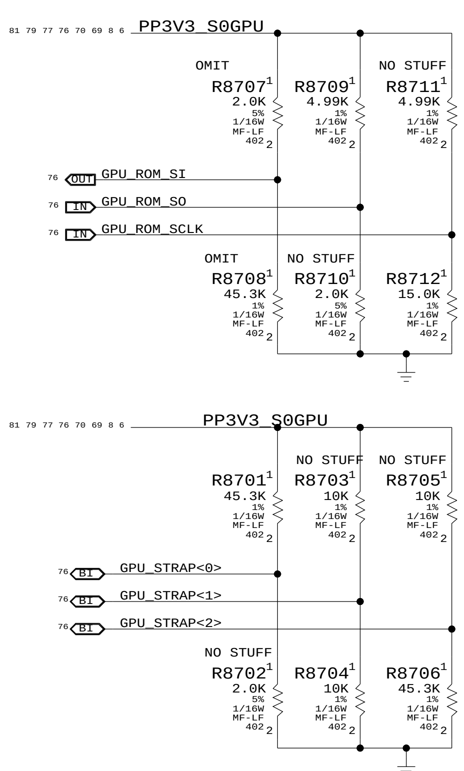
Unused signals

Unused I2C Buses



I2CS ties into SMBus connection page
(I2CS requires pullups even if not used)

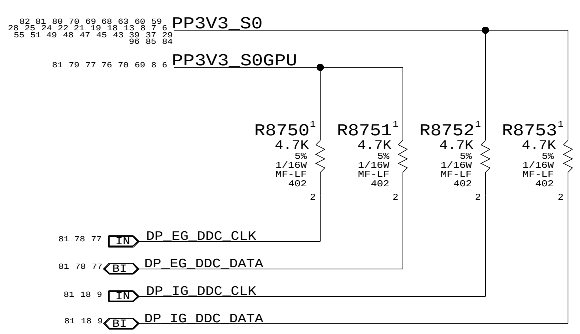
Config Straps



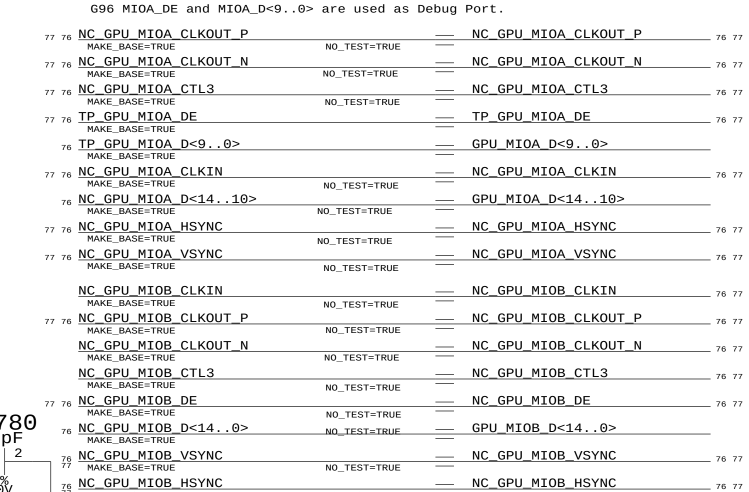
Physical Strapping Pin	Strapping Bit 3	Strapping Bit 2	Strapping Bit 1	Strapping Bit 0
ROM_SO	XCLK_277	TVMODE[2]	TVMODE[1]	TVMODE[0]
ROM_SCLK	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLLEN_TERM100
ROM_SI	RAMCFG[3]	RAMCFG[2]	RAMCFG[1]	RAMCFG[0]
STRAP 2	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP 1	3GIO_PADCFG[3]	3GIO_PADCFG[2]	3GIO_PADCFG[1]	3GIO_PADCFG[0]
STRAP 0	USER[3]	USER[2]	USER[1]	USER[0]

Strap	S1/S2 Bit[3:0]	PU/PD	Rval
0	0000	PD	5k
1	0001	PD	10k
2	0010	PD	15k
3	0011	PD	20k
4	0100	PD	25k
5	0101	PD	30k
6	0110	PD	35k
7	0111	PD	45k
8	1000	PU	5k
9	1001	PU	10k
A	1010	PU	15k
B	1011	PU	20k
C	1100	PU	25k
D	1101	PU	30k
E	1110	PU	35k
F	1111	PU	45k

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0378	1	RES,MTL FILM,1/16W,45.3K,1,0402,SMD,LF	R8708		VRAM_512_SAMSUNG
114S0368	1	RES,MTL FILM,1/16W,35.7K,1,0402,SMD,LF	R8708		VRAM_512_HYNIX
114S0361	1	RES,MTL FILM,1/16W,30.1K,1,0402,SMD,LF	R8708		VRAM_512_QIMONDA
114S0343	1	RES,MTL FILM,1/16W,20.0K,1,0402,SMD,LF	R8708		VRAM_256_SAMSUNG
114S0331	1	RES,MTL FILM,1/16W,15.0K,1,0402,SMD,LF	R8708		VRAM_256_HYNIX
114S0378	1	RES,MTL FILM,1/16W,45.3K,1,0402,SMD,LF	R8707		VRAM_1024_SAMSUNG
114S0361	1	RES,MTL FILM,1/16W,30.1K,1,0402,SMD,LF	R8707		VRAM_1024_QIMONDA

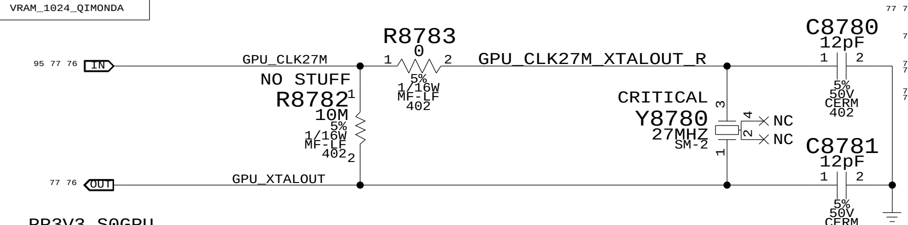


G96 MIOA_DE and MIOA_D<9..0> are used as Debug Port.

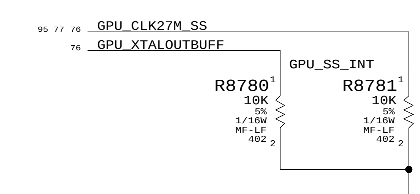


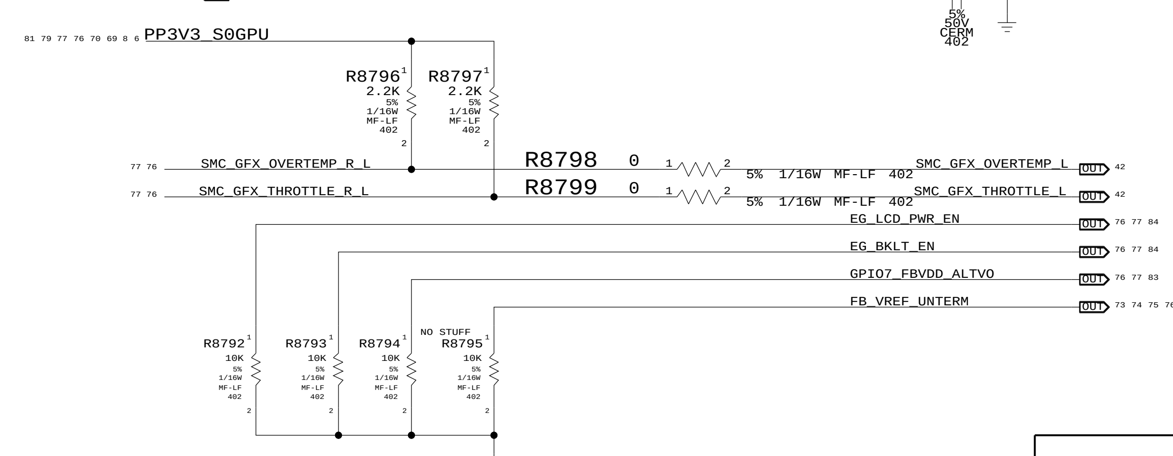
G96 HDCP ROM APN is 341S2272, blank device is 335S0574.

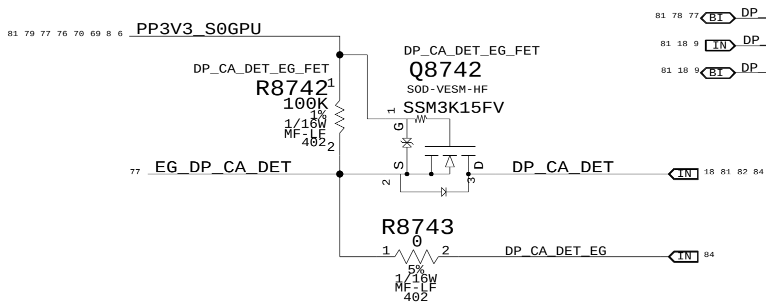
GPU 27MHz Crystal



Unused Clocks







Isolation FETs for DP MUX inputs

G96 GPIOs & Straps

SYNC_MASTER=MUXGFX SYNC_DATE=07/09/2008

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

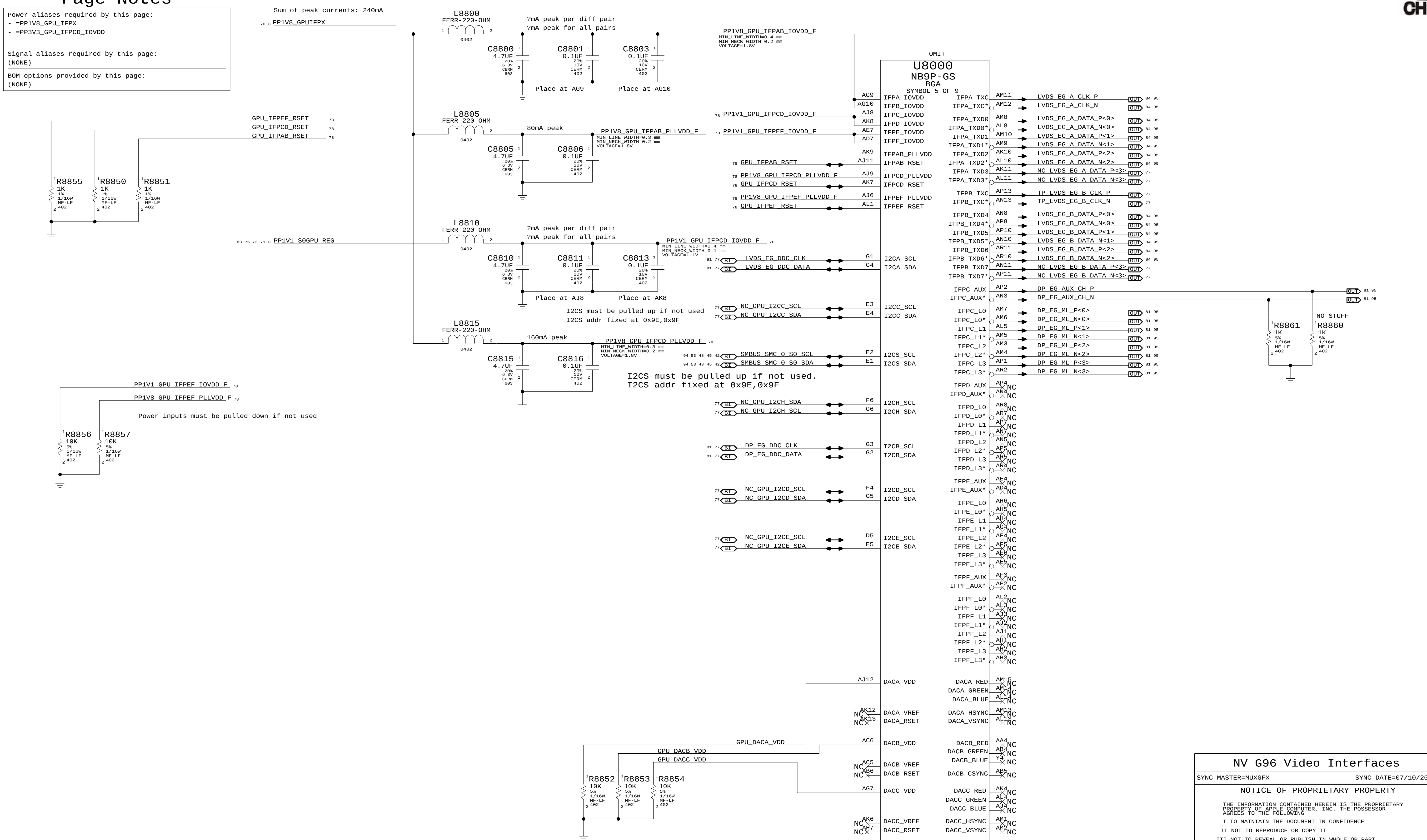
SCALE NONE | SHT 77 OF 97

Page Notes

Power aliases required by this page:
- =PP1V8_GPU_IFPX
- =PP3V3_GPU_IFPCD_IOVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



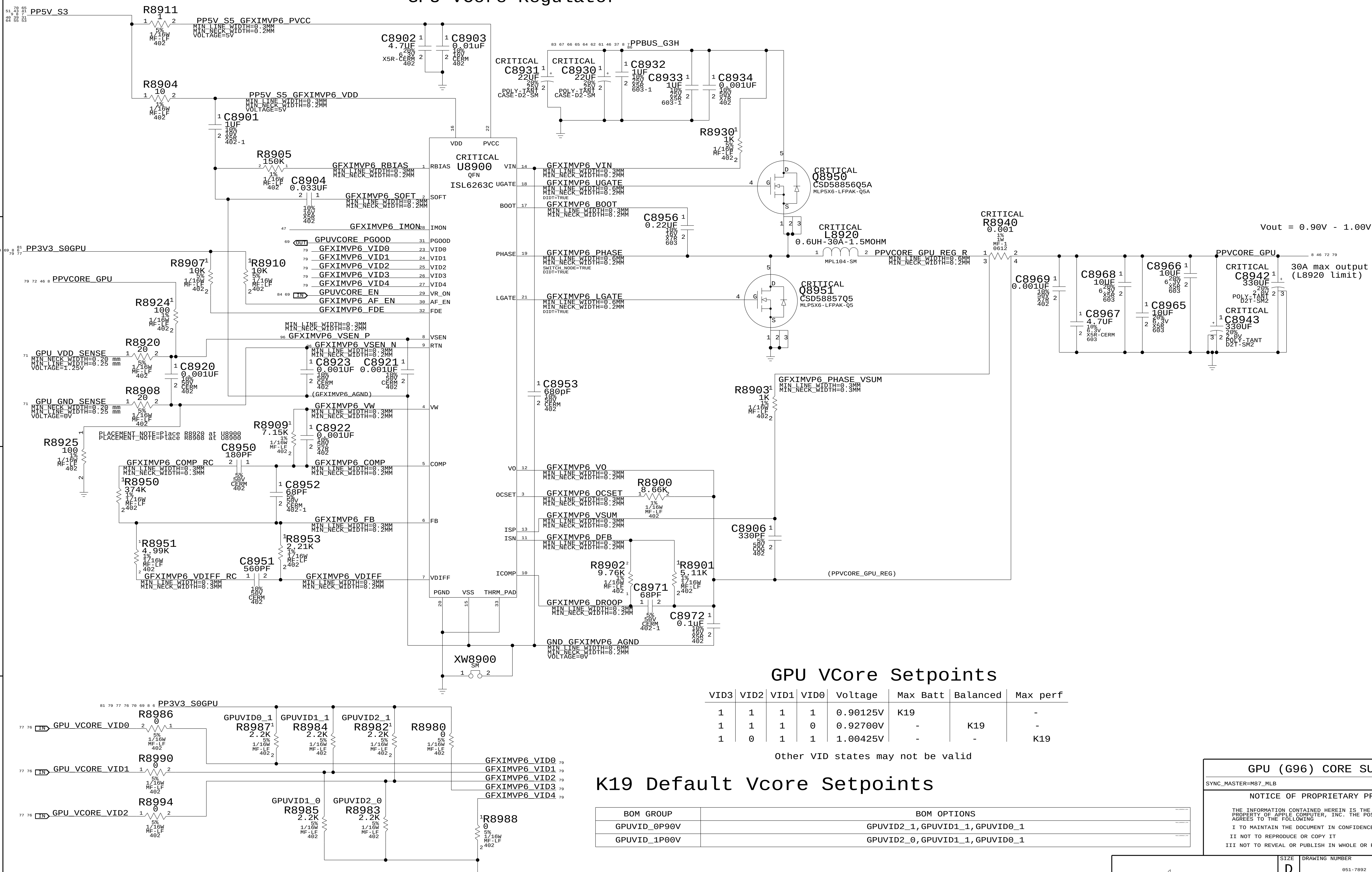
NV G96 Video Interfaces

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 78	OF 97

GPU VCore Regulator



GPU VCore Setpoints

VID3	VID2	VID1	VID0	Voltage	Max Batt	Balanced	Max perf
1	1	1	1	0.90125V	K19		-
1	1	1	0	0.92700V	-	K19	-
1	0	1	1	1.00425V	-	-	K19

Other VID states may not be valid

K19 Default Vcore Setpoints

BOM GROUP	BOM OPTIONS
GPUVID_0P90V	GPUVID2_1,GPUVID1_1,GPUVID0_1
GPUVID_1P00V	GPUVID2_0,GPUVID1_1,GPUVID0_1

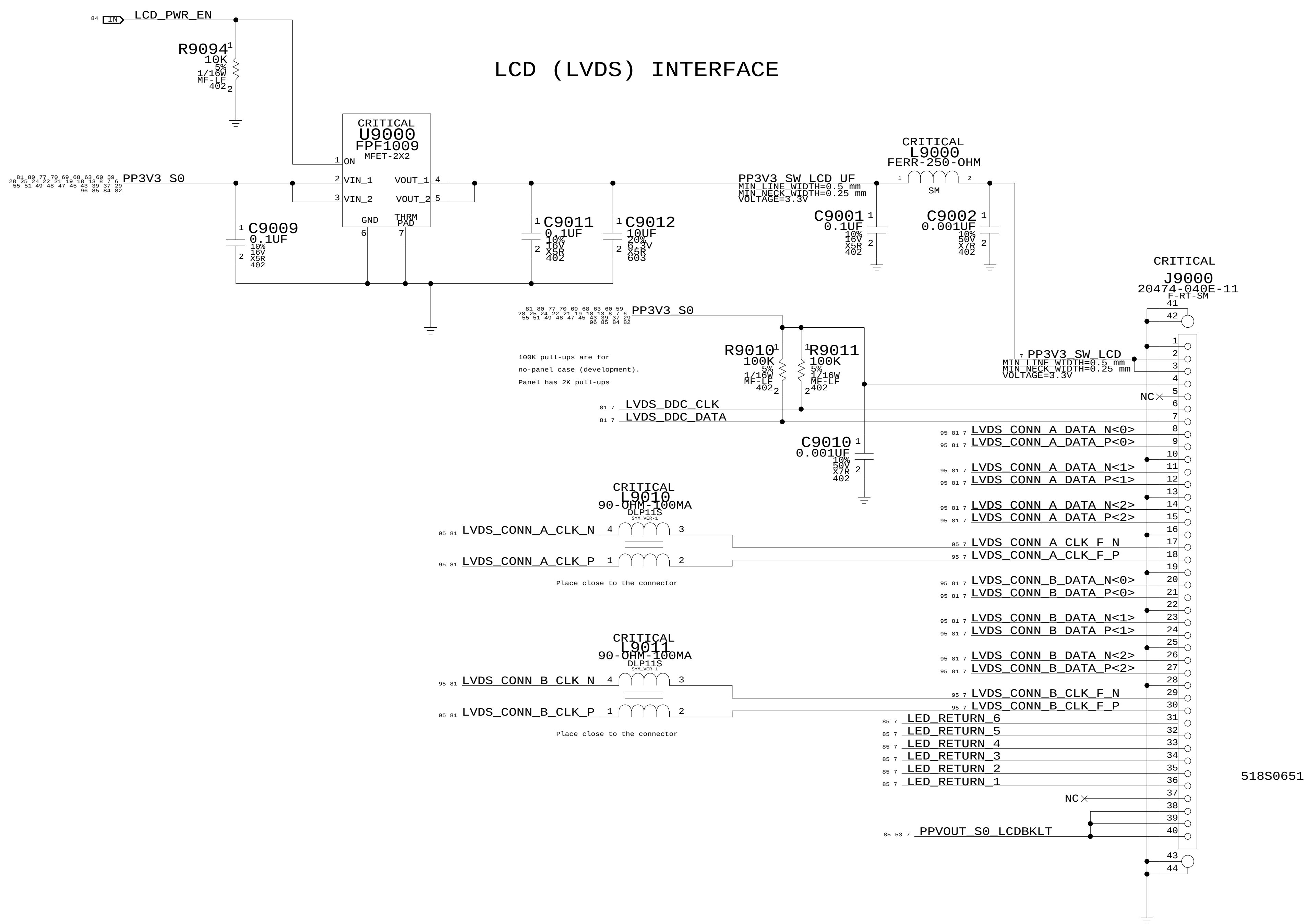
GPU (G96) CORE SUPPLY

SYNC_MASTER=M87_MLB SYNC_DATE=10/17/2007

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0
SCALE NONE | SHT 79 OF 97

LCD (LVDS) INTERFACE



LVDS Display Connector

SYNC_MASTER=DDR SYNC_DATE=12/19/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

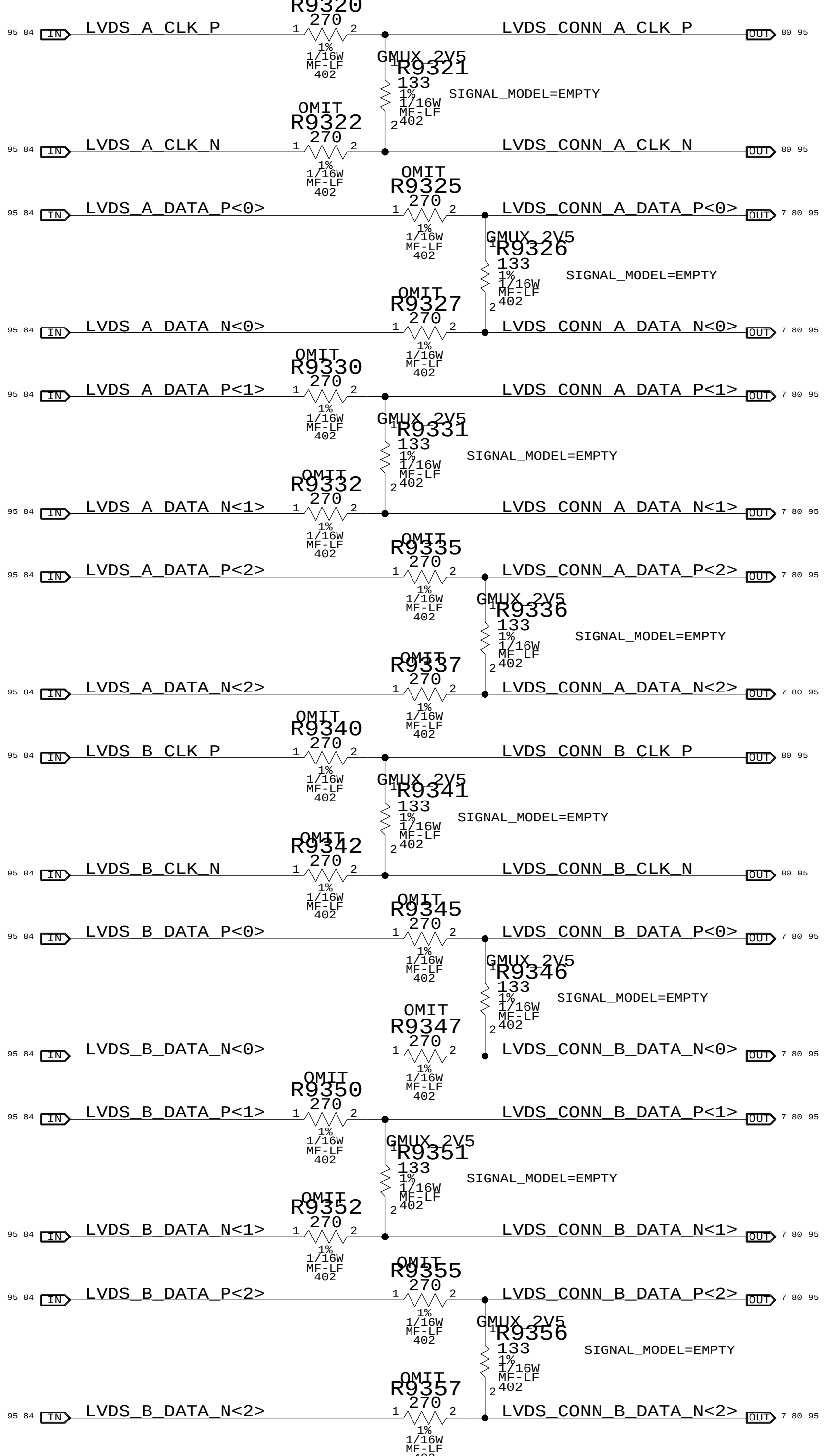
APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE: NONE	SHT 80	OF 97

LVDS Transmitter Termination

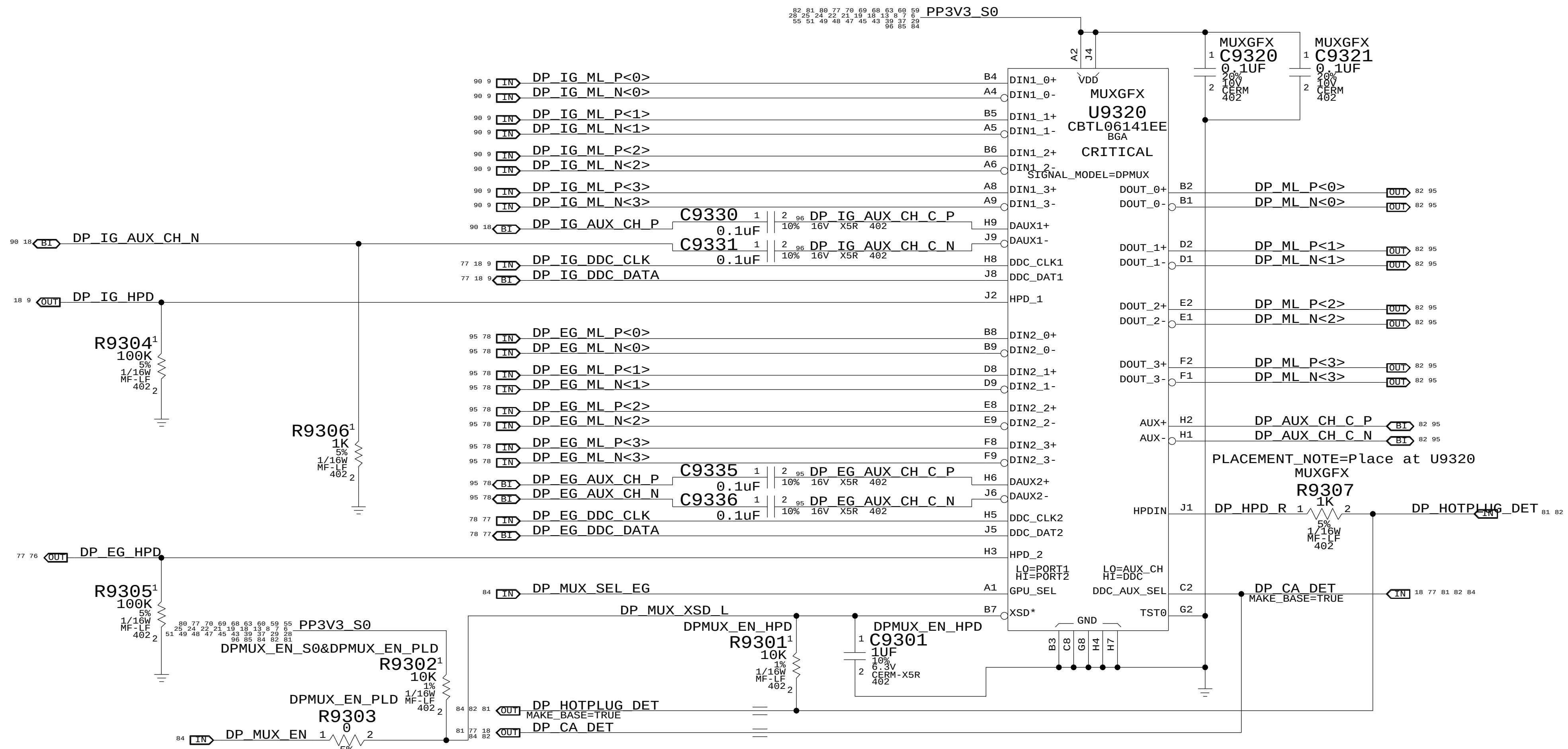
All emulated LVDS outputs require this termination

PLACEMENT_NOTE=Place at U9600 (All 24 resistors)

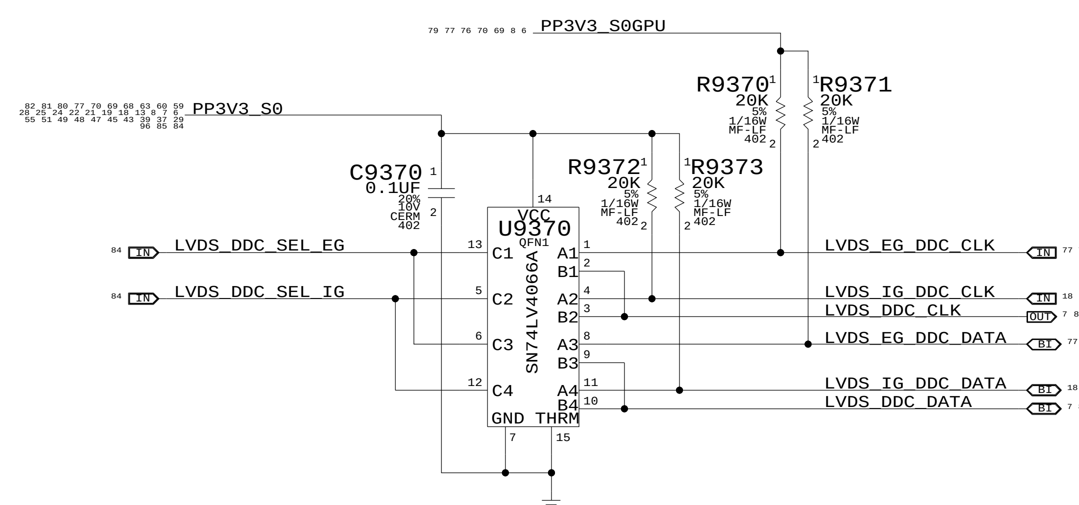


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
114S0517	16	RES,MTL FILM,270 OHM,1%,1/16W,0402,SMD,L	[illegible]		GMUX_2V5
114S0174	16	RES,MTL FILM,1/16W,357 OHM,1,0402,SMD,LF	[illegible]		GMUX_1V8

DisplayPort Mux



LVDS DDC MUX



Muxed Graphics Support

SYNC_MASTER=AMASON_M98_MLB SYNC_DATE=12/05/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

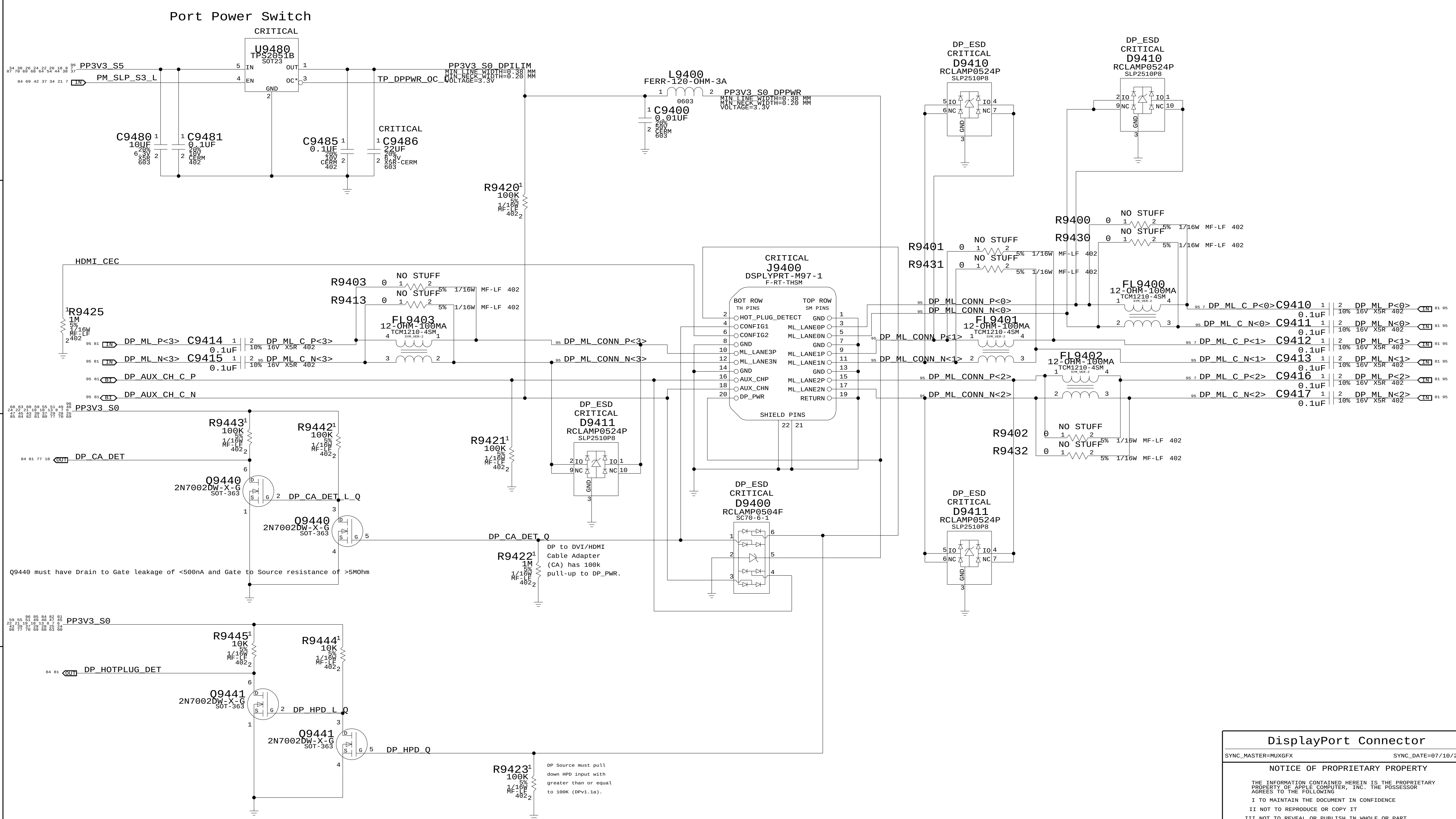
II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE NONE SHT 81 OF 97



DisplayPort Connector

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

NOTICE OF PROPRIETARY PROPERTY

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING

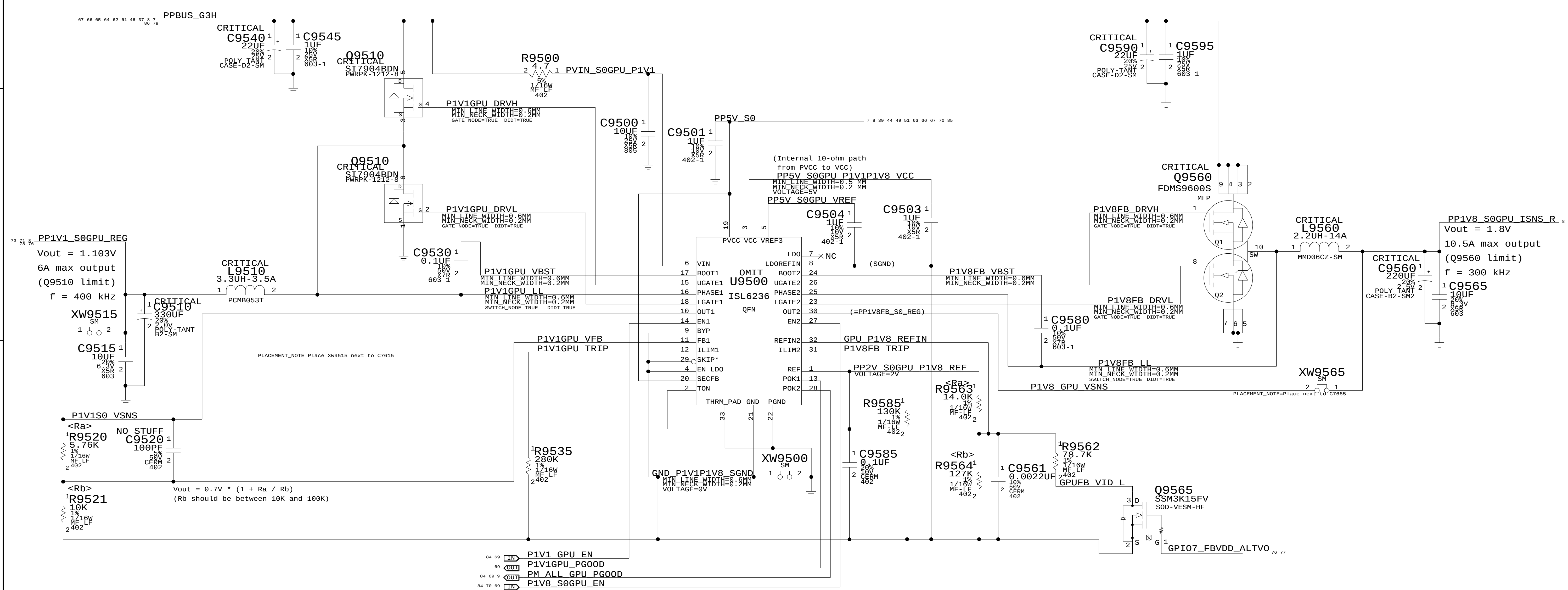
I TO MAINTAIN THE DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 82	OF 97



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S2312	1	IC,ISL6236,DUAL PWM CTRL,QFN32	U9500	CRITICAL	

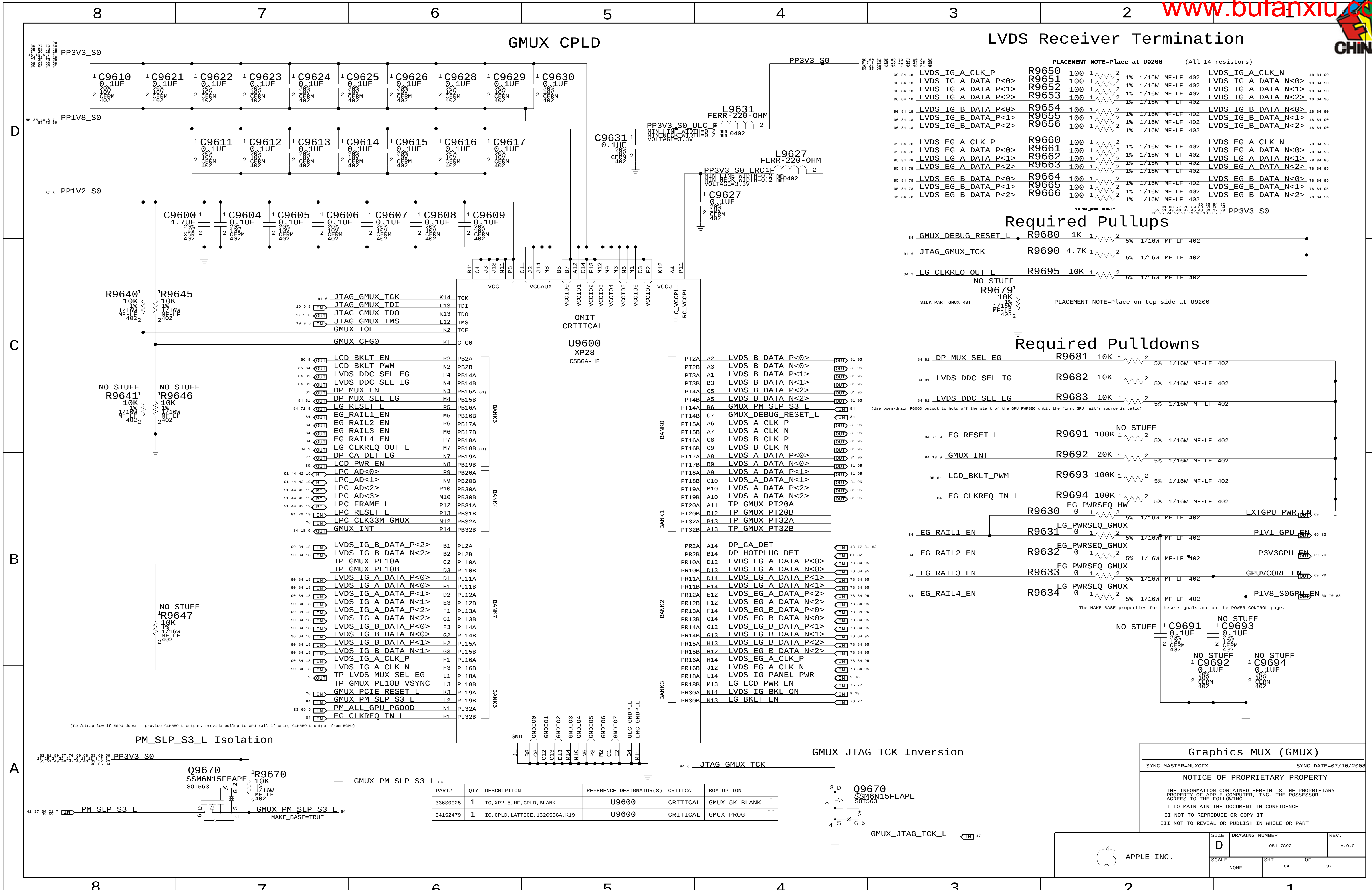
1.1V / 1V8 FB Power Supply

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE NONE SHT 83 OF 97



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
336S0025	1	IC,XP2-5,HF,CPLD,BLANK	U9600	CRITICAL	GMUX_5K_BLANK
341S2479	1	IC,CPLD,LATTICE,132CSBGA,K19	U9600	CRITICAL	GMUX_PROG

Graphics MUX (GMUX)

SYNC_MASTER=MUXGFX SYNC_DATE=07/10/2008

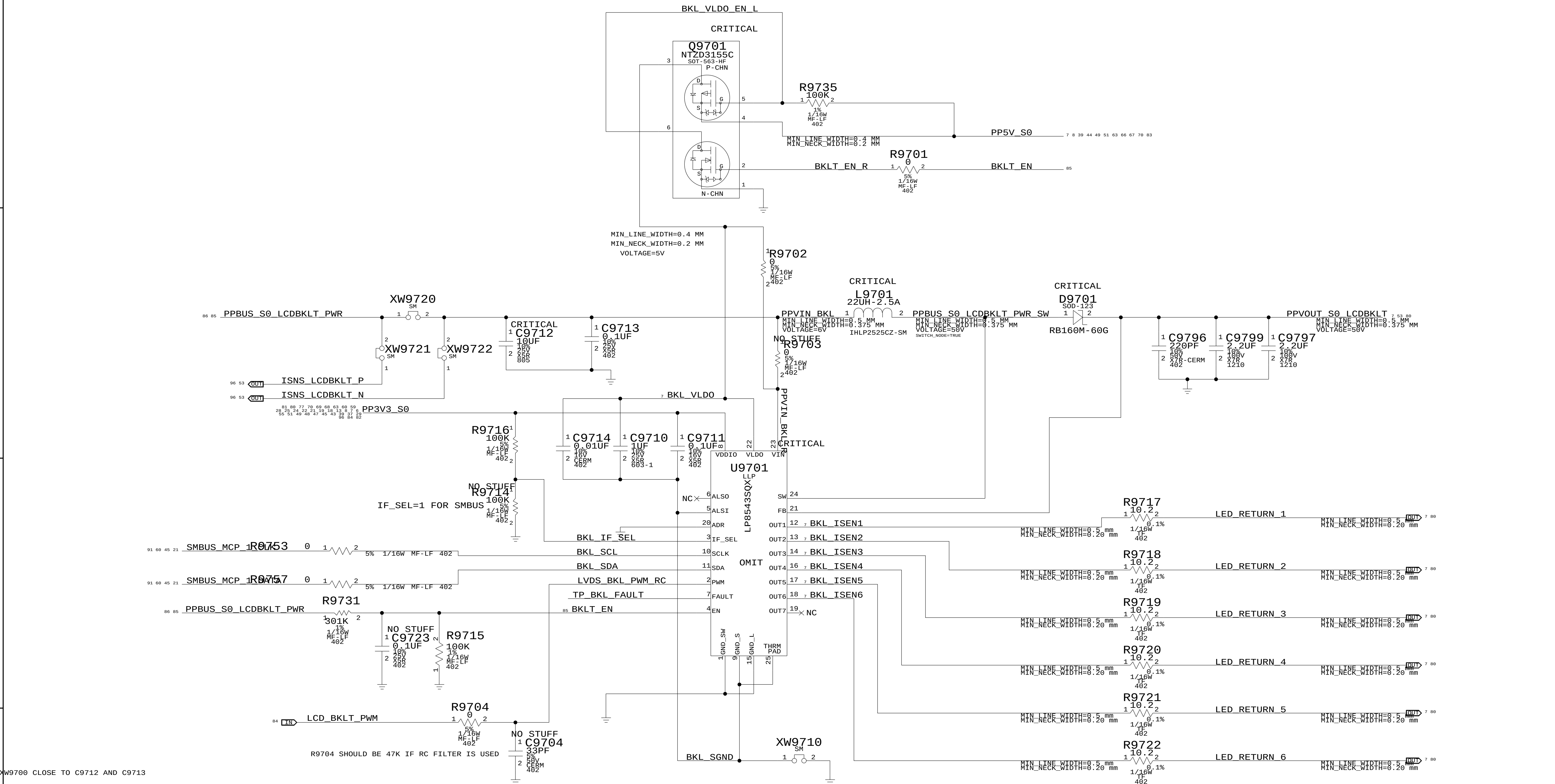
NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE | SHT 84 OF 97

*L9701, D9701, C9796, C9797, C9799, C9712 AND C9713 SHOULD ALL BE PLACED NEAR EACHOTHER.
*PPVOUT_S0_LCDBKLT_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.
* LVDS_IG_BKL_PWM SHOULD BE AWAY FROM BOOST CIRCUIT



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S2670	1	IC,LP8543,WHT LED BKLT,PROD	U9701	CRITICAL	

LCD BACKLIGHT DRIVER

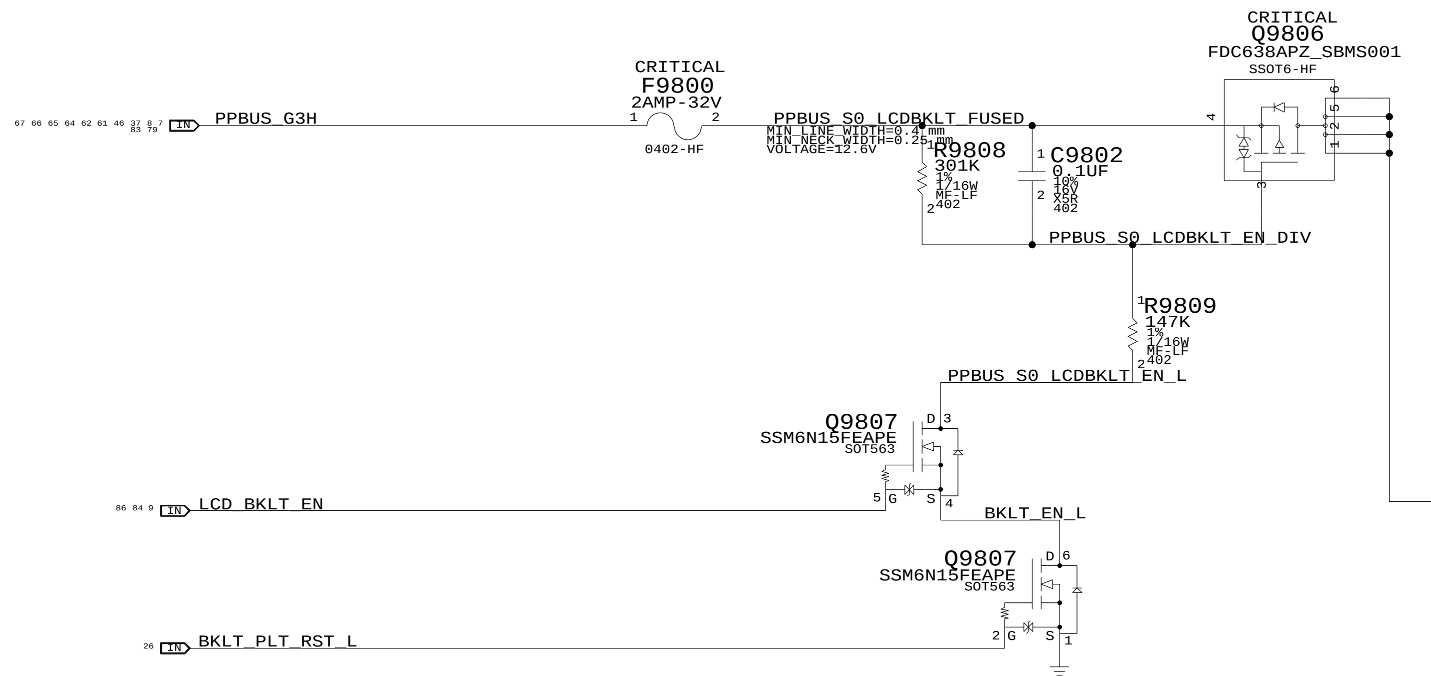
SYNC_MASTER=DDR SYNC_DATE=12/12/2008

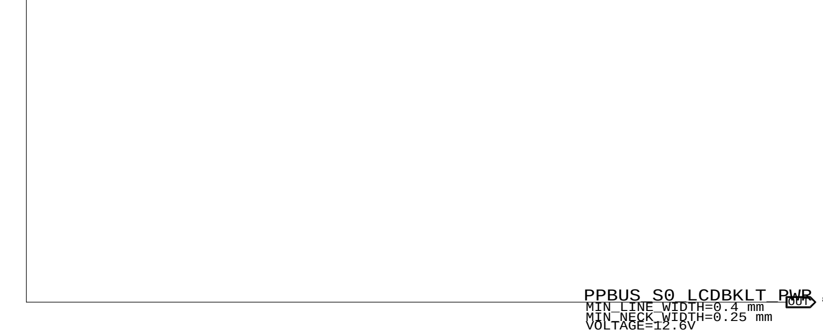
NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

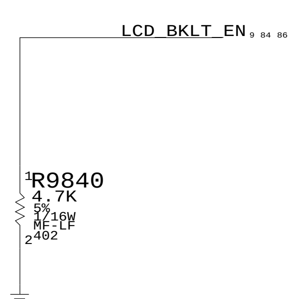
SCALE NONE | SHT 85 OF 97





PPBUS S0 LCDBkLT FET

MOSFET	FDC638APZ
CHANNEL	P-TYPE
RDS(ON)	43 mOhm @4.5V
LOADING	0.4 A (EDP)



LCD Backlight Support

SYNC_MASTER=YITE_M98_MLB SYNC_DATE=07/02/2008

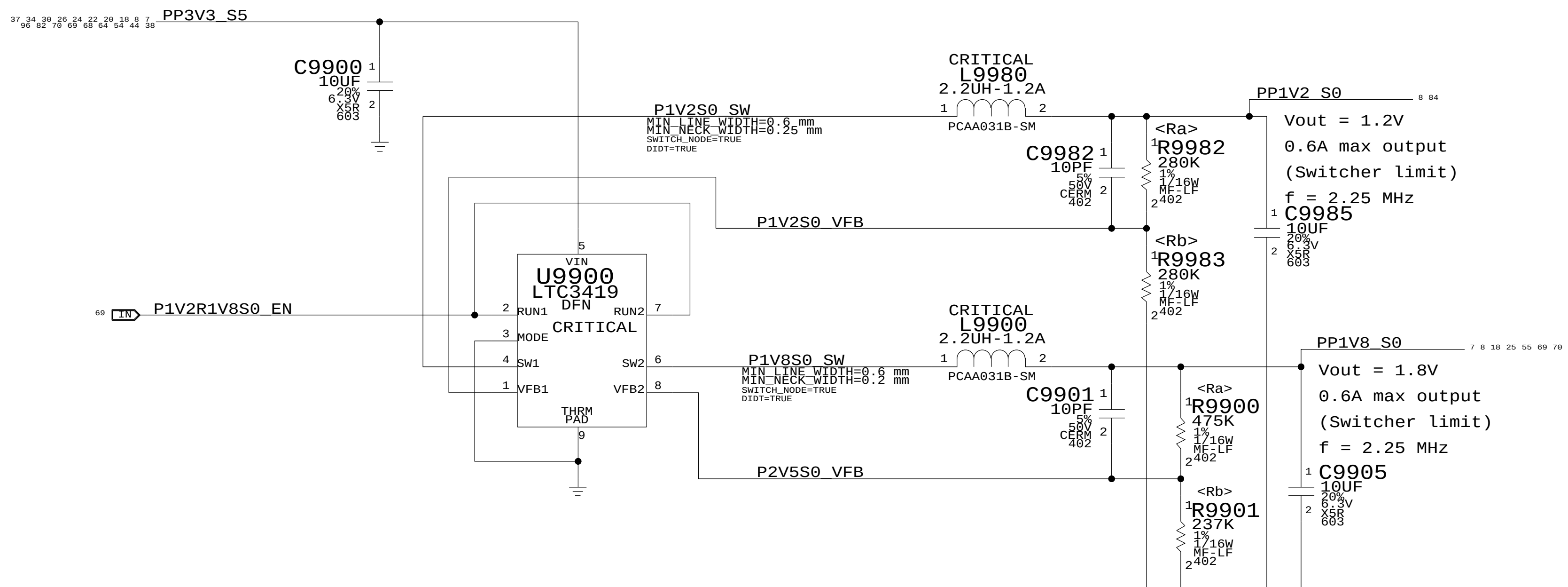
NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	86	97

1.8V/1.2V S0 SWITCHER



Vout = 0.6V * (1 + Ra/Rb)

Misc Power Supplies

SYNC_MASTER=MUXGFX SYNC_DATE=02/01/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 87	OF 97

FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2x_DIELECTRIC	?
FSB_DSTB	*	=3x_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?
FSB_ADSTB	*	=2x_DIELECTRIC	?
FSB_1X	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.
Signals within each 4x group should be matched within 5 ps of strobe.
DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 300 ps.
Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.
DSTB# complementary pairs are spaced normally and are NOT routed as differential pairs.

FSB 2X signals / groups shown in signal table on right.
Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 300 ps.
Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.
Signals within each 1x group should be matched to CPU clock, +0/-1000 mils.

Design Guide recommends each strobe/signal group is routed on the same layer.
Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	25 MIL	?
CPU_GTLREF	*	25 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SR DG recommends at least 25 mils, >50 mils preferred

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?

Most CPU signals with impedance requirements are 55-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

Group	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
FSB 4X Signal Groups	FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB_D_L<15..0>	7 10 14
	FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB_DINV_L<0>	7 10 14
	FSB_DSTB0	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_P<0>	7 10 14
	FSB_DSTB0	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_N<0>	7 10 14
	FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB_D_L<31..16>	7 10 14
	FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB_DINV_L<1>	7 10 14
	FSB_DSTB1	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_P<1>	7 10 14
	FSB_DSTB1	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_N<1>	7 10 14
	FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB_D_L<47..32>	7 10 14
	FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB_DINV_L<2>	7 10 14
	FSB_DSTB2	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_P<2>	7 10 14
	FSB_DSTB2	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_N<2>	7 10 14
	FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB_D_L<63..48>	7 10 14
	FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB_DINV_L<3>	7 10 14
	FSB_DSTB3	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_P<3>	7 10 14
	FSB_DSTB3	FSB_DSTB_50S	FSB_DSTB	FSB_DSTB_L_N<3>	7 10 14
FSB 2X Signals	FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB_A_L<16..3>	7 10 14
	FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB_REQ_L<4..0>	10 14
	FSB_ADSTB0	FSB_50S	FSB_ADSTB	FSB_ADSTB_L<0>	7 10 14
	FSB_ADDR_GROUP1	FSB_50S	FSB_ADDR	FSB_A_L<35..17>	7 10 14
	FSB_ADSTB1	FSB_50S	FSB_ADSTB	FSB_ADSTB_L<1>	7 10 14
FSB 1X Signals	FSB_1X	FSB_50S	FSB_1X	FSB_ADS_L	7 10 14
	FSB_BREQ0_L	FSB_50S	FSB_1X	FSB_BREQ0_L	9 10 14
	FSB_BREQ1_L	FSB_50S	FSB_1X	FSB_BREQ1_L	14
	FSB_1X	FSB_50S	FSB_1X	FSB_BNR_L	10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_BPRI_L	10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_DBSY_L	10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_DEFER_L	10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_DRDY_L	10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_HIT_L	7 10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_HITM_L	7 10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_LOCK_L	7 10 14
	FSB_CPURST_L	FSB_50S	FSB_1X	FSB_CPURST_L	9 10 13 14
	FSB_1X	FSB_50S	FSB_1X	FSB_RS_L<2..0>	10 14
	FSB_1X	FSB_50S	FSB_1X	FSB_TRDY_L	10 14
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_A20M_L	10 14
	CPU_BSEL	CPU_50S	CPU_AGTL	CPU_BSEL<2..0>	9 10
	CPU_FERR_L	CPU_50S	CPU_8MIL	CPU_FERR_L	10 14
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_IGNNE_L	10 14
	CPU_INIT_L	CPU_50S	CPU_AGTL	CPU_INIT_L	10 14
	CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU_INTR	9 10 14
	CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU_NMI	9 10 14
	CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	10 14 43 63
	CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	10 13 14
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_SMI_L	10 14
	CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_STPCLK_L	10 14
	PM_THRMTRIP_L	CPU_50S	CPU_8MIL	PM_THRMTRIP_L	10 14 43
	FSB_CPUSLP_L	CPU_50S	CPU_AGTL	FSB_CPUSLP_L	10 14
	CPU_FROM_SB	CPU_50S	CPU_AGTL	CPU_DPSLP_L	10 14
	CPU_DPRSTP_L	CPU_50S	CPU_AGTL	CPU_DPRSTP_L	9 10 14 63
	CPU_ASYNC	CPU_50S	CPU_AGTL	FSB_DPWR_L	10 14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_VDD	14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_GND	14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_VCC	14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_GND	14
	FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB_CLK_CPU_P	10 14
	FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB_CLK_CPU_N	10 14
	FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB_CLK_ITP_P	13 14
	FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB_CLK_ITP_N	13 14
	FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB_CLK_MCP_P	14
	FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB_CLK_MCP_N	14
	CPU_IERR_L	CPU_50S		CPU_IERR_L	10
	PM_DPRSLPVR	CPU_50S	CPU_AGTL	PM_DPRSLPVR	21 63
	(See above)	CPU_50S	CPU_AGTL	IMVP_DPRSLPVR	63
	CPU_GTLREF	CPU_50S	CPU_GTLREF	CPU_GTLREF	10 27
	CPU_COMP	CPU_50S	CPU_COMP	CPU_COMP<3>	10
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<2>	10
	CPU_COMP	CPU_50S	CPU_COMP	CPU_COMP<1>	10
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<0>	10
	XDP_TDI	CPU_50S	CPU_ITP	XDP_TDI	6 10 13
	XDP_TDO	CPU_50S	CPU_ITP	XDP_TDO	6 10
	XDP_TMS	CPU_50S	CPU_ITP	XDP_TMS	6 10 13
	XDP_TCK	CPU_50S	CPU_ITP	XDP_TCK	6 10 13
	XDP_TRST_L	CPU_50S	CPU_ITP	XDP_TRST_L	6 10 13
	XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM_L<4..0>	10 13
	XDP_BPM_L5	CPU_50S	CPU_ITP	XDP_BPM_L<5>	10 13
	(FSB_CPURST_L)	CPU_50S	CPU_ITP	XDP_CPURST_L	13
		CPU_50S	CPU_8MIL	CPU_VID<6..0>	9 11
		CPU_50S	CPU_8MIL	IMVP6_VID<6..0>	9 63
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	11 63
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	11 63
	(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_P	63
	(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_N	63

CPU/FSB Constraints

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE: NONE	SHT 88	OF 97

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_40S_VDD	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_70D_VDD	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_2OTHER	*	25 MIL	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_DATA	*	*	MEM_2OTHER
MEM_DQS	*	*	MEM_2OTHER

Need to support MEM_*-style wildcards!

DDR2:

DQ signals should be matched within 20 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, no inter-pair matching requirement.
All DQS pairs should be matched within 100 ps of clocks.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 140 ps.
A/BA/cmd signals should be matched within 75 ps, no CLK matching requirement.
All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).
DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

DDR3:

DQ signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching shoulw be within 180 ps
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
A/BA/cmd signals should be matched within 5 ps of CLK pairs.
All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).
DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.3
SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MEM_COMP	*	Y	7 MIL	7 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_MEM_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.3.4

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
MEM_A_CLK	MEM_70D_VDD	MEM_CLK	MEM_A_CLK_P<5..0>	15 28
MEM_A_CLK	MEM_70D_VDD	MEM_CLK	MEM_A_CLK_N<5..0>	15 28
MEM_A_CNTL	MEM_40S_VDD	MEM_CTRL	MEM_A_CKE<3..0>	15 28
MEM_A_CNTL	MEM_40S_VDD	MEM_CTRL	MEM_A_CS_L<3..0>	15 28
MEM_A_CNTL	MEM_40S_VDD	MEM_CTRL	MEM_A_ODT<3..0>	15 28
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM_A_A<14..0>	15 28
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM_A_BA<2..0>	15 28
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM_A_RAS_L	15 28
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM_A_CAS_L	15 28
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM_A_WE_L	15 28
MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM_A_DQ<7..0>	15 28
MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM_A_DQ<15..8>	15 28
MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM_A_DQ<23..16>	15 28
MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM_A_DQ<31..24>	15 28
MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM_A_DQ<39..32>	15 28
MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM_A_DQ<47..40>	15 28
MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM_A_DQ<55..48>	15 28
MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM_A_DQ<63..56>	15 28
MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM_A_DM<0>	15 28
MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM_A_DM<1>	15 28
MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM_A_DM<2>	15 28
MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM_A_DM<3>	15 28
MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM_A_DM<4>	15 28
MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM_A_DM<5>	15 28
MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM_A_DM<6>	15 28
MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM_A_DM<7>	15 28
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM_A_DQS_P<0>	15 28
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM_A_DQS_N<0>	15 28
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM_A_DQS_P<1>	15 28
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM_A_DQS_N<1>	15 28
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM_A_DQS_P<2>	15 28
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM_A_DQS_N<2>	15 28
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM_A_DQS_P<3>	15 28
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM_A_DQS_N<3>	15 28
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM_A_DQS_P<4>	15 28
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM_A_DQS_N<4>	15 28
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM_A_DQS_P<5>	15 28
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM_A_DQS_N<5>	15 28
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM_A_DQS_P<6>	15 28
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM_A_DQS_N<6>	15 28
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM_A_DQS_P<7>	15 28
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM_A_DQS_N<7>	15 28
MEM_B_CLK	MEM_70D_VDD	MEM_CLK	MEM_B_CLK_P<5..0>	15 29
MEM_B_CLK	MEM_70D_VDD	MEM_CLK	MEM_B_CLK_N<5..0>	15 29
MEM_B_CNTL	MEM_40S_VDD	MEM_CTRL	MEM_B_CKE<3..0>	15 29
MEM_B_CNTL	MEM_40S_VDD	MEM_CTRL	MEM_B_CS_L<3..0>	15 29
MEM_B_CNTL	MEM_40S_VDD	MEM_CTRL	MEM_B_ODT<3..0>	15 29
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM_B_A<14..0>	15 29
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM_B_BA<2..0>	15 29
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM_B_RAS_L	15 29
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM_B_CAS_L	15 29
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM_B_WE_L	15 29
MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM_B_DQ<7..0>	15 29
MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM_B_DQ<15..8>	15 29
MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM_B_DQ<23..16>	15 29
MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM_B_DQ<31..24>	15 29
MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM_B_DQ<39..32>	15 29
MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM_B_DQ<47..40>	15 29
MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM_B_DQ<55..48>	15 29
MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM_B_DQ<63..56>	15 29
MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM_B_DM<0>	15 29
MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM_B_DM<1>	15 29
MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM_B_DM<2>	15 29
MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM_B_DM<3>	15 29
MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM_B_DM<4>	15 29
MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM_B_DM<5>	15 29
MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM_B_DM<6>	15 29
MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM_B_DM<7>	15 29
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM_B_DQS_P<0>	15 29
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM_B_DQS_N<0>	15 29
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM_B_DQS_P<1>	15 29
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM_B_DQS_N<1>	15 29
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM_B_DQS_P<2>	15 29
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM_B_DQS_N<2>	15 29
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM_B_DQS_P<3>	15 29
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM_B_DQS_N<3>	15 29
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM_B_DQS_P<4>	15 29
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM_B_DQS_N<4>	15 29
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM_B_DQS_P<5>	15 29
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM_B_DQS_N<5>	15 29
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM_B_DQS_P<6>	15 29
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM_B_DQS_N<6>	15 29
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM_B_DQS_P<7>	15 29
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM_B_DQS_N<7>	15 29
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_VDD	16
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_GND	16

Memory Constraints

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	89	97

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	13.1 MM	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?
MCP_PEX_COMP	*	8 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.4

Analog Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CRT_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CRT	*	=4:1_SPACING	?
CRT_2CRT	*	=STANDARD	?
CRT_2CLK	*	50 MIL	?
CRT_2SWITCHER	*	250 MIL	?
CRT_SYNC	*	16 MIL	?
MCP_DAC_COMP	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CRT	CRT	*	CRT_2CRT

CRT signal single-ended impedence varies by location:
- 37.5-ohm from MCP to first termination resistor.
- 50-ohm from first to second termination resistor.
- 75-ohm from output of three-pole filter to connector (if possible).

R/G/B signals should be matched as close as possible and < 10 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.1 & 2.5.2.

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	20 MIL	20 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DIsplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=4x_DIELECTRIC	?
SATA_TERMP	*	8 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=3x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.7.1.

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	PCIE_90D	PCIE	PEG_R2D_P<15..0>	71
	PCIE_90D	PCIE	PEG_R2D_N<15..0>	71
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_C_P<15..0>	9 71
	PCIE_90D	PCIE	PEG_R2D_C_N<15..0>	9 71
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_P<15..0>	9 71
	PCIE_90D	PCIE	PEG_D2R_N<15..0>	9 71
	PCIE_90D	PCIE	PEG_D2R_C_P<15..0>	71
	PCIE_90D	PCIE	PEG_D2R_C_N<15..0>	71
	PCIE_90D	PCIE	PCIE_MINI_R2D_P	7 31 96
	PCIE_90D	PCIE	PCIE_MINI_R2D_N	7 31 96
PCIE_MINI_R2D	PCIE_90D	PCIE	PCIE_MINI_R2D_C_P	17 31
	PCIE_90D	PCIE	PCIE_MINI_R2D_C_N	17 31
PCIE_MINI_D2R	PCIE_90D	PCIE	PCIE_MINI_D2R_P	7 17 31
	PCIE_90D	PCIE	PCIE_MINI_D2R_N	7 17 31
	PCIE_90D	PCIE	PCIE_FW_R2D_P	36
	PCIE_90D	PCIE	PCIE_FW_R2D_N	36
PCIE_FW_R2D	PCIE_90D	PCIE	PCIE_FW_R2D_C_P	17 36
	PCIE_90D	PCIE	PCIE_FW_R2D_C_N	17 36
PCIE_FW_D2R	PCIE_90D	PCIE	PCIE_FW_D2R_P	17 36
	PCIE_90D	PCIE	PCIE_FW_D2R_N	17 36
	PCIE_90D	PCIE	PCIE_FW_D2R_C_P	36
	PCIE_90D	PCIE	PCIE_FW_D2R_C_N	36
	PCIE_90D	PCIE	PCIE_EXCARD_R2D_P	96
	PCIE_90D	PCIE	PCIE_EXCARD_R2D_N	96
PCIE_EXCARD_R2D	PCIE_90D	PCIE	TP_PCIE_EXCARD_R2D_C_P	9 17
	PCIE_90D	PCIE	TP_PCIE_EXCARD_R2D_C_N	9 17
PCIE_EXCARD_D2R	PCIE_90D	PCIE	TP_PCIE_EXCARD_D2R_P	9 17
	PCIE_90D	PCIE	TP_PCIE_EXCARD_D2R_N	9 17
MCP_PE0_REFCLK	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_P	17 71
	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_N	17 71
MCP_PE1_REFCLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_P	17 31
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_N	17 31
MCP_PE2_REFCLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P	17 36
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N	17 36
MCP_PE3_REFCLK	CLK_PCIE_100D	CLK_PCIE	TP_PCIE_CLK100M_EXCARD_P	9 17
	CLK_PCIE_100D	CLK_PCIE	TP_PCIE_CLK100M_EXCARD_N	9 17
MCP_PEX_CLK_COMP		MCP_PEX_COMP	MCP_PEX_CLK_COMP	17
CRT_RED	CRT_50S	CRT	NC_CRT_IG_R_C_PR	18 25
CRT_GREEN	CRT_50S	CRT	NC_CRT_IG_G_Y_Y	18 25
CRT_BLUE	CRT_50S	CRT	NC_CRT_IG_B_COMP_PB	18 25
CRT_SYNC	CRT_50S	CRT_SYNC	NC_CRT_IG_HSYNC	18 25
CRT_SYNC	CRT_50S	CRT_SYNC	NC_CRT_IG_VSYNC	18 25
MCP_DAC_RSET		MCP_DAC_COMP	NC_MCP_TV_DAC_RSET	18 25
MCP_DAC_VREF		MCP_DAC_COMP	NC_MCP_TV_DAC_VREF	18 25
TMDS_IG_TXC	DP_100D	DISPLAYPORT	TMDS_IG_TXC_P	
TMDS_IG_TXC	DP_100D	DISPLAYPORT	TMDS_IG_TXC_N	
TMDS_IG_TXD	DP_100D	DISPLAYPORT	TMDS_IG_TXD_P<2..0>	
TMDS_IG_TXD	DP_100D	DISPLAYPORT	TMDS_IG_TXD_N<2..0>	
DP_ML	DP_100D	DISPLAYPORT	DP_IG_ML_P<3..0>	9 81
DP_ML	DP_100D	DISPLAYPORT	DP_IG_ML_N<3..0>	9 81
DP_AUX_CH	DP_100D	DISPLAYPORT	DP_IG_AUX_CH_P	18 81
DP_AUX_CH	DP_100D	DISPLAYPORT	DP_IG_AUX_CH_N	18 81
MCP_HDMI_RSET	MCP_DV_COMP		MCP_HDMI_RSET	18 25
MCP_HDMI_VPROBE	MCP_DV_COMP		MCP_HDMI_VPROBE	18 25
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_P	18 84
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_N	18 84
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<2..0>	18 84
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<2..0>	18 84
LVDS_IG_A_DATA3	LVDS_100D	LVDS	NC_LVDS_IG_A_DATAP<3>	9 18
LVDS_IG_A_DATA3	LVDS_100D	LVDS	NC_LVDS_IG_A_DATAN<3>	9 18
LVDS_IG_B_CLK	LVDS_100D	LVDS	NC_LVDS_IG_B_CLKP	9 18
LVDS_IG_B_CLK	LVDS_100D	LVDS	NC_LVDS_IG_B_CLKN	9 18
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<2..0>	18 84
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<2..0>	18 84
LVDS_IG_B_DATA3	LVDS_100D	LVDS	NC_LVDS_IG_B_DATAP<3>	9 18
LVDS_IG_B_DATA3	LVDS_100D	LVDS	NC_LVDS_IG_B_DATAN<3>	9 18
MCP_IFPAB_RSET	MCP_DV_COMP		MCP_IFPAB_RSET	18 25
MCP_IFPAB_VPROBE			MCP_IFPAB_VPROBE	18 25
SATA_HDD_R2D	SATA_100D	SATA	SATA_HDD_R2D_C_P	20 39
	SATA_100D	SATA	SATA_HDD_R2D_C_N	20 39
	SATA_100D	SATA	SATA_HDD_R2D_P	7 39
	SATA_100D	SATA	SATA_HDD_R2D_N	7 39
SATA_HDD_D2R	SATA_100D	SATA	SATA_HDD_D2R_P	20 39
	SATA_100D	SATA	SATA_HDD_D2R_N	20 39
	SATA_100D	SATA	SATA_HDD_D2R_C_P	7 39
	SATA_100D	SATA	SATA_HDD_D2R_C_N	7 39
SATA_ODD_R2D	SATA_100D	SATA	SATA_ODD_R2D_C_P	20 39
	SATA_100D	SATA	SATA_ODD_R2D_C_N	20 39
	SATA_100D	SATA	SATA_ODD_R2D_P	7 39
	SATA_100D	SATA	SATA_ODD_R2D_N	7 39
SATA_ODD_D2R	SATA_100D	SATA	SATA_ODD_D2R_P	20 39
	SATA_100D	SATA	SATA_ODD_D2R_N	20 39
	SATA_100D	SATA	SATA_ODD_D2R_C_P	7 39
	SATA_100D	SATA	SATA_ODD_D2R_C_N	7 39
MCP_SATA_TERMP		SATA_TERMP	MCP_SATA_TERMP	20

MCP Constraints 1

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 90	OF 97

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.8.

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.9.1.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.10.1.

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.11.1.

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.12.1.

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.13.

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.14.

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
MCP_DEBUG	PCI_55S	PCI	MCP_DEBUG<7..0>	13 19
PCI_AD	PCI_55S	PCI	PCI_AD<23..8>	
PCI_AD24	PCI_55S	PCI	PCI_AD<24>	
PCI_AD	PCI_55S	PCI	PCI_AD<31..25>	
PCI_AD	PCI_55S	PCI	PCI_PAR	
PCI_C_BE_L	PCI_55S	PCI	PCI_C_BE_L<3..0>	
PCI_CNTL	PCI_55S	PCI	PCI_IRDY_L	
PCI_CNTL	PCI_55S	PCI	PCI_DEVSEL_L	
PCI_CNTL	PCI_55S	PCI	PCI_PERR_L	
PCI_CNTL	PCI_55S	PCI	PCI_SERR_L	
PCI_CNTL	PCI_55S	PCI	PCI_STOP_L	
PCI_CNTL	PCI_55S	PCI	PCI_TRDY_L	
PCI_CNTL	PCI_55S	PCI	PCI_FRAME_L	
PCI_REQ0_L	PCI_55S	PCI	PCI_REQ0_L	19
PCI_GNT0_L	PCI_55S	PCI	PCI_GNT0_L	
PCI_REQ1_L	PCI_55S	PCI	PCI_REQ1_L	19
PCI_GNT1_L	PCI_55S	PCI	PCI_GNT1_L	
PCI_INTW_L	PCI_55S	PCI	PCI_INTW_L	
PCI_INTX_L	PCI_55S	PCI	PCI_INTX_L	
PCI_INTY_L	PCI_55S	PCI	PCI_INTY_L	
PCI_INTZ_L	PCI_55S	PCI	PCI_INTZ_L	
MCP_PCI_CLK2	CLK_PCI_55S	CLK_PCI	PCI_CLK33M_MCP_R	19
	CLK_PCI_55S	CLK_PCI	PCI_CLK33M_MCP	19
LPC_AD	LPC_55S	LPC	LPC_AD<3..0>	19 42 44 84
LPC_FRAME_L	LPC_55S	LPC	LPC_FRAME_L	19 42 44 84
LPC_RESET_L	LPC_55S	LPC	LPC_RESET_L	19 26 84
MCP_LPC_CLK0	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	19 26
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	26 42
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	26 44
USB_EXTA	USB_90D	USB	USB_EXTA_P	20 40
	USB_90D	USB	USB_EXTA_N	20 40
	USB_90D	USB	USB_EXTA_MUXED_P	
	USB_90D	USB	USB_EXTA_MUXED_N	
USB_MINI	USB_90D	USB	NC_USB_MINIP	9 20
	USB_90D	USB	NC_USB_MININ	9 20
USB_EXTD	USB_90D	USB	NC_USB_EXTDP	9 20
	USB_90D	USB	NC_USB_EXTDN	9 20
USB_CAMERA	USB_90D	USB	USB_CAMERA_P	7 20 31
	USB_90D	USB	USB_CAMERA_N	7 20 31
USB_BT	USB_90D	USB	USB_BT_P	7 20 31
	USB_90D	USB	USB_BT_N	7 20 31
USB_TPAD	USB_90D	USB	USB_TPAD_P	20 50
	USB_90D	USB	USB_TPAD_N	20 50
USB_IR	USB_90D	USB	USB_IR_P	20 41
	USB_90D	USB	USB_IR_N	20 41
USB_EXTB	USB_90D	USB	USB_EXTB_P	20 40
	USB_90D	USB	USB_EXTB_N	20 40
USB_EXCARD	USB_90D	USB	NC_USB_EXCARDP	9 20
	USB_90D	USB	NC_USB_EXCARDN	9 20
USB_EXTC	USB_90D	USB	NC_USB_EXTCP	9 20
	USB_90D	USB	NC_USB_EXTCN	9 20
MCP_USB_RBIAS	MCP_USB_RBIAS		MCP_USB_RBIAS_GND	20
SMBUS_MCP_0_CLK	SMB_55S	SMB	SMBUS_MCP_0_CLK	13 21 28 29 45
SMBUS_MCP_0_DATA	SMB_55S	SMB	SMBUS_MCP_0_DATA	13 21 28 29 45
SMBUS_MCP_1_CLK	SMB_55S	SMB	SMBUS_MCP_1_CLK	21 45 60 85
SMBUS_MCP_1_DATA	SMB_55S	SMB	SMBUS_MCP_1_DATA	21 45 60 85
HDA_BIT_CLK	HDA_55S	HDA	HDA_BIT_CLK	21 55
	HDA_55S	HDA	HDA_BIT_CLK_R	21
HDA_SYNC	HDA_55S	HDA	HDA_SYNC	21 55
	HDA_55S	HDA	HDA_SYNC_R	21
HDA_RST_L	HDA_55S	HDA	HDA_RST_R_L	21
	HDA_55S	HDA	HDA_RST_L	21 55
HDA_SDIN0	HDA_55S	HDA	HDA_SDIN0	21 55
	HDA_55S	HDA	HDA_SDIN_CODEC	
HDA_SDOUT	HDA_55S	HDA	HDA_SDOUT	21 55
	HDA_55S	HDA	HDA_SDOUT_R	21
MCP_HDA_PULLDN_COMP		MCP_HDA_COMP	MCP_HDA_PULLDN_COMP	21
MCP_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK_R	21 26
	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK	26 42
SPI_CLK	SPI_55S	SPI	SPI_CLK_R	21 44
	SPI_55S	SPI	SPI_CLK	54
SPI_MOSI	SPI_55S	SPI	SPI_MOSI_R	21 44
	SPI_55S	SPI	SPI_MOSI	54
SPI_MISO	SPI_55S	SPI	SPI_MISO	21 44
	SPI_55S	SPI	SPI_MISO_R	54
SPI_CS0	SPI_55S	SPI	SPI_CS0_R_L	21 44
	SPI_55S	SPI	SPI_CS0_L	

MCP Constraints 2

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 91	OF 97

MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MII_COMP	*	=STANDARD	7.5 MIL	7.5 MIL	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	12 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_MDI_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	25 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
MCP_MII_COMP	MCP_MII_COMP		MCP_MII_COMP_VDD	18
MCP_MII_COMP	MCP_MII_COMP		MCP_MII_COMP_GND	18
MCP_CLK25M_BUF0	ENET_MII_55S	MCP_BUF0_CLK	MCP_CLK25M_BUF0_R	18 34
	ENET_MII_55S	MCP_BUF0_CLK	RTL8211_CLK25M_CKXTAL1	33 34
ENET_INTR_L	ENET_MII_55S	ENET_MII	ENET_INTR_L	
ENET_MDIO	ENET_MII_55S	ENET_MII	ENET_MDIO	18 33
ENET_MDC	ENET_MII_55S	ENET_MII	ENET_MDC	18 33
ENET_PWRDWN_L	ENET_MII_55S	ENET_MII	ENET_PWRDWN_L	
	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK_R	33
ENET_RXCLK	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK	18 33
	ENET_MII_55S	ENET_MII	ENET_RXD_R<3..0>	33
ENET_RXD	ENET_MII_55S	ENET_MII	ENET_RXD<0>	18 33
ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET_RXD<3..1>	18 33
ENET_RXD	ENET_MII_55S	ENET_MII	ENET_RX_CTRL	18 33
ENET_TXCLK	ENET_MII_55S	ENET_MII	ENET_CLK125M_TXCLK	18 33
ENET_TXD0	ENET_MII_55S	ENET_MII	ENET_TXD<0>	18 33
ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TXD<3..1>	18 33
ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TX_CTRL	18 33
	ENET_MII_55S	ENET_MII	ENET_RESET_L	18 33
ENET_MDI	ENET_MDI_100D	ENET_MDI	ENET_MDI_P<3..0>	33 35
	ENET_MDI_100D	ENET_MDI	ENET_MDI_N<3..0>	33 35

Ethernet Constraints

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE D | DRAWING NUMBER 051-7892 | REV. A.0.0

SCALE NONE |

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

SD CARD INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
FW_P0_TPA	FW_110D	FW_TP	NC_FW0_TPAP	36 38
FW_P0_TPA	FW_110D	FW_TP	NC_FW0_TPAN	36 38
FW_P0_TPB	FW_110D	FW_TP	NC_FW0_TPBP	36 38
FW_P0_TPB	FW_110D	FW_TP	NC_FW0_TPBN	36 38
FW_P1_TPA	FW_110D	FW_TP	FW_PORT1_TPA_P	36 38
FW_P1_TPA	FW_110D	FW_TP	FW_PORT1_TPA_N	36 38
FW_P1_TPB	FW_110D	FW_TP	FW_PORT1_TPB_P	36 38
FW_P1_TPB	FW_110D	FW_TP	FW_PORT1_TPB_N	36 38

Port 2 Not Used

SD CARD NET PROPERTIES

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
I32	SD_DATA	SD_55S	SD_INTERFACE	SD_D<0>	7 32
I31	SD_DATA	SD_55S	SD_INTERFACE	SD_D<1>	7 32
I30	SD_DATA	SD_55S	SD_INTERFACE	SD_D<2>	7 32
I29	SD_DATA	SD_55S	SD_INTERFACE	SD_D<3>	7 32
I28	SD_DATA	SD_55S	SD_INTERFACE	SD_D<4>	7 32
I27	SD_DATA	SD_55S	SD_INTERFACE	SD_D<5>	7 32
I26	SD_DATA	SD_55S	SD_INTERFACE	SD_D<6>	7 32
I25	SD_DATA	SD_55S	SD_INTERFACE	SD_D<7>	7 32
I23	SD_CLK	SD_55S	SD_INTERFACE	SD_CLK	7 32
I24	SD_CMD	SD_55S	SD_INTERFACE	SD_CMD	7 32

FireWire Constraints

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	93	97

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
SMBUS_SMC_A_S3_SCL	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	7 31 42 45 51
SMBUS_SMC_A_S3_SDA	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	7 31 42 45 51
SMBUS_SMC_B_S0_SCL	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	42 45 48
SMBUS_SMC_B_S0_SDA	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	42 45 48
SMBUS_SMC_0_S0_SCL	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	42 45 48 53 78
SMBUS_SMC_0_S0_SDA	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	42 45 48 53 78
SMBUS_SMC_BSA_SCL	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	7 42 45 61 62
SMBUS_SMC_BSA_SDA	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	7 42 45 61 62
SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	27 39 42 45
SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	27 39 42 45

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
CHGR_CSI	1TO1_DIFFPAIR		CHGR_CSI_P	62
	1TO1_DIFFPAIR		CHGR_CSI_N	62
CHGR_CSO	1TO1_DIFFPAIR		CHGR_CSO_P	62
	1TO1_DIFFPAIR		CHGR_CSO_N	62

SMC Constraints

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0

SCALE	SHT	OF
NONE	94	97

GDDR3 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR3_40R55SE	*	=55_OHM_SE	=40_OHM_SE	0.095 MM	12.7 MM	=STANDARD	=STANDARD
GDDR3_40SE	*	=40_OHM_SE	=40_OHM_SE	0.095 MM	=40_OHM_SE	=STANDARD	=STANDARD
GDDR3_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	0.095 MM	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR3_CLK	*	=2.5:1_SPACING	?
GDDR3_CMD	*	=2.5:1_SPACING	?
GDDR3_DATA	*	=2.5:1_SPACING	?
GDDR3_DQS	*	=2.5:1_SPACING	?

From T18 MXM:

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be within 150 ps. Inter-pair matching should be within 150 ps.
DIsplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

MUXGFX Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I148	LVDS_A_CLK	LVDS_100D	LVDS	LVDS_A_CLK_P	81 84
I149	LVDS_A_CLK	LVDS_100D	LVDS	LVDS_A_CLK_N	81 84
I199	LVDS_A_DATA	LVDS_100D	LVDS	LVDS_A_DATA_P<2..0>	81 84
I198	LVDS_A_DATA	LVDS_100D	LVDS	LVDS_A_DATA_N<2..0>	81 84
I152	LVDS_B_CLK	LVDS_100D	LVDS	LVDS_B_CLK_P	81 84
I153	LVDS_B_CLK	LVDS_100D	LVDS	LVDS_B_CLK_N	81 84
I201	LVDS_B_DATA	LVDS_100D	LVDS	LVDS_B_DATA_P<2..0>	81 84
I200	LVDS_B_DATA	LVDS_100D	LVDS	LVDS_B_DATA_N<2..0>	81 84
I183		LVDS_100D	LVDS	LVDS_CONN_A_CLK_F_P	7 80
I182		LVDS_100D	LVDS	LVDS_CONN_A_CLK_F_N	7 80
I184		LVDS_100D	LVDS	LVDS_CONN_B_CLK_F_P	7 80
I185		LVDS_100D	LVDS	LVDS_CONN_B_CLK_F_N	7 80
I190		LVDS_100D	LVDS	LVDS_CONN_A_CLK_P	80 81
I191		LVDS_100D	LVDS	LVDS_CONN_A_CLK_N	80 81
I192		LVDS_100D	LVDS	LVDS_CONN_A_DATA_P<2..0>	7 80 81
I193		LVDS_100D	LVDS	LVDS_CONN_A_DATA_N<2..0>	7 80 81
I194		LVDS_100D	LVDS	LVDS_CONN_B_CLK_P	80 81
I195		LVDS_100D	LVDS	LVDS_CONN_B_CLK_N	80 81
I196		LVDS_100D	LVDS	LVDS_CONN_B_DATA_P<2..0>	7 80 81
I197		LVDS_100D	LVDS	LVDS_CONN_B_DATA_N<2..0>	7 80 81
I161	DP_ML	DP_100D	DISPLAYPORT	DP_ML_C_P<3..0>	7 82
I160		DP_100D	DISPLAYPORT	DP_ML_C_N<3..0>	82
I155	DP_ML	DP_100D	DISPLAYPORT	DP_ML_P<3..0>	81 82
I157		DP_100D	DISPLAYPORT	DP_ML_N<3..0>	81 82
I202	DP_ML	DP_100D	DISPLAYPORT	DP_ML_CONN_P<3..0>	82
I203		DP_100D	DISPLAYPORT	DP_ML_CONN_N<3..0>	82
I159	DP_AUX_CH	DP_100D	DISPLAYPORT	DP_AUX_CH_C_P	81 82
I158	DP_AUX_CH	DP_100D	DISPLAYPORT	DP_AUX_CH_C_N	81 82

GDDR3 FB A/B Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
FB_A_CLK_P	GDDR3_80D	GDDR3_CLK	FB_A_CLK_P<0>	73 74
	GDDR3_80D	GDDR3_CLK	FB_A_CLK_N<0>	73 74
FB_B_CLK_P	GDDR3_80D	GDDR3_CLK	FB_A_CLK_P<1>	73 74
	GDDR3_80D	GDDR3_CLK	FB_A_CLK_N<1>	73 74
FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_A_MA<1..0>	73 74
FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_A_MA<12..6>	73 74
FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_A_BA<2..0>	73 74
FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_A_RAS_L	73 74
FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_A_CAS_L	73 74
FB_AB_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_A_WE_L	73 74
FB_AB_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB_A_UCKE	
FB_AB_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB_A_LCKE	
FB_AB_CS0	GDDR3_40R55SE	GDDR3_CMD	FB_A_LCS0_L	
FB_AB_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB_A_DRAM_RST	73 74
FB_A_CMD	GDDR3_40SE	GDDR3_CMD	FB_A_LMA<5..2>	73 74
FB_B_CMD	GDDR3_40SE	GDDR3_CMD	FB_A_UMA<5..2>	73 74
FB_A_WDQS0	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<0>	73 74
FB_A_WDQS1	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<1>	73 74
FB_A_WDQS2	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<2>	73 74
FB_A_WDQS3	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<3>	73 74
FB_A_RDQS0	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<0>	73 74
FB_A_RDQS1	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<1>	73 74
FB_A_RDQS2	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<2>	73 74
FB_A_RDQS3	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<3>	73 74
FB_A_DQ_BYTE0	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<7..0>	73 74
FB_A_DQ_BYTE1	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<15..8>	73 74
FB_A_DQ_BYTE2	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<23..16>	73 74
FB_A_DQ_BYTE3	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<31..24>	73 74
FB_A_DQM0	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<0>	73 74
FB_A_DQM1	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<1>	73 74
FB_A_DQM2	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<2>	73 74
FB_A_DQM3	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<3>	73 74
FB_B_WDQS0	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<4>	73 74
FB_B_WDQS1	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<5>	73 74
FB_B_WDQS2	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<6>	73 74
FB_B_WDQS3	GDDR3_40SE	GDDR3_DQS	FB_A_WDQS<7>	73 74
FB_B_RDQS0	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<4>	73 74
FB_B_RDQS1	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<5>	73 74
FB_B_RDQS2	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<6>	73 74
FB_B_RDQS3	GDDR3_40SE	GDDR3_DQS	FB_A_RDQS<7>	73 74
FB_B_DQ_BYTE0	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<39..32>	73 74
FB_B_DQ_BYTE1	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<47..40>	73 74
FB_B_DQ_BYTE2	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<55..48>	73 74
FB_B_DQ_BYTE3	GDDR3_40SE	GDDR3_DATA	FB_A_DQ<63..56>	73 74
FB_B_DQM0	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<4>	73 74
FB_B_DQM1	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<5>	73 74
FB_B_DQM2	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<6>	73 74
FB_B_DQM3	GDDR3_40SE	GDDR3_DATA	FB_A_DQM_L<7>	73 74

G96 Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	(CK505_DOT96)	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M	76 77
	CK505_CLK27MSS	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M_SS	76 77
	LVDS_EG_A_CLK	LVDS_100D	LVDS	LVDS_EG_A_CLK_P	78 84
	LVDS_EG_A_CLK	LVDS_100D	LVDS	LVDS_EG_A_CLK_N	78 84
	LVDS_EG_A_DATA	LVDS_100D	LVDS	LVDS_EG_A_DATA_P<2..0>	78 84
	LVDS_EG_A_DATA	LVDS_100D	LVDS	LVDS_EG_A_DATA_N<2..0>	78 84
	LVDS_EG_B_DATA	LVDS_100D	LVDS	LVDS_EG_B_DATA_P<2..0>	78 84
	LVDS_EG_B_DATA	LVDS_100D	LVDS	LVDS_EG_B_DATA_N<2..0>	78 84
I142	DP_ML	DP_100D	DISPLAYPORT	DP_EG_ML_P<3..0>	78 81
I144	DP_ML	DP_100D	DISPLAYPORT	DP_EG_ML_N<3..0>	78 81
I145	DP_AUX_CH	DP_100D	DISPLAYPORT	DP_EG_AUX_CH_P	78 81
I143	DP_AUX_CH	DP_100D	DISPLAYPORT	DP_EG_AUX_CH_N	78 81
I139		DP_100D	DISPLAYPORT	DP_EG_AUX_CH_C_P	81
I138		DP_100D	DISPLAYPORT	DP_EG_AUX_CH_C_N	81

GDDR3 FB C/D Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
FB_C_CLK_P	GDDR3_80D	GDDR3_CLK	FB_B_CLK_P<0>	73 75
	GDDR3_80D	GDDR3_CLK	FB_B_CLK_N<0>	73 75
FB_D_CLK_P	GDDR3_80D	GDDR3_CLK	FB_B_CLK_P<1>	73 75
	GDDR3_80D	GDDR3_CLK	FB_B_CLK_N<1>	73 75
FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_B_MA<1..0>	73 75
FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_B_MA<12..6>	73 75
FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_B_BA<2..0>	73 75
FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_B_RAS_L	73 75
FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_B_CAS_L	73 75
FB_CD_CMD	GDDR3_40R55SE	GDDR3_CMD	FB_B_WE_L	73 75
FB_CD_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB_B_UCKE	
FB_CD_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB_B_LCKE	
FB_CD_CS0	GDDR3_40R55SE	GDDR3_CMD	FB_B_LCS0_L	
FB_CD_CMD_PD	GDDR3_40R55SE	GDDR3_CMD	FB_B_DRAM_RST	73 75
FB_C_CMD	GDDR3_40SE	GDDR3_CMD	FB_B_LMA<5..2>	73 75
FB_D_CMD	GDDR3_40SE	GDDR3_CMD	FB_B_UMA<5..2>	73 75
FB_C_WDQS0	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<0>	73 75
FB_C_WDQS1	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<1>	73 75
FB_C_WDQS2	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<2>	73 75
FB_C_WDQS3	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<3>	73 75
FB_C_RDQS0	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<0>	73 75
FB_C_RDQS1	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<1>	73 75
FB_C_RDQS2	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<2>	73 75
FB_C_RDQS3	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<3>	73 75
FB_C_DQ_BYTE0	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<7..0>	73 75
FB_C_DQ_BYTE1	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<15..8>	73 75
FB_C_DQ_BYTE2	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<23..16>	73 75
FB_C_DQ_BYTE3	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<31..24>	73 75
FB_C_DQM0	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<0>	73 75
FB_C_DQM1	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<1>	73 75
FB_C_DQM2	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<2>	73 75
FB_C_DQM3	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<3>	73 75
FB_D_WDQS0	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<4>	73 75
FB_D_WDQS1	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<5>	73 75
FB_D_WDQS2	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<6>	73 75
FB_D_WDQS3	GDDR3_40SE	GDDR3_DQS	FB_B_WDQS<7>	73 75
FB_D_RDQS0	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<4>	73 75
FB_D_RDQS1	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<5>	73 75
FB_D_RDQS2	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<6>	73 75
FB_D_RDQS3	GDDR3_40SE	GDDR3_DQS	FB_B_RDQS<7>	73 75
FB_D_DQ_BYTE0	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<39..32>	73 75
FB_D_DQ_BYTE1	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<47..40>	73 75
FB_D_DQ_BYTE2	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<55..48>	73 75
FB_D_DQ_BYTE3	GDDR3_40SE	GDDR3_DATA	FB_B_DQ<63..56>	73 75
FB_D_DQM0	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<4>	73 75
FB_D_DQM1	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<5>	73 75
FB_D_DQM2	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<6>	73 75
FB_D_DQM3	GDDR3_40SE	GDDR3_DATA	FB_B_DQM_L<7>	73 75

GPU (G96) CONSTRAINTS

SYNC_MASTER=MUXGFX SYNC_DATE=02/18/2008

APPLE INC.	SIZE	DRAWING NUMBER	REV.
	D	051-7892	A.0.0
	SCALE NONE	SHT 95	OF 97

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
PP1V8_MEM	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_GTLREF	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM
FSB_DSTB	FSB_DSTB	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for GMCH fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_70D	BOTTOM			0.127 MM	6.35 MM		

Graphics ,SATA Constraint Relaxations

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
LVDS_100D	BGA	100_DIFF_BGA
DP_100D	BGA	100_DIFF_BGA
SATA_100D	BGA	100_DIFF_BGA

PGA CONSTRAINT RELAXATIONS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PGA_50SE	*	Y	=50_OHM_SE	0.073 MM	=50_OHM_SE	=1:1_DIFFPAIR	0.073 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FSB_50S	PGA	PGA_50SE
FSB_DSTB_50S	PGA	PGA_50SE
CPU_50S	PGA	PGA_50SE

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FSB_DATA	*	PGA	PGA_CPU
FSB_DSTB	*	PGA	PGA_CPU
FSB_ADDR	*	PGA	PGA_CPU
FSB_ADSTB	*	PGA	PGA_CPU
FSB_1X	*	PGA	PGA_CPU
CPU_AGTL	*	PGA	PGA_CPU
CPU_8MIL	*	PGA	PGA_CPU

K19 Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
		ENET_MDI_100D	ENETCONN	ENETCONN_P<3..0>	35
		ENET_MDI_100D	ENETCONN	ENETCONN_N<3..0>	35
I146		SATA_100D	SATA	SATA_ODD_R2D_UF_P	39
I145		SATA_100D	SATA	SATA_ODD_R2D_UF_N	39
I144		SATA_100D	SATA	SATA_ODD_D2R_UF_P	7 39
I142		SATA_100D	SATA	SATA_ODD_D2R_UF_N	7 39
I143		SATA_100D	SATA	SATA_HDD_D2R_UF_P	39
I140		SATA_100D	SATA	SATA_HDD_D2R_UF_N	39
I141		SATA_100D	SATA	SATA_HDD_R2D_UF_P	39
I139		SATA_100D	SATA	SATA_HDD_R2D_UF_N	39
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GFXIMVP6_VSEN_P	79
		SENSE_1TO1_55S	SENSE	GFXIMVP6_VSEN_N	79
I124	CPUTHMSNS_D2_DP	THERM_1TO1_55S	THERM	CPUTHMSNS_D2_P	48
I125		THERM_1TO1_55S	THERM	CPUTHMSNS_D2_N	48
I127	CPU_THERMD_DP	THERM_1TO1_55S	THERM	CPU_THERMD_P	10 48
I126		THERM_1TO1_55S	THERM	CPU_THERMD_N	10 48
I128	GPUTHMSNS_D_DP	THERM_1TO1_55S	THERM	GPUTHMSNS_D_P	48
I130		THERM_1TO1_55S	THERM	GPUTHMSNS_D_N	48
I129	GPU_THERMD_DP	THERM_1TO1_55S	THERM	GPU_TDIODE_P	48 76 77
		THERM_1TO1_55S	THERM	GPU_TDIODE_N	48 76 77
I138	MCPTHMSNS_D_DP	THERM_1TO1_55S	THERM	MCPTHMSNS_D_P	48
I137		THERM_1TO1_55S	THERM	MCPTHMSNS_D_N	48
I135	MCP_THERMD_DP	THERM_1TO1_55S	THERM	MCP_THMDIODE_P	21 48
I136		THERM_1TO1_55S	THERM	MCP_THMDIODE_N	21 48
I156	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPUVTTISNS_R_P	47
I157		SENSE_1TO1_55S	SENSE	CPUVTTISNS_R_N	47
I153	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	GPUISENS_P	47
I152		SENSE_1TO1_55S	SENSE	GPUISENS_N	47
I151	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPUVTT_ISNS_P	47 67
I150		SENSE_1TO1_55S	SENSE	CPUVTT_ISNS_N	47 67
I158	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V8GPU_P	47
I147		SENSE_1TO1_55S	SENSE	P1V8GPU_N	47
I186	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_CPU_P	46
I185		SENSE_1TO1_55S	SENSE	ISNS_CPU_N	46
			SB_POWER	PP3V3_S5	82 87 37 38 44 54 64 68 69 70 7 8 18 20 22 24 26 30 34
			SB_POWER	PP3V3_S0	48 49 51 55 59 60 63 68 6 7 8 13 18 19 21 22 24 25 28 29 37 39 43 45 47 69 70 77 80 81 82 84 85
I134	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	P1V8GPUISNS_R_P	47
I133		SENSE_1TO1_55S	SENSE	P1V8GPUISNS_R_N	47
I214	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_AIRPORT_P	31 53
I209		SENSE_1TO1_55S	SENSE	ISNS_AIRPORT_N	31 53
I215	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_AIRPORT_R_P	53
I216		SENSE_1TO1_55S	SENSE	ISNS_AIRPORT_R_N	53
I218	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_1V5_S3_R_P	53
I217		SENSE_1TO1_55S	SENSE	ISNS_1V5_S3_R_N	53
I210	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_1V5_S3_P	53 65
I211		SENSE_1TO1_55S	SENSE	ISNS_1V5_S3_N	53 65
I212	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_LCDBKLT_P	53 85
I213		SENSE_1TO1_55S	SENSE	ISNS_LCDBKLT_N	53 85
I219	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_LCDBKLT_R_P	
I220		SENSE_1TO1_55S	SENSE	ISNS_LCDBKLT_R_N	

K19 Specific Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
I164	(PCIE_EXCARD)	PCIE_90D	PCIE	PCIE_EXCARD_R2D_P	90
I162	(PCIE_EXCARD)	PCIE_90D	PCIE	PCIE_EXCARD_R2D_N	90
I163	(PCIE_MINI)	PCIE_90D	PCIE	PCIE_MINI_R2D_P	7 31 90
I161	(PCIE_MINI)	PCIE_90D	PCIE	PCIE_MINI_R2D_N	7 31 90
I160		CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_CONN_P	7 31
I159		CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_CONN_N	7 31
I168		1TO1_DIFFPAIR		CHGR_CSI_R_P	62
I166		1TO1_DIFFPAIR		CHGR_CSI_R_N	62
I167		1TO1_DIFFPAIR		CHGR_CSO_R_P	46 62
I165		1TO1_DIFFPAIR		CHGR_CSO_R_N	46 62
I182	(USB_EXTA)	USB_90D	USB	USB2_EXTA_MUXED_P	40
I181	(USB_EXTA)	USB_90D	USB	USB2_EXTA_MUXED_N	40
I179	(USB_EXTA)	USB_90D	USB	USB2_LT1_P	40
I180	(USB_EXTA)	USB_90D	USB	USB2_LT1_N	40
I177	(USB_EXTD)	USB_90D	USB	USB_TPAD_R_P	50
I178	(USB_EXTD)	USB_90D	USB	USB_TPAD_R_N	50
I176	(USB_CAMERA)	USB_90D	USB	USB_CAMERA_CONN_P	7 31
I175	(USB_CAMERA)	USB_90D	USB	USB_CAMERA_CONN_N	7 31
I174		USB_90D	USB	CONN_USB2_BT_P	7 31
I172		USB_90D	USB	CONN_USB2_BT_N	7 31
I173		USB_90D	USB	USB_LT2_P	40
I171		USB_90D	USB	USB_LT2_N	40
I169	USB_CARDREADER	USB_90D	USB	USB_CARDREADER_P	9 20 32
I170		USB_90D	USB	USB_CARDREADER_N	9 20 32
I183		DP_100D	DISPLAYPORT	DP_IG_AUX_CH_C_P	81
I184		DP_100D	DISPLAYPORT	DP_IG_AUX_CH_C_N	81
I198	SPK_OUT	DIFFPAIR	AUDIO	SPKRCONN_L_OUT_P	7 58 59
I197		DIFFPAIR	AUDIO	SPKRCONN_L_OUT_N	7 58 59
I201	SPK_OUT	DIFFPAIR	AUDIO	SPKRCONN_S_OUT_P	7 58 59
I200		DIFFPAIR	AUDIO	SPKRCONN_S_OUT_N	7 58 59
I199	SPK_OUT	DIFFPAIR	AUDIO	SPKRCONN_R_OUT_P	7 58 59
I202		DIFFPAIR	AUDIO	SPKRCONN_R_OUT_N	7 58 59
I206		DIFFPAIR	AUDIO	SPKRAMP_L_OUT_P	58
I207		DIFFPAIR	AUDIO	SPKRAMP_L_OUT_N	58
I204		DIFFPAIR	AUDIO	SPKRAMP_R_OUT_P	58
I205		DIFFPAIR	AUDIO	SPKRAMP_R_OUT_N	58
I208		DIFFPAIR	AUDIO	SPKRAMP_S_OUT_P	58
I203		DIFFPAIR	AUDIO	SPKRAMP_S_OUT_N	58
I223	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_ODD_P	39 53
I224		SENSE_1TO1_55S	SENSE	ISNS_ODD_N	39 53
I226	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_ODD_R_P	53
I225		SENSE_1TO1_55S	SENSE	ISNS_ODD_R_N	53
I221	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HDD_P	39 53
I222		SENSE_1TO1_55S	SENSE	ISNS_HDD_N	39 53
I227	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HDD_R_P	53
I228		SENSE_1TO1_55S	SENSE	ISNS_HDD_R_N	53
			GND	GND	

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S_VDD OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D_VDD OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_90D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
USB_90D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MEM_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MII_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_USB_RBIAS OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	ISL4,ISL9 OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S_VDD OVERRIDE	ISL3,ISL10 OVERRIDE	N OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D OVERRIDE	ISL4,ISL9 OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D_VDD OVERRIDE	ISL3,ISL10 OVERRIDE	N OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE

Ground-referenced memory signals (DQ,DQM,DQS) MAY route on ISL9 (VDD-referenced plane)but not next to VDD island.
Forces power-referenced memory signals (CLK,ADDR,CTRL) to not route on ISL3, ISL4 & ISL10(GND-referenced planes).

Project Specific Constraints

SYNC_MASTER=MUXGFX SYNC_DATE=02/21/2008

NOTICE OF PROPRIETARY PROPERTY

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 96	OF 97

K19 Board-Specific Spacing & Physical Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,PGA	MM	15.5.1

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	33.6 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
BGA_P3MM	*	=DEFAULT	?
PGA_CPU	*	0.073 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P2MM
CLK_FSB	*	BGA	BGA_P2MM
CLK_PCIE	*	BGA	BGA_P2MM
CLK_SLOW	*	BGA	BGA_P2MM
FSB_DSTB	FSB_DSTB	BGA	BGA_P3MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
1.8:1_SPACING	*	0.18 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

NOTE:From T18 MLB, changed to reflect M99 stackup.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.135 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.250 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3,ISL4	Y	0.160 MM	0.160 MM		0.175 MM	0.175 MM
70_OHM_DIFF	ISL9,ISL10	Y	0.160 MM	0.160 MM		0.175 MM	0.175 MM
70_OHM_DIFF	ISL2,ISL11	Y	0.170 MM	0.170 MM		0.150 MM	0.150 MM
70_OHM_DIFF	TOP,BOTTOM	Y	0.170 MM	0.095 MM		0.150 MM	0.150 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3,ISL4	Y	0.125 MM	0.125 MM		0.180 MM	0.180 MM
80_OHM_DIFF	ISL9,ISL10	Y	0.125 MM	0.125 MM		0.180 MM	0.180 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.140 MM	0.140 MM		0.190 MM	0.190 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.140 MM	0.095 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL4	Y	0.102 MM	0.102 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL9,ISL10	Y	0.102 MM	0.102 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.115 MM	0.115 MM		0.230 MM	0.230 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.095 MM		0.230 MM	0.230 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL4	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL9,ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL2,ISL11	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3,ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9,ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3,ISL4	Y	0.077 MM	0.077 MM		0.330 MM	0.330 MM
110_OHM_DIFF	ISL9,ISL10	Y	0.077 MM	0.077 MM		0.330 MM	0.330 MM
110_OHM_DIFF	ISL2,ISL11	Y	0.077 MM	0.077 MM		0.330 MM	0.330 MM
110_OHM_DIFF	TOP,BOTTOM	Y	0.077 MM	0.077 MM		0.330 MM	0.330 MM

PCB Rule Definitions

SYNC_MASTER=M99_MLB SYNC_DATE=01/22/2008

APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7892	A.0.0
SCALE NONE	SHT 97	OF 97